Ankicaster — Game Design Document & Implementation Specification

An Anki add-on school-of-magic game, inspired by the base game of Spellcaster University

9 November 2026

Version: GDD v1.0, expanding approved Concept Draft 8 **Date:** 9 November 2026 **Platform:** Anki desktop add-on (Anki 2.1.66+ / 23.x / 24.x / 25.x, Qt6), Python + aqt/PyQt6, HTML/JS front end in an `AnkiWebView` **Inspiration credit:** Ankicaster's school-building loop, room-and-furniture logic, card draft, creatures and alchemy are *inspired by* the base game of *Spellcaster University* (Sneaky Yak Studio / Lion's Shade). No DLC content is used. Ankicaster contains no assets, text, names of specific cards, or code from that game; every room, item, creature and number in this document is original to Ankicaster.

How to use this document

This document is two things at once:

1. **A game design document** — what Ankicaster is, why it is fun, and how every system works, with concrete content (rooms, furniture, creatures, potions, events, spells) and numbers.
2. **An implementation specification** — the data model, algorithms, hooks, files, milestones and tests an AI coding assistant (or a human developer) needs to build a working, genuinely playable first version.

If you are an AI builder, read **Section 1 (Instructions for the AI Builder)** first, then **Section 2 (Binding Rules Register)**. Every other section can be implemented in the milestone order given in **Section 39**. Content catalogs (rooms, furniture, creatures, etc.) are written as tables so they can be converted directly into the JSON content files listed in **Section 38.15**.

Conventions used throughout:

- **"Anki card"** always means a flashcard in the user's collection. **"Draft card"** always means an in-game card drawn from a draft deck. Code must never mix the two (`anki_card_id` vs `draft_card_id`).
- **"Review"** means one honest answer to an Anki card that Anki logged, made after Ankicaster was installed (full definition in Section 5.2).
- **"Bell"** is the game's simulation tick: one bell rings every **10 honest reviews**. The school only "lives" (needs, production, hazards) when bells ring. Real-calendar days matter only for the systems the concept explicitly ties to days (Veil upkeep, teacher delay, intervals).
- **"Day"** means the Anki scheduling day (respecting the user's "next day starts at" setting), not the midnight clock.
- **Rule IDs** like **BR-07** refer to the Binding Rules Register (Section 2). Never implement anything that breaks a BR rule.
- All numbers in this document are **starting values to tune**. This is stated once here and again in the Economy section (Section 33); it is not repeated on every line.

Table of contents

1. Instructions for the AI Builder
2. Binding Rules Register
3. Vision, Pillars and Player Fantasy
4. Glossary
5. The Honest Mana Economy
6. Core Loops and the Shape of a Session

7. The Player Wizard
8. The Five Schools of Magic
9. Students and Card Linking
10. Skill, Mastery and the Master Title
11. Careers and Faculty
12. Enrollment, Recruitment and the Enrollment Tracker
13. Retirement and Alumni
14. Student Traits, Personalities and Needs
15. The School Map, Construction and Room Quality
16. Room Catalog
17. Furniture and Decor Catalog
18. The Veil and Upkeep
19. Hazards and Wards
20. Misbehavior
21. Food, Gardens and the Greenhouse
22. Creatures and Taming (Bestiary)
23. Injuries, Health and Healers
24. Alchemy and Potions
25. Materials, Essences and Artifacts
26. The Card Draft
27. The Rift and Expeditions
28. Random School Events
29. Wizard Spells and Abilities
30. Game Modes: Timed Term and Endless School; Reset Rules
31. Spellbooks and Alumni Bonuses (Meta Progression)
32. Achievements and Milestones
33. Economy Balance Tables
34. Progression Pacing
35. Tutorial and Onboarding
36. UI/UX Specification
37. Art Direction and Audio
38. Technical Implementation Specification
39. Build Milestones and Acceptance Criteria
40. Testing Plan
41. Appendix A — Formula Reference
42. Appendix B — Design Decisions Made in This GDD (for review)

1. Instructions for the AI Builder

Read this section before writing any code. It tells you what to build first, what must never break, and how to make decisions when this document is silent.

1.1 What you are building

Ankicaster is an Anki desktop add-on. While the user studies normally in Anki, every honest review they answer earns **neutral mana**. In a separate game window (opened from Anki's **Tools** menu or a toolbar button) the user spends that mana to train their own wizard or witch, build a magic school room by room, furnish it, enroll students who are each *linked to a set of the user's own young Anki cards*, raise those students to mastery as the linked cards mature, give masters careers, tame creatures, brew potions, keep the protective **Veil** paid, and eventually send adventurers through **the Rift**.

The game is cozy, idle-flavoured and slow-burning. It is meant to be opened for two to ten minutes before or after a study session, and to make the user feel that their real learning is building something visible.

1.2 Build order (do it in this order)

Build in vertical slices. Each slice must be playable and must pass its acceptance criteria (Section 39) before you start the next.

Order	Slice	Why first
1	**Mana ledger** from honest reviews (revlog-verified, post-install only, undo-safe) and the post-review toast	Everything depends on it; it is where the guardrails live
2	**Save system** (SQLite in `user_files/`, per Anki profile, migrations)	Nothing else is safe without durable state
3	**Wizard and training sessions** (orb sessions, 25×n reviews per level, buffs at 6–10)	The idle early game; playable on day one
4	**School grid and construction** (standard plans, materials, mana cost, build time in reviews)	Gives the user something to do with mana
5	**Students, card linking, enrollment tracker, skill and mastery**	The heart of the concept
6	**Daily tick, Veil upkeep and break handling**	Required before anything can "go wrong"
7	Needs, food, refectory, dorms, greenhouse	Makes the school feel alive
8	Careers and faculty limit, teacher delay	Turns masters into a second engine
9	Furniture and room quality	The main decorating/optimizing toy
10	Card draft	Variety and surprise

Order	Slice	Why first
11	Creatures and taming	Collection fun
12	Hazards, wards, misbehavior, injuries, infirmary	Light pressure
13	Alchemy and potions	Crafting layer
14	The Rift	Late-game adventure
15	Events, spells, achievements, game modes, meta progression	Long-term retention

The **MVP slice** that should be shown to the user first is orders 1–6 plus the Dormitory, Refectory, one Classroom per school, Faculty Quarters, Storage, Woodlot, Quarry and Vegetable Patch. It must already be fun: training, building, enrolling, watching skill bars rise.

1.3 Guardrails you must never break

These come from the user's approved concept and are non-negotiable. They are also listed with IDs in Section 2. If any instruction anywhere else seems to conflict with these, these win.

1. **Never write to the Anki collection.** No scheduling changes, no edits to cards, notes, decks, tags, flags, revlog or config. All collection access is read-only (`SELECT` only). Do not call any scheduler method that changes state. Do not add, bury, suspend, reschedule, or "set due date". Do not hook into the answer path in a way that changes the answer.
2. **Mana comes only from honest reviews made after install.** Past history grants nothing, ever: no backfill, no "welcome bonus based on your collection", no retroactive mana for imported revlogs.
3. **The game never rewards review count, review speed, adding cards, or pressing Good over Again.** All answer buttons earn identical mana. There are no streaks, no combo multipliers, no "do 200 reviews today" goals, no achievements for review totals, no bonuses for answering fast, no rewards for new cards added. Training and construction progress by reviews because the concept says so, but nothing escalates the reward for doing *more* reviews per day.
4. **Breaks never delete progress.** If the user stops studying for a day, a month or a year, nothing is lost: no characters, card links, titles, buildings, furniture, creatures, wizard levels, spellbooks, alumni or banked mana are deleted. The school "sleeps".
5. **Card intervals are read, never written.** Student skill, mastery and retirement are derived from `cards.ivl` (days) as Anki reports it.
6. **Cards are never shared between characters.** One Anki card can be linked to at most one character at a time.
7. **No deaths.** Characters and creatures can be injured or cursed on a Rift expedition, or escape for a while; they always recover or come back.
8. **The add-on must never crash or slow down reviewing.** All game work triggered from a review hook must finish in under 5 ms on the hook thread; anything heavier is deferred (`mw.progress.single_shot` / `QTimer`) and wrapped in `try/except` that logs and continues. A game bug must never block the user from studying.

1.4 How to make decisions this document does not cover

- Prefer the option that is **kinder to a user who took a break**.

- Prefer **data-driven content** (JSON in `content/`) over hard-coded content, so numbers can be tuned without code changes.
- Prefer **deterministic, seeded randomness** (Section 38.11) so bugs are reproducible and save-scumming gains nothing.
- Prefer **fewer, clearer numbers** on screen. Every modifier a player sees must show its source in a tooltip.
- If a feature would require writing to the collection, **drop the feature**.

1.5 Definition of “workable and genuinely fun” for v1

A first version is acceptable when a user can install it, study as usual, and over the first week: see mana arrive after reviews; train their wizard to level 5 in a school; build at least six room types and furnish them; enroll students whose skill bars visibly follow their real cards; get a creature; meet at least three random events; and never once be told they did something wrong by studying honestly. Over the first two months they should reach their first master and see the Rift on the horizon.

2. Binding Rules Register

Every rule below is a decision from approved Concept Draft 8 and is binding. This GDD expands them but does not change them. Code reviews and tests should cite these IDs.

ID	Rule	Where expanded
BR-01	There is one **neutral mana pool**. Mana is earned only from honest reviews done **after install**. Past Anki history grants nothing.	5
BR-02	Ankicaster **never alters Anki scheduling** or any collection data.	1.3, 38.4
BR-03	The game **never rewards review count or speed, adding cards, or pressing Good over Again**.	5.4
BR-04	**Breaks never delete progress.**	18.5, 38.10
BR-05	The player has their **own wizard or witch**. The early game is idle-style.	7
BR-06	**Training sessions** (pondering an orb, studying a grimoire) cost mana and finish after N reviews.	7.3
BR-07	Wizard **level n needs 25×n reviews** (25 to 250; 1,375 reviews to reach level 10). Maximum level **10 per school**.	7.4
BR-08	Wizard **level 5 in a school unlocks teaching that school**.	7.5
BR-09	Wizard levels **6–10 give percentage buffs** per level: **Arcane +2% mana**; **Nature +3% food, herbs and taming**; **Light +3% recovery and teaching**; **Shadow −3% Veil cost and −3% hazards and misbehavior**; **Alchemy −3% build time and materials, +3% potion yield**.	7.6
BR-10	**Five schools:** Arcane, Nature, Light, Shadow, Alchemy.	8
BR-11	**Careers:** teacher, scholar, gardener, tamer, healer, warder, alchemist, builder; **adventurer** from any school.	11
BR-12	Each **student links to the 50 youngest free Anki cards**. The number is a setting (**default 50**) and **applies to all students**.	9, 38.14
BR-13	**Cards are never shared** between characters.	9.3
BR-14	**Card age = review interval.**	9.2
BR-15	**Skill tracks intervals and falls on lapses.**	10.1

ID	Rule	Where expanded
BR-16	**Mastery** happens when **all** of a student's cards are **over 30 days plus** a **teacher delay = 30 − the average interval of the teacher's 20 teaching cards**, in **real days**; the delay is **0 once all 20 teaching cards are over 30 days**.	10.3
BR-17	The **master title is permanent**.	10.5
BR-18	**Careers add the 20 youngest free cards** each. (Young free cards are taken first; older free cards are used only when not enough young ones exist. This fallback is a GDD decision.)	9.4, 11.2
BR-19	**Career switching** is only for **masters of 2+ schools**; **old career cards are kept**.	11.5
BR-20	**Second-school enrollment adds a fresh card set.**	12.6
BR-21	**Retirement** happens when **all** of a character's cards pass **retirement age** (setting, default **90** days).	13
BR-22	**Faculty limit** comes from **wizard levels, housing, classrooms and student count**.	11.6
BR-23	The **enrollment tracker** shows **free young cards vs needed** and **blocks enrollment** when there are not enough.	12.3
BR-24	**Veil upkeep** is daily mana that scales with **school size (buildings, rooms, grounds)**, **not headcount**. It is lowered by **Shadow wizard levels, warders, some creatures, Arcane research and draft cards**.	18.2
BR-25	**Unpaid Veil thins**: hazards and **Rift breaches** become more likely; **nothing is deleted**.	18.4, 19.4
BR-26	**While away**, banked mana auto-pays the Veil until the bank falls to **10% of what it was at the last review**; then payments stop and the **Veil holds where it is (it does not thin)**.	18.5
BR-27	**The Rift opens at 3 masters.** Only **adventurers** go. **Teams of 1–3** (3 is best). **Five destinations**, one per tier (**easy / mild / medium / hard / deadly**), with a **chance meter**, odds, modifiers, a "when it goes wrong" table, and **no deaths: nobody dies or is lost; the worst case is a long infirmary recovery**.	27

ID	Rule	Where expanded
BR-31	**Rift destinations.** Skill needed = the **average interval of the party's adventurer career cards**. Safe odds = the chance of returning unscathed for a team of three at exactly that skill. Best odds = the cap. Mossy Hollow: Easy, 0 days, 85%, 95%, 50 reviews, herbs, timber, common eggs. Whispering Fen: Mild, 10 days, 75%, 90%, 100 reviews, rare herbs, owls and sprites. Crystal Caverns: Medium, 20 days, 65%, 85%, 150 reviews, crystals, stone, golem cores. Ashen Wastes: Hard, 30 days, 55%, 80%, 250 reviews, fire creatures, rare materials. The Shattered Spire: Deadly, 45 days, 45%, 70%, 400 reviews, rare cards, artifacts, legendary eggs. **Mana cost grows with each tier.**	27.3, 27.4
BR-32	**Rift odds modifiers:** each 5 days of skill below needed −10%; each 5 days above +5%, up to the cap; team of two −10%; going alone −25%; Rift Elixir +10%, up to the cap; Thin Veil at home −5%.	27.5
BR-33	**When it goes wrong** (Easy–Medium / Hard–Deadly): minor injury with full rewards 60% / 40%; serious injury with half rewards 25% / 30%; curse on one adventurer 10% / 20%; retreat with nothing 5% / 10%.	27.6
BR-34	**Timed-term reset detail:** buildings, materials, unspent mana, students, faculty and creatures reset (their cards become free again). Wizard levels, alumni bonuses and spellbooks carry over. **Anki cards are never touched.**	30.4
BR-35	**Settings defaults:** cards per student **50** (applies to all students), retirement age **90** days, exam timer **off**.	30.2, 38.14
BR-28	Systems in scope: **food/greenhouse; creatures/taming; injuries/healers; misbehavior; hazards/wards; construction (blueprint + materials + mana + build time in reviews); potions; student needs; card draft**.	14–26
BR-29	**Two modes: timed term** (exam timer optional) vs **endless school**.	30

ID	Rule	Where expanded
BR-30	**Reset rules:** in timed terms **creatures and characters reset**; **wizard levels, alumni bonuses and spellbooks carry over**.	30.4

BR-01 to BR-35 have been checked against the Draft 8 page. Where this GDD adds detail inside a binding rule (for example, Rift mana costs or the young-first career fallback), the addition is marked as a GDD decision and listed in Appendix B. # 3. Vision, Pillars and Player Fantasy

3.1 Elevator pitch

Your flashcards are your students. Every honest review you do in Anki pours a little neutral mana into your tower. Spend it to train your own wizard or witch through five schools of magic, raise a castle-school room by room, and enroll students who are each bound to fifty of your youngest cards. As your real memories mature, your students grow into masters, take careers, tame creatures, brew potions, and finally step through the Rift. Take a week off and the school simply sleeps under its Veil; nothing is lost.

3.2 Player fantasy

- **“I am a wizard, and I am getting better.”** The player’s own character is the first thing that grows, quickly and visibly, before any students exist.
- **“My learning is building something.”** The school is a physical, cross-section castle that gets taller, fuller and more beautiful as the user studies. Each student’s skill bar is a mirror of real cards.
- **“I run a magic school.”** Classic school-sim pleasures inspired by Spellcaster University’s base game: picking rooms from a draft, fitting them into a grid, furnishing them for quality, keeping students fed, rested and out of trouble, collecting creatures, brewing potions, and sending a party into a dangerous place for loot.
- **“It never nags me.”** No streaks, no guilt, no lost progress. The Veil protects the school while the user is away.

3.3 Design pillars

Pillar	Meaning	Concrete test
Honest study is the only fuel	Mana is a by-product of normal reviewing. The game must never change *how* a user studies.	A user who presses Again on every lapse earns exactly the same mana as one who presses Good.
The cards are the clock	Student progress, mastery and retirement come from real card intervals; the game reads them, never writes them.	A student’s skill bar can be reproduced from `cards.ivl` alone.
Cozy pressure, never punishment	Hazards, misbehavior and hunger create small, readable problems that are fun to fix. Nothing permanent is ever taken away.	No code path deletes a character, card link, building, creature or banked mana.

Pillar	Meaning	Concrete test
A school worth looking at	Rooms and furniture visibly change the castle; quality tiers have different art; creatures wander.	A screenshot of week four looks very different from day one.
Short visits, long arcs	Every visit has a 2-minute action (collect, queue, place) and every month has a milestone (master, creature, Rift).	Each session ends with at least one "next thing" queued.

3.4 What Ankicaster borrows from Spellcaster University (inspiration only)

Spellcaster University base-game idea	Ankicaster translation
Side-on cross-section school built room by room, with footprints, connections and "air above" rooms	School map grid with footprints, support, connections and sky-access rooms (Section 15)
Draw three cards, keep one, prices rising per draw	Draft decks per school plus a Foundation deck, draw-3-keep-1, prices that rise with recent draws (Section 26)
Duplicate room cards level a room (+10% efficiency per level)	Duplicate blueprints level rooms to L2/L3 (+10% per level) (Section 15.7)
Item slots and furniture that change room effectiveness	Required and optional furniture with room stats and quality tiers (Section 17)
Five schools of magic, each with its own colored mana	Five schools; mana is neutral (BR-01), but each school's students produce a school **essence** material (Section 25)
Students with traits, needs (hunger, tiredness, boredom, health, sanity) and misbehavior	Students with personalities, traits, and needs (Hunger, Rest, Leisure, Health, Composure) (Section 14)
Graduation into Futures with permanent alumni bonuses that carry across a campaign	Retirement and term-end graduation into alumni bonuses that carry over (Sections 13, 31)
Familiars and creatures that give buffs	Bestiary of 32 creatures with pens, taming, diets and escape trouble (Section 22)
Ingredients combined three at a time into potions	Alchemy lab recipes of three ingredients, discovered by experiment (Section 24)
Dungeon delves with a party of students	The Rift: teams of 1–3 adventurers, five destinations from Easy to Deadly, gated by how well the user really remembers the adventurers' cards (Section 27)
A looming clock (the Dark Lord)	Replaced by the gentle Veil upkeep and the optional exam timer of a Timed Term (Sections 18, 30)
Spell Books chosen between maps based on stars earned	Spellbooks earned from term charters and Endless milestones (Section 31)

What Ankicaster deliberately does **not** borrow: a hard fail clock in Endless mode, death as an outcome, gold and wages, faction reputation diplomacy, and anything from DLCs.

3.5 Target user and session shape

The reference user does **100–200 reviews per day** in sessions of 15–45 minutes and adds new cards regularly (roughly 10–25 new cards a day). The game must also be pleasant for a user at 30 reviews/day (slower, but nothing breaks) and must not become trivially exploitable at 800 reviews/day (the only gain from more reviews is the same linear mana everyone gets).

4. Glossary

Term	Definition
Honest review	An answer to an Anki card logged in the revlog after install, made through normal studying, that passes the honesty checks in Section 5.2.
Mana	The single neutral currency, earned only from honest reviews. Stored in the **mana bank**.
Bell	Simulation tick; rings once per 10 honest reviews. Needs, production, yields, hazards and misbehavior are evaluated per bell.
Day tick	Processing that runs once per Anki day rollover: Veil upkeep, interval refresh, mastery and retirement checks, applicant board refresh, daily events.
Wizard	The player's own character (wizard or witch, player's choice of name, look and pronouns). Has a level 0–10 in each school.
Training session	A mana-paid activity on the wizard ("Ponder an Orb" for school levels, "Study a Grimoire" for spells) that completes after N honest reviews.
School	One of Arcane, Nature, Light, Shadow, Alchemy. Also refers to the player's whole institution ("the school"); context disambiguates, and the UI calls the institution "your Academy" to avoid confusion.
Academy	The player's institution: castle grid plus grounds.
Character	Any student, faculty member or alumnus other than the wizard.
Student	A character enrolled in a school, linked to a study set of Anki cards.
Master	A character who has mastered at least one school. Title is permanent.
Career	A job taken by a master (teacher, scholar, gardener, tamer, healer, warder, alchemist, builder, adventurer). Adds 20 career cards.
Faculty	Characters holding careers.
Card link	The binding between one Anki card and one character, in one **card set** (study set or career set).
Free card	An eligible Anki card not linked to any character.
Young card	A card whose interval is ≤ 30 days (setting `young_threshold_days`).
Skill	0–100 score per study set derived from linked card intervals.
Teacher delay	Real days a mastery-ready student must wait, derived from the teacher's career-card intervals (BR-16).
Retirement age	Interval every linked card must exceed for a character to retire (default 90 days).

Term	Definition
Alumni	Retired characters (and, in Timed Terms, term-end graduates). Grant permanent alumni bonuses.
Veil	The protective enchantment over the Academy. Costs daily mana upkeep based on Academy size. Has a **strength** 0–100.
Hazard	A Veil-related magical problem (e.g., wild growth, poltergeist).
Breach	A serious hazard possible only when the Veil is thin: something gets in.
Misbehavior	Student-caused trouble (pranks, spills, unauthorized summoning).
Room	A placed building unit on the grid or grounds, with footprint, level, furniture, cleanliness and efficiency.
Plot	A grounds room (outdoor), e.g., Woodlot, Quarry, Paddock.
Furniture	Placeable items inside rooms; required or optional; add room stats.
Room stats	Comfort, Focus, Order, Aura, Charm.
Quality	A room's furniture score, giving a tier (Bare → Splendid) and an efficiency bonus.
Blueprint	A draft card or standard plan that permits building a room.
Materials	Timber, Stone, Glass, Cloth (common) and rare materials (Crystal, Moonsilk, Starmetal, etc.).
Essences	School-specific materials produced by students' lessons: Insight (Arcane), Herbs (Nature), Radiance (Light), Umbra (Shadow), Reagents (Alchemy).
Provisions / Greens / Fish / Nectar	Food types. Provisions feed people; creatures eat various foods and essences.
Draft	Pay mana to draw 3 draft cards from a deck and keep 1.
Rift	The late-game expedition system that opens at 3 masters.
Expedition	A Rift trip by a team of 1–3 adventurers to one destination (each destination has its own tier); resolves after that destination's number of reviews.
Chance meter	The displayed chance that a planned expedition returns safely (unscathed, full rewards).
Renown	The Academy's standing; earned from masters, alumni, decor, achievements. Unlocks applicant quality and draft rarity weights. Never affects mana.
Timed Term	A mode with a charter (goals), an optional exam timer, and a reset at the end.
Endless School	A mode with no reset.
Spellbook	A permanent meta-progression passive, carried across terms.

5. The Honest Mana Economy

5.1 Role and player goal

Mana is the only currency that comes from outside the game. It is the bridge between real study and the Academy. The player's goal is to spend a steady, honest trickle of mana well: training the wizard, building, drafting, keeping the Veil paid, and funding expeditions. The design goal is that the *only* way to get mana is to study the way you would have studied anyway.

5.2 What counts as an honest review

A review grants mana if and only if **all** of these are true. The check is performed against the **revlog** (the source of truth), not against the UI event alone.

#	Check	Rule	Reason
H1	Post-install	`revlog.id` (a millisecond timestamp) is greater than `install_ms`, recorded the first time Ankicaster ever runs on this profile.	BR-01: past history grants nothing.
H2	Real answer	`revlog.ease` is 1, 2, 3 or 4.	Excludes manual reschedules, "set due date", resets and other non-answers.
H3	Study entry type	`revlog.type` is learn (0), review (1), relearn (2) or filtered/early review (3). Any other type (manual, rescheduled, future types) is ignored.	Only genuine study answers.
H4	Not already counted	`revlog.id` is not already in the mana ledger.	Idempotence across hooks, sync catch-up and restarts.
H5	Not a mash	`revlog.time` (answer time in ms) ≥ `min_answer_ms` (setting, default **800 ms**). Entries below this are logged as "too quick to count" and grant nothing.	Discourages button-mashing without rewarding speed; normal answers are never affected.
H6	Per-card daily ceiling	At most `max_counted_answers_per_card_per_day` (default **6**) answers to the same Anki card count per Anki day.	Stops "review the same card in a loop" exploits; ordinary learning steps stay below this.
H7	Card still exists	The card id exists in `cards` at the time of counting (or existed at the time of the hook).	Deleted-card spam cannot farm mana.

All four answer buttons are equal. **Again, Hard, Good and Easy grant identical mana** (BR-03). The answer time affects nothing except check H5.

5.3 How much mana

Quantity	Starting value
Base mana per honest review	**10**
Arcane wizard buff (levels 6–10)	+2% per level, up to **+10%** (BR-09)
Other mana multipliers (spellbooks, alumni)	Listed in Section 31; combined with Arcane they are capped at **+25%** total
Mana bank cap	None
Fractional mana	Tracked as a float; displayed rounded down

Formula: `mana_gain = mana_per_review × (1 + min(0.25, arcane_bonus + other_bonuses))`.

There is deliberately **no** daily bonus, first-review-of-the-day bonus, streak, combo, accuracy bonus, speed bonus, “review goal” chest or new-card bonus (BR-03).

5.4 Anti-incentive guarantees (BR-03)

Temptation the game must not create	How Ankicaster avoids it
Press Good instead of Again to keep a student’s skill up	Skill falls when the interval falls, which is the honest outcome of a lapse, but mana is identical for every button and **no student, building or creature is ever harmed by a lapse**. The UI explicitly says “Lapses are part of learning — your student will relearn with you.” There is no penalty event triggered by lapses.
Do more reviews to unlock things faster	Rewards scale linearly with honest reviews and nothing else. There are no review-count achievements, no daily review goals and no escalating multipliers.
Rush through reviews	No speed reward. Too-fast answers (< 0.8 s) earn nothing.
Add lots of cards to get more students	Adding cards earns nothing. More young cards only *permit* enrollment (BR-23); they do not grant mana or resources. The enrollment tracker never says “add more cards”; it says “students will be able to enroll as you learn new cards”.
Use custom study / review-ahead to farm	Allowed (they are real reviews) but capped by check H6, and early reviews do not raise intervals much, so they cannot speed mastery.
Suspend or delete hard young cards to speed mastery	Handled by the linking rules in Section 9.6 (suspended cards stay linked; deleted cards are replaced by the next youngest free card).

5.5 Where mana comes from in practice

1. **Live reviews on desktop:** detected through `gui_hooks.reviewer_did_answer_card`. The hook only records the card id and schedules a deferred revlog check (Section 38.6).
2. **Reviews done on other devices (AnkiMobile, AnkiDroid, AnkiWeb) after install:** these are honest post-install reviews. When the setting `count_reviews_from_other_devices` is on (default **on**), after each sync and on profile open the add-on scans the revlog for entries newer than the ledger watermark and newer than `install_ms`, applies H1–H7, grants mana and advances bells. A toast says “+370 mana from 37 reviews on other devices”.

3. Nothing else. No potion, creature, event, spellbook or alumni bonus creates mana from nothing; they can only multiply honest review mana (capped at +25%) or reduce costs.

5.6 Undo

If the user presses Undo in the reviewer (`gui_hooks.review_did_undo`) or uses Edit → Undo, Anki removes the revlog entry. On the undo hook (and on every ledger reconciliation) the add-on checks whether each recently counted revlog id still exists. For each missing one it reverses exactly what that review granted: the mana, one review of progress on each session, construction, taming and expedition that the review advanced, and the bell counter. If the mana has already been spent, the shortfall becomes **mana debt** that is paid off from the next earnings; the bank never displays a negative number, and nothing already bought is taken back.

5.7 Mana sinks (overview)

Sink	Typical size	Section
Wizard training sessions (orbs)	80 × level (80–800 per level)	7
Grimoire study (spells)	150–1,200 per spell	29
Construction	25–2,500 per room	15–16
Furniture	20–900 per item	17
Veil upkeep	~40–900 per day depending on Academy size	18
Card draft	120–600 per draw	26
Spell casting	30–400 per cast	29
Rift expedition launch	150–1,200 per expedition	27
Applicant board reroll	100	12
Recipe research, creature lures, misc.	50–300	22, 24

5.8 Design purpose and tensions

The economy has one faucet and many sinks. The faucet is honest and linear; the fun is in allocation. Early on, nearly all mana goes into training and the first rooms. In the mid game the Veil becomes a meaningful fixed cost that makes the player think about Academy size, warding, and the Shadow school. Late game, the draft and the Rift absorb surplus. Because the Veil auto-pays while away only down to 10% of the bank (BR-26), a large bank is a buffer for holidays, not just a score.

6. Core Loops and the Shape of a Session

6.1 Nested loops

Loop	Time scale	What happens	Player decisions
A. Review loop	Seconds	User answers an Anki card → honest-review check → +10 mana → every 10th review rings a **bell**.	None (the user just studies).
B. Bell loop	Every 10 reviews	Active training, constructions, taming and expeditions advance (they advance per review; bells just batch the UI). Needs decay and are serviced; gardens, plots and lessons produce; hazards, misbehavior and creature escape rolls happen; meals every 5th bell.	None in real time; outcomes queue as notices.
C. Visit loop	A 2–10 minute game visit, usually after a study session	Collect notices; queue training; start/finish builds; place furniture; enroll applicants; assign careers; brew; tame; resolve events; plan expeditions.	Almost all decisions live here.
D. Day loop	Each Anki day rollover	Veil upkeep is charged; intervals are refreshed from Anki; skill recomputed; mastery countdowns advance; retirements checked; applicant board refreshes; daily event roll.	Respond to the day's summary.
E. Arc loop	Weeks to months	First students (week 1), first creature (week 1–2), all five schools teachable (weeks 3–6), first master (weeks 6–10), the Rift (weeks 8–14), first retirement (months 4–7).	Long-term specialization and layout choices.
F. Meta loop	Terms / forever	Timed Term ends with a charter score, alumni, spellbook choice and reset; Endless School grows forever and earns spellbooks at milestones.	Mode choice; spellbook picks.

6.2 Feedback structure

Positive loops (growth):

1. Reviews → mana → wizard levels → buffs (Arcane +mana, Alchemy −build cost) → more efficient spending.
2. Reviews → mana → rooms → more beds and seats → more students → more essences → more furniture and potions → higher room quality → better yields.
3. Real learning → mature intervals → masters → careers (gardeners, builders, alchemists…) → stronger production → more Academy.
4. Masters → the Rift → rare materials, blueprints, creatures and artifacts → late-game rooms.
5. Retirements → alumni bonuses → permanently easier future play.

Negative loops (pressure, all gentle):

1. Academy size → Veil upkeep (BR-24). Building more costs more every day.
2. More students → more food, beds, leisure needed → service shortfalls → lower morale → more misbehavior.
3. Unpaid or thin Veil → hazards and breaches → injuries and dirty rooms → healer and warder demand.
4. Faculty limit (BR-22) caps how many careers can run at once.
5. The enrollment tracker (BR-23) caps students by the user's own supply of young cards.
6. Draft prices rise with recent draws from the same deck.

6.3 The shape of a typical visit (mid game)

1. **Arrival summary** (modal card, dismissible): "Since your last visit: 142 reviews, +1,420 mana, 14 bells. Built: Common Room. Training: Nature 7 (61/175). Events: 2 waiting. Veil: Strong (paid 410)."
2. **Notices tray:** finished builds, mastery achieved, creature tamed, expedition returned, hazards resolved automatically, items waiting.
3. **Decide:** queue the next orb, place the furniture you drafted, enroll an applicant (tracker shows 63 free young cards, need 50), brew two Mending Draughts, answer an event.
4. **Leave something running:** a construction, a training session, a taming, maybe an expedition. The next study session will move all of them.

6.4 Illustrative first week (not measured data)

Day 1: the user installs, creates a witch named Hazel, and studies 140 cards. The first 8 reviews buy "Ponder the Verdant Orb" (Nature 1, 80 mana); it completes at review 33. By the end of the session Hazel is Nature 2 and halfway through Nature 3, and the user has placed a Vegetable Patch. Day 2: Nature 4 (finished at review 258 overall), a Dormitory is under construction, and a Mossback Tortoise wanders onto the grounds. Day 3: Nature 5 at review 383 overall. The Nature Classroom plan unlocks, the tracker shows 71 free young cards, and the first student, a gentle apprentice called Tamsin Reed, is linked to 50 of them. Day 6: two more students; a Refectory; Arcane training begins. Day 7: the first random event (an owl delivers a misaddressed letter).

7. The Player Wizard

7.1 Role and player goal

The wizard (or witch, or any title the player chooses) is the player's avatar and the first thing that grows. In the idle-style early game (BR-05) there are no students: the wizard ponders orbs, gains levels, and lays the first stones. Later the wizard becomes the Academy's headmaster: a stand-in teacher, a builder when no builders exist, a spellcaster for emergencies, and the source of the five school buffs that shape the whole economy.

7.2 Creation

On first launch the player creates their character (all choices changeable later in the Wizard screen except the founding date):

Field	Options
Name	Free text, 1–24 characters
Title	Wizard, Witch, Mage, Sorcerer, Sorceress, Warlock, Magus, or custom
Pronouns	she/her, he/him, they/them, custom
Appearance	Body shape (4), skin tone (8), hair style (12), hair colour (10), robe colour (12), hat (8, including "no hat"), staff (6)
Sigil	Choose from 24 sigils; used as the Academy crest
Academy name	Free text, default "Ankicaster Academy"
Mode	Endless School (default) or Timed Term (Section 30)

7.3 Training sessions (BR-06)

Training happens at the **Wizard's Study** (a pre-built room). There are two kinds of session:

Session	What it trains	Cost	Completes after	Requirements
Ponder an Orb	+1 level in one school	**80 × n** mana for level n	**25 × n** honest reviews (BR-07)	Current level n−1 in that school; the school's Orb is on the Study's orb stand (all five Orbs are part of the starting kit).
Study a Grimoire	Learn one wizard spell	Listed per spell (Section 29), 150–1,200 mana	Listed per spell, 40–250 honest reviews	The grimoire item (from the draft, the Library, or the Rift) and the spell's school-level requirement.

Rules:

1. **One active session at a time.** Only the active session advances; each honest review adds 1 progress point (Alchemy buffs do **not** affect training; nothing speeds training, so that BR-07's

numbers always hold).

2. **Queue.** The Study holds a queue: 1 queued session at Study level 1, 2 at level 2, 4 at level 3. When the active session finishes, the next queued session starts automatically **if it is already paid**. Mana is paid when a session is queued.
3. **Pause and switch.** The player can pause the active session and start another; paused progress is kept forever (BR-04). There is no penalty for switching.
4. **Cancel.** Cancelling a session with zero progress refunds 100% of its mana. Cancelling a session with progress is not allowed (the player can pause instead), so mana and progress can never be lost.
5. **Completion** triggers a level-up animation in the Study and a toast ("Nature 6 — your gardens grow 3% more").
6. Breaks: sessions simply wait; nothing decays.

7.4 Level costs (BR-07)

Level	Reviews needed	Cumulative reviews	Mana cost	Cumulative mana
1	25	25	80	80
2	50	75	160	240
3	75	150	240	480
4	100	250	320	800
5	125	375	400	1,200
6	150	525	480	1,680
7	175	700	560	2,240
8	200	900	640	2,880
9	225	1,125	720	3,600
10	250	1,375	800	4,400

All five schools to level 10: 6,875 reviews and 22,000 mana.

7.5 What each level unlocks (per school)

Level	Unlock
1	That school's **draft deck** opens (common cards). The school's tier-1 grimoire appears in the Library shop.
2	That school's **utility room standard plan**: Arcane → Library; Nature → Greenhouse; Light → Infirmary; Shadow → Warding Chamber; Alchemy → Alchemy Lab.
3	That school's **creature habitat** plan (Arcane → Crystal Vivarium; Nature → Meadow Paddock and Tide Tank; Light → Aviary; Shadow → Gloom Hutch; Alchemy → Ember Pit). Uncommon draft cards. Tier-2 grimoires.
4	Rare draft cards begin to appear. The school's **special room** blueprint can be drafted (Observatory, Menagerie, Sunlit Sanctum, Detention Hall, Workshop).

Level	Unlock
5	**Teaching unlocked (BR-08):** the school's **Classroom** standard plan; students can enroll in that school; masters of that school may take that school's careers; the wizard can personally teach one classroom of that school. Tier-3 grimoires.
6	First buff step (Section 7.6).
7	Buff step. That school's **Advanced Hall** blueprint can be drafted. Epic draft cards begin to appear.
8	Buff step. Tier-4 grimoires.
9	Buff step. Legendary draft cards can appear.
10	Buff step. Title **"Archmagus of [School]"** (cosmetic aura on the wizard sprite). The school's **capstone spell** grimoire becomes available.

Wizard-wide unlocks by **total level** (sum across schools, 0–50):

Total level	Unlock
3	Grounds expansion plan I (+4 plots)
8	Castle expansion plan I (+2 columns, +2 floors)
15	Study level 2 (queue 2)
20	Grounds expansion plan II (+4 plots)
25	Castle expansion plan II (+2 columns, +2 floors)
30	Study level 3 (queue 4)
40	Castle expansion plan III (+2 columns, +2 floors)
50	Title "Grand Magus"; final cosmetic tower spire

7.6 Buffs at levels 6–10 (BR-09)

Each level from 6 to 10 adds one step. Five steps maximum per school.

School	Per level (6–10)	At level 10	Exactly what it modifies
Arcane	+2% mana	+10% mana	Multiplies honest-review mana (inside the +25% global cap).
Nature	+3% food, herbs and taming	+15%	Food output of plots/greenhouses/ponds/orchards; Herbs from all sources; taming progress per review (×1.15 at L10).
Light	+3% recovery and teaching	+15%	Recovery: injury healing speed and need recovery from service rooms. Teaching: teacher delay is multiplied by (1 − 0.03 × steps), and classroom essence yields +3% per step.
Shadow	−3% Veil cost, −3% hazards and misbehavior	−15%	Daily Veil upkeep; per-bell hazard chance; per-bell misbehavior chance.

School	Per level (6–10)	At level 10	Exactly what it modifies
Alchemy	−3% build time and materials, +3% potion yield	−15% / +15%	Reviews required for constructions and furniture crafting; material costs of rooms and furniture; potions produced per brew (fractional yield accumulates).

Buff percentages are additive within a school and multiplicative with other sources unless stated in Appendix A.

7.7 The wizard's roles in the Academy

Role	Rule
Stand-in teacher	For each school at L5+, the wizard can personally teach **one** classroom of that school. Wizard-taught students use the **wizard teaching delay**: `20 − 2 × (L − 5)` days (L5: 20, L10: 10), reduced by Light's teaching buff.
Builder	The wizard always occupies construction slot 1 and advances it at 1 point per honest review.
Tamer of commons	With Nature L3+, the wizard can tame common creatures at 50% speed when no tamer is free.
Spellcaster	The wizard casts known spells (Section 29) for mana.
Headmaster	The wizard's total level contributes to the faculty limit (Section 11.6).

The wizard never gets injured, never misbehaves, and never goes through the Rift; they are the safe heart of the Academy.

7.8 Design purpose and tensions

Because BR-07 fixes training at 25×n reviews and only one session runs at a time, the wizard's full growth takes about 6,875 reviews (roughly 7 weeks at 150 reviews a day if the player trains continuously and can afford it). In practice mana competes with construction and the Veil, so most players will finish their fifth school in months 2–3, about when the first masters appear. The key early decision is order: Nature first (food and taming), Alchemy first (cheaper building), Arcane first (mana), Shadow first (cheaper Veil) or Light first (faster teaching). The tutorial recommends Nature first but does not enforce it.

8. The Five Schools of Magic

8.1 Overview

School	Theme	Colour	Essence produced by lessons	Careers	Utility room (L2)	Habitat (L3)	Special room (L4 draft)
Arcane	Stars, glyphs, time and pure magic	Deep violet and silver	**Insight**	Scholar	Library	Crystal Vivarium	Observatory
Nature	Growth, beasts, weather	Moss green and amber	**Herbs**	Gardener, Tamer	Greenhouse	Meadow Paddock, Tide Tank	Menagerie
Light	Healing, courage, warmth	Gold and white	**Radiance**	Healer	Infirmary	Aviary	Sunlit Sanctum
Shadow	Wards, secrets, discipline, night	Indigo and teal	**Umbra**	Warder	Warding Chamber	Gloom Hutch	Detention Hall
Alchemy	Transmutation, craft, potions	Copper and vermilion	**Reagents**	Alchemist, Builder	Alchemy Lab	Ember Pit	Workshop

Teacher and **Adventurer** careers are available from every school.

8.2 School identities

Arcane is the economist’s school. Its wizard buff is the only mana multiplier; its essence (Insight) powers draft rerolls, recipe research and spell study discounts; scholars run the Library and Observatory, giving foresight on events and extra Rift Fortune (never better odds). Arcane creatures tend to manipulate luck and information.

Nature is the provider. Its buff feeds people and creatures and speeds taming; gardeners and tamers keep the Academy fed and collect the largest share of the bestiary. Herbs are the most common potion ingredient.

Light is the carer. Its buff shortens teacher delay and speeds recovery; healers run the Infirmary and make injuries trivial; Radiance improves morale-oriented furniture and sanctum rituals. Light is the school that makes mastery arrive sooner once cards are mature.

Shadow is the guardian. Its buff is the only one that reduces the Veil’s daily cost, and it suppresses hazards and misbehavior. Warders reinforce the Veil and handle breaches. Umbra is used for wards and some potent potions.

Alchemy is the maker. Its buff cuts build time and material costs and raises potion yields; alchemists brew, builders add construction slots and maintain cleanliness. Reagents appear in most furniture recipes beyond the basics.

8.3 Lessons and essence yields

Each student enrolled in a school and seated in a **staffed** classroom of that school produces essence every bell:

```
essence_per_bell = 0.20 × room_efficiency × morale_factor × (1 + teaching_buff)
```

where `morale_factor` is 0.6 at morale 0 rising linearly to 1.2 at morale 100 (Section 14.6). A full basic classroom (5 students, efficiency 1.0, morale 70) yields about 1.0 essence per bell — roughly 15 per day at 150 reviews. Advanced Halls yield ×1.5.

Lessons never affect skill or mastery: those come only from card intervals (BR-14 to BR-16). Lessons are how students contribute to the *game* economy while their real cards mature.

8.4 Cross-school synergies (design intent)

Pair	Synergy
Nature + Alchemy	Herbs + Reagents are the two most common potion inputs; gardeners feed the alchemy lab.
Shadow + Arcane	Wards and foresight: Observatory warnings let warders pre-empt hazards.
Light + Nature	Healers and tamers: injured creatures and students recover in the same Infirmary wing.
Alchemy + Arcane	Cheaper buildings plus Insight-driven draft rerolls make the best layouts.
Light + Shadow	Morale plus Order: the lowest-misbehavior combination.

9. Students and Card Linking

9.1 Role and player goal

Students are the emotional centre of Ankicaster: each one is a little person whose growth is literally the growth of a slice of the user's memory. The player's goal is to enroll students when the Academy can house and seat them, keep them well (needs, morale), and watch them become masters as their linked cards mature.

9.2 Card sets

Every character holds one or more **card sets**:

Set kind	Size	Created when	Used for
Study set (one per school enrolled)	`cards_per_student` (setting, default **50**, applies to **all** students, BR-12)	The character enrolls in a school	Skill and mastery of that school
Career set (one per career ever taken)	**20** (BR-18)	The character takes a career	Career proficiency; teacher delay if the career is Teacher

Card age is the review interval (BR-14), read from Anki. For each linked card the game uses the **honest interval**:

```
honest_ivl = max(0, min(cards.ivl, last_answer_ivl))
```

- `cards.ivl` is the card's current interval in days (0 for new and learning cards).
- `last_answer_ivl` is the `revlog.ivl` of the card's most recent revlog entry of type 0–3 (a real answer); negative values (seconds, used for learning steps) count as 0.
- The `min` means intervals stretched by manual rescheduling tools ("set due date", bulk reschedules) do not count until the card is genuinely answered at that interval, while intervals *shortened* by rescheduling are respected. This is a read-only calculation and is controlled by the setting `honest_intervals_only` (default on).

9.3 Eligible and free cards

A card is **eligible** for linking when all of these hold:

1. It belongs to a deck inside the linking scope (setting `deck_scope`: "all decks" by default, or a list of deck ids; subdecks included).
2. It is not suspended (`queue != -1`). Buried cards are eligible (burying is temporary).
3. It has been studied at least once (`type` in 1 learning, 2 review, 3 relearning). New, never-studied cards are excluded unless the setting `link_new_cards` is on (default off), because a student bound to cards the user has not started would be stuck.
4. It is not already linked to any character (BR-13). A card is linked to at most one character and one set at a time.

A **free young card** is an eligible card with `honest_ivl ≤ young_threshold_days` (default 30).

9.4 The linking algorithm (“youngest free cards”)

```
function pick_youngest_free(n, require_young):
    candidates = eligible cards not in card_links
    if require_young: candidates = candidates where honest_ivl <= young_threshold_days
    order candidates by honest_ivl ascending,
                        then by first_studied_at descending (most recently started first),
                        then by card id ascending (stable tie-break)
    if avoid_siblings (setting, default on):
        walk the ordered list, skipping a card if another card of the same note
        is already chosen for this set, unless skipping would leave fewer than n
    return the first n cards
```

- **Students** (study sets) use `require_young = true`. If fewer than `n` free young cards exist, enrollment is blocked (BR-23); partial sets are never created.
- **Careers** take the 20 youngest free cards (BR-18): first `require_young = true`; if fewer than 20 free young cards exist, the remaining slots fall back to the youngest free older cards (`require_young = false`). The fallback is shown on the career screen (“12 young + 8 older cards”).
- **Changing `cards_per_student`** applies to every student (BR-12), not just new ones. At the next day tick, each current study set is resized. When the number rises, each set is topped up with the youngest free young cards; if there are not enough, the set fills as cards become free, and the enrollment tracker shows the shortfall. When it falls, each set releases its most recently linked cards. Master titles already earned are permanent (BR-17). A student who was not yet a master is re-checked against the new set size.
- Linking is done in a single database transaction in the add-on’s own save file; the Anki collection is only read.

9.5 Refreshing intervals

Intervals are read in bulk with read-only SQL (`SELECT id, ivl, type, queue, nid FROM cards WHERE id IN (...)`, chunked 500 ids at a time) and the latest real answer per card from the revlog. Refresh happens:

- at every day tick,
- when the game window opens (if the last refresh is more than 10 minutes old),
- for the single answered card right after each honest review (cheap, deferred),
- after a sync.

The result is cached in `card_links.cached_ivl` with a timestamp; the UI never queries Anki directly.

9.6 Special cases (deleted, suspended, reset, moved)

Situation	Detection	Rule	Why
Card deleted	Linked id missing from `cards` at refresh	Link marked `missing`. At the next day tick it is replaced by the youngest free young card (students) or youngest free card (careers). Until replaced, the slot counts as interval 0. If no replacement exists, the slot waits and the UI shows "1 slot waiting for a card".	Deleting cards never speeds mastery or retirement, and never deletes the character.
Card suspended	`queue == -1`	Stays linked and counts at its current honest interval. If it remains suspended for more than `suspended_release_days` (default 30) it is released and replaced as for deleted cards. The UI shows a moon icon on suspended links.	Suspending hard cards cannot speed anything up; long-term suspension frees the slot so the character is never stuck forever.
Card reset / forgotten	`type == 0` or interval drops to 0	Stays linked; skill falls honestly. If the card stays new (never re-studied) for more than 30 days it is released and replaced.	Honest; never punitive.
Card moved to another deck	—	Stays linked. Deck scope applies only to new links.	Reorganising decks never disturbs students.
Card rescheduled manually	Revlog entry of type outside 0–3	Honest-interval rule (Section 9.2) applies.	Prevents instant mastery by "set due date".
Note type changed / card regenerated with a new id	Old id missing	Treated as deleted.	Simple and safe.
Collection fully replaced (full sync download, restore backup)	Many links missing at once (>25% of all links)	The game pauses replacement, shows a dialog: "Many linked cards are missing. If you restored a backup or are mid-sync, choose Wait. Choose Relink to replace them." Nothing is deleted either way.	Prevents mass relinking during a temporary state.
Manual swap	Player action on the character card	The player may release a **suspended, missing or new** card in exchange for the youngest free young card. Mature cards cannot be swapped.	Gives control without enabling exploits.

9.7 Student identity

Attribute	Notes
Name	Generated from a fantasy name list (given name + family name); editable
Appearance	Generated sprite parameters (body, skin, hair, robe trim in their school's colour, accessory)
Personality	One of eight (Section 14.2)
Traits	1–3 traits (Section 14.3)
School(s)	Current study school; mastered schools
Card sets	Study and career sets with live skill bars
Needs	Hunger, Rest, Leisure, Health, Composure (0–100)
Morale	Derived from needs and modifiers
Assignments	Bed, classroom seat, workplace (if faculty), expedition (if adventurer)
Status	Student, Master (resident), Faculty, Away (Rift), Injured, Alumnus
History	Enrolled day, mastery days, career history, notable events (a short chronicle)

The character card in the Roster shows the study set as a **"constellation"** of 50 small dots, each dot coloured by the linked card's interval (grey = learning, then a gradient to gold at 30+ days). Hovering a dot shows the card's front text (first 60 characters, read-only) and interval, so the user can see *which* of their cards the student is struggling with.

9.8 Design purpose and tensions

Linking to the *youngest* free cards means every new student is genuinely a beginner and mirrors what the user is learning *right now*. Because cards are never shared, the user's supply of young cards (not their review count) is the natural limit on how many students the Academy can hold, and the enrollment tracker makes that limit visible and fair. Enrollment needs no extra effort from the user beyond normal study.

10. Skill, Mastery and the Master Title

10.1 Skill (BR-15)

For each card set:

```
skill = round(100 × mean over slots of min(honest_ivl, 30) / 30)
```

Missing slots count as 0. Skill therefore rises as intervals grow and **falls when a lapse shortens a card's interval**, exactly mirroring the user's real memory. There is no other input: no game item, potion, spell or creature changes skill.

Skill	Band	Sprite cue
0–19	Novice	Plain robe, L-plate badge
20–39	Apprentice	Robe trim
40–59	Journeyman	Sash
60–79	Adept	Cape
80–99	Expert	Glowing trim
Ready	Mastery-ready (all cards > 30 days)	Sparkling outline; countdown badge

A second, thinner bar shows **seasoning**: `round(100 × mean(min(honest_ivl, retirement_age) / retirement_age))` across **all** of the character's cards. It shows progress toward retirement (Section 13).

Lapse messaging. When a linked card lapses, the character's bubble shows a gentle line ("Oops — I'll get that one next time!") at most once per character per day. The game never shows a red warning, never removes anything, and never asks the user to avoid pressing Again.

10.2 Mastery-ready

A student is **mastery-ready** in a school when **every** slot of that study set is filled and has `honest_ivl > 30` days (strictly over 30, BR-16). With default Anki settings this is usually 5–9 weeks after the youngest card was first studied, longer if cards lapse.

10.3 Teacher delay (BR-16)

Mastery also requires waiting a **teacher delay** in real days after becoming ready.

Who teaches the student	Delay formula (days)
A **Teacher** (career)	`max(0, 30 − mean(honest_ivl of the teacher's 20 teaching cards))`; **0 once all 20 teaching cards are over 30 days**
The **wizard** (stand-in)	`20 − 2 × (L − 5)` where L is the wizard's level in that school (20 at L5, 10 at L10)
Nobody (unstaffed classroom)	30

Then the Light teaching buff applies: `delay × (1 − 0.03 × light_steps)`, rounded **up** to whole days (Light L10 → ×0.85).

The teacher's 20 teaching cards are that teacher's **career set** for the Teacher career. A brand-new teacher has young teaching cards, so their delay starts near 30 days and shrinks as *their* cards mature: experienced teachers really do graduate students faster.

10.4 Mastery countdown

At each day tick, for each student:

```
if mastery_ready(student, school):
    student.ready_days += elapsed_days       # real days since the last tick (usually 1)
    D = current_teacher_delay(student)        # recomputed daily
    if student.ready_days >= ceil(D):
        grant_mastery(student, school)
else:
    pass                                      # ready_days is kept (paused), never reset
```

- `ready_days` **pauses** (does not reset) if a lapse makes the student not-ready; it resumes when they are ready again. This is the gentle reading of the concept: a lapse costs time, never past progress.
- Away days count as real days. Intervals do not change while the user is away, so a student who was ready when the user left keeps counting toward mastery.
- If the teacher changes, the next day's D uses the new teacher.

10.5 The master title (BR-17)

On mastery the student gains the title **Master of [School]** permanently. A later lapse never removes it. The ceremony is a short scene in the Entrance Hall (robes change colour, the crest above the door gets a new star). Effects:

Effect	Value
Renown	+10
Counts toward the Rift's 3-master threshold (BR-27)	Yes (counts characters, not titles)
Career eligibility	That school's careers + Teacher + Adventurer
Resident tutor bonus (if no career)	+5% essence yield to that school's classrooms, up to 3 tutors per school
Study set	Stays linked and keeps maturing toward retirement

Multi-school titles: 2 schools "Double Master", 3 "Triple Master", 4 "Grand Master", 5 "Paragon". A Paragon adds +1 Rift Fortune to any expedition they join (Section 27.9; Fortune never changes the odds).

10.6 After mastery: options

1. **Take a career** (Section 11) if the faculty limit allows.
2. **Enroll in a second school** (Section 12.6): adds a fresh 50-card study set (BR-20).
3. **Remain a resident master** (tutor bonus, can still be an adventurer later).

11. Careers and Faculty

11.1 Role and player goal

Masters become the Academy's workforce. Careers are the mid-game engine: each one automates or boosts a system (teaching, food, creatures, healing, wards, potions, construction, expeditions). The player's goal is to pick careers that fix their current bottleneck while staying within the faculty limit.

11.2 Taking a career (BR-18)

Requirements: the character is a master of a qualifying school; the faculty limit (Section 11.6) has room; a workplace slot of the right kind exists; a faculty bed is free (Faculty Quarters). On acceptance the character gains a **career set of the 20 youngest free cards** (BR-18): the youngest free *young* cards (honest interval ≤ 30 days) are taken first, and only if there are not 20 free young cards does the set fall back to the youngest free older cards. If fewer than 20 free cards exist at all, the career cannot start (the career screen says how many are free, and how many of them are young).

11.3 Career proficiency

Each career's output is scaled by the skill of its own career set:

`proficiency_factor = 0.75 + 0.5 × career_skill / 100` (0.75 at skill 0, 1.25 at skill 100)

So new faculty work at three-quarters strength and improve as their 20 cards mature. For Teachers, career skill also drives teacher delay (Section 10.3).

11.4 Career catalog

Career	Qualifying school(s)	Workplace	Slots per workplace	Effect (at proficiency 1.0)
Teacher	Any mastered school (teaches that school)	Classroom or Advanced Hall of that school	1	Staffs the classroom (full essence yield, ×0.5 when unstaffed). Students in that room use this teacher's delay. +5% room efficiency at career skill 100.

Career	Qualifying school(s)	Workplace	Slots per workplace	Effect (at proficiency 1.0)
Scholar	Arcane	Library, Observatory, Scriptorium	2 / 1 / 1	+1 Insight per bell (Library); event foresight and +1 Rift Fortune per scholar in an Observatory (max 2); crafts scrolls (Scriptorium); −10% recipe research cost per scholar (max −30%).
Gardener	Nature	Greenhouse (1), or up to 3 garden plots	1	+50% output of the assigned greenhouse or plots; unlocks Rare Seed crops; reduces wild-growth hazard on assigned rooms by 50%.
Tamer	Nature	Any creature habitat or the Menagerie	1 per 2 habitats	Tames creatures (Section 22.4) at 1.0 progress per review; −50% escape chance in assigned habitats; recaptures escaped creatures.
Healer	Light	Infirmary, Sunlit Sanctum	1 per infirmary, 1 per sanctum	Heals injuries at ×2 speed; cures Hexed injuries; +1 Composure per bell to everyone in the Sanctum.
Warder	Shadow	Warding Chamber, Detention Hall	1 / 1	+5 Veil strength per active day; Veil cost −3% each (max 3 warders); −10% hazard and misbehavior chance each (max −40% combined); auto-resolves breaches; supervises Detention.
Alchemist	Alchemy	Alchemy Lab	1	Required to brew advanced potions; brewing time ×0.5; −50% lab explosion chance; +1 potion per 4 brews.

Career	Qualifying school(s)	Workplace	Slots per workplace	Effect (at proficiency 1.0)
Builder	Alchemy	Workshop	2	Adds one construction slot each (max 4 slots including the wizard's); a builder's slot advances 1.25 per review; idle builders clean 4 cleanliness per bell across the Academy.
Adventurer	Any (BR-11)	Training Yard (resting), the Rift (on expedition)	3 per yard	Goes on Rift expeditions (Section 27). The mean interval of the adventurer's 20 career cards is their **Rift skill** in days, which sets the odds against each destination's skill requirement. Drilled teams get +1 Rift Fortune per Training Yard level.

11.5 Career switching (BR-19)

- Only **masters of two or more schools** may switch to a different career.
- **Old career cards are kept.** The character keeps every career set they have ever had (they keep maturing toward retirement), and the new career adds a new set of 20.
- Masters of a single school may take a **sabbatical** (leave their workplace, stay faculty-housed) and later **resume the same career** with the same set; they may not take a different career.
- Switching is instant and free in mana; its real cost is 20 more cards bound to the character, which also delays that character's retirement.

11.6 Faculty limit (BR-22)

The number of characters holding active careers may not exceed:

```
faculty_limit = min(A, B, C, D)
```

Term	Source	Formula
A	Wizard levels	`1 + floor(total_wizard_levels / 4)` (max 13 at total 50)
B	Housing	Faculty beds (2 per Faculty Quarters, +1 per level above 1)
C	Classrooms	`1 + 2 × number_of_classrooms` (basic and advanced halls of any school)
D	Student count	`1 + floor(current_students / 3)`

The Faculty panel shows all four terms, highlights the smallest, and explains it (“Your limit is 3 because you have 6 students. Each 3 students support one more faculty member.”). Workplace slots are a separate, per-career constraint.

11.7 Design purpose and tensions

Careers convert matured memory into automation, but each career binds 20 more cards to a character, delaying their retirement and using up free cards that could have enrolled students. The faculty limit stops the Academy from becoming all staff and no students. Teachers are the most important early career because they unlock more classrooms’ full yields and, as their cards mature, shrink teacher delay to zero.

12. Enrollment, Recruitment and the Enrollment Tracker

12.1 Role and player goal

Recruitment is where the Academy grows. The player chooses which applicants to admit into which school, within the limits of beds, seats, and, above all, the user's supply of free young cards.

12.2 Requirements to enroll a new student in school X

#	Requirement	Shown in
1	Wizard level 5 in X (BR-08)	Recruitment screen tab is locked otherwise
2	At least `cards_per_student` (50) **free young cards** (BR-23)	Enrollment tracker
3	A free student bed (Dormitory or Grand Dormitory)	Capacity chip
4	A free seat in a classroom of school X	Capacity chip
5	60 mana ("robes, wand and a welcome cake")	Button

12.3 The enrollment tracker (BR-23)

The tracker is a permanent widget at the top of the Recruitment screen and a small chip on the Academy map:

```
Free young cards   63 / 50 needed   ✔ one student can enroll
──────────────────────────────────────────────────
 Eligible young cards (≤30 d) .......... 412
 Linked to characters .................. 349
 Free young cards ...................... 63
 Beds free 2 · Arcane seats free 1 · Nature seats free 3
```

- When free young cards < 50, the Enroll buttons are disabled and the tracker reads, for example: **"38 / 50 free young cards. New students can enroll when more of your cards are in their first month."** It never suggests adding cards (BR-03).
- The tracker recalculates on window open, after each day tick, and after linking/unlinking.
- Clicking the tracker opens a breakdown by deck (read-only), so the user understands which decks feed enrollment.

12.4 The applicant board

At each day tick the board offers **3 applicants per teachable school** (max 15). Each applicant shows: name, portrait, personality, traits (1–3), a one-line backstory, and a "fit" hint (e.g., "Would love a Greenhouse"). Unchosen applicants leave at the next day tick (no penalty). The player may

reroll one school's applicants for 100 mana (or 15 Insight). Renown (Section 32.4) improves applicant quality: at higher Renown, more applicants have positive traits and fewer have troublesome ones.

Applicant trait roll weights:

Renown	Positive traits	Neutral/quirky	Troublesome
0–49	40%	35%	25%
50–149	50%	32%	18%
150–299	58%	30%	12%
300+	65%	28%	7%

12.5 Seating and beds

- A student occupies **one bed** and **one seat** in a classroom of their current study school. If a classroom is demolished or locked by a hazard, students keep their enrollment and are "between rooms" (their essence yield is 0 until reseated; nothing else is affected).
- Beds and seats are auto-assigned (nearest free), and the player can drag a student to another seat/bed in the Roster.

12.6 Second-school enrollment (BR-20)

A character who is a master of at least one school may enroll in another teachable school. This adds a **fresh study set of 50 free young cards** (the same tracker check applies) and requires a seat in that school's classroom. They keep their bed (student dorm or faculty bed) and any career. A character studies at most one new school at a time.

12.7 Design purpose and tensions

Enrollment is self-limiting in a way that reflects real study: an Academy can only hold as many beginners as the user is genuinely beginning. Users who add few new cards will have a smaller, older, more masterful Academy; users who learn lots of new material will run a bustling school. Both are valid. Beds and seats make construction choices matter; the applicant board adds personality and choice.

13. Retirement and Alumni

13.1 Rule (BR-21)

A character **retires** when **every card in every one of their card sets** has `honest_ivl > retirement_age_days` (setting, default **90**). The check runs at each day tick. Retirement is automatic and celebratory, not a loss.

13.2 What happens on retirement

1. A farewell scene in the Entrance Hall; the character's portrait is hung in the **Alumni Gallery** (a panel of the Roster, and a decor wall in the Trophy Hall if built).
2. All of their card links are **released** (the cards become free; being mature, they will not count as young).
3. Their bed, seat and workplace free up.
4. The Academy gains an **alumni bonus** chosen from that character's history (Section 31.3) and +5 Renown.
5. The alumnus is kept forever in the save (name, portrait, titles, chronicle). In a Timed Term, alumni bonuses carry over (BR-30).

13.3 Alumni visits

Each day tick there is a 3% chance (max one per day) that an alumnus visits, bringing a small gift linked to their career (e.g., a retired gardener brings 20 Greens and a Rare Seed; a retired adventurer brings a Rift map fragment). Visits are pure upside.

13.4 Expected timing

With default Anki settings, a card typically passes 90 days about 4–6 months after first being studied, so a student who never takes a career can retire around months 5–8 after enrollment; faculty retire later because their career sets are younger. A retirement-age setting of 60 makes retirements more frequent; 180 makes them rarer and grander (the alumni bonus is scaled by `retirement_age / 90`, clamped 0.67–2.0, so the setting is a style choice rather than an exploit).

14. Student Traits, Personalities and Needs

14.1 Role and player goal

Personalities and traits make each student a character; needs make the Academy a place. Needs never affect skill or mastery (those are card intervals only). They affect **morale**, which drives essence yields, misbehavior, career output and Rift odds. The player's goal is to keep services ahead of headcount and to place the right furniture for the right people.

14.2 Personalities (one per character)

Personality	Flavour	Need modifiers	Misbehavior ×	Loves (morale +5 if the Academy has it)	Dislikes (morale −5 when adjacent/assigned)	Rift note
Bookish	Lives in the stacks	Leisure decays ×0.8 if a Library exists	×0.6	Library	Noisy rooms (Workshop next to dorm)	+1 Rift Fortune at the Crystal Caverns
Mischievous	Pranks with a heart of gold	Leisure decays ×1.3	×2.0	Common Room	Detention Hall	+1 Rift Fortune at the Shattered Spire
Gentle	Soft-spoken, loved by animals	Composure decays ×0.8	×0.7	Any creature habitat	Training Yard	+4 taming speed as tamer
Bold	First through any door	Rest decays ×1.2	×1.2	Training Yard	Bare rooms (quality tier Bare)	+1 Rift Fortune on every expedition
Dreamy	Head in the clouds	Hunger decays ×0.9	×0.9	Observatory	Windowless rooms (no Glass furniture)	+1 Rift Fortune at the Whispering Fen
Steadfast	Calm and reliable	All needs ×0.9	×0.5	Sanctum	—	Recovers from Rift injuries 25% faster
Social	Knows everyone	Leisure decays ×1.2; Composure ×0.8 when Common Room exists	×1.1	Common Room, Refectory	Dorms below Cozy	+2 team bonus when in a team of 3
Curious	Pokes everything	Composure decays ×1.1	×1.3	Alchemy Lab, Menagerie	—	+1 Rift Fortune at Mossy Hollow and the Ashen Wastes

14.3 Traits (1–3 per character)

Positive

Trait	Effect
Green Thumb	+20% output when gardener or when seated in a Nature room
Beast Whisperer	+25% taming speed; creatures on their floor escape 30% less
Night Owl	Rest decays 20% slower; +5% hazard detection (hazards on their floor resolve faster)
Tidy	Rooms they use lose cleanliness 30% slower
Brave	+1 Rift Fortune as adventurer; all injuries heal 15% faster
Kind Heart	+3 Composure per bell to roommates (dorm)
Iron Stomach	Hunger decays 30% slower
Lucky	+1 loot roll on successful expeditions; −20% chance of being hit by hazards
Quick Hands	Builder or alchemist output +15%
Steady Hand	Alchemy Lab explosion chance −50% when present
Sunny	+5 morale; +2 Composure per bell to classmates
Warded	Immune to Hexed injuries
Star-Touched	+10% Insight yield; Observatory foresight +1 day
Well-Read	+10% essence yield in any classroom with a Bookshelf
Hardy	Healing time −30%

Neutral / quirky

Trait	Effect
Collector	Occasionally (2%/day) brings back a trinket (random common material)
Early Riser	Rest decays faster (×1.1) but Leisure slower (×0.9)
Picky Eater	Needs a Kitchen for full Hunger recovery (else target −20)
Homesick	−5 morale for the first 7 days after enrolling; +5 after 30 days
Chatterbox	+Leisure for roommates, −Focus (−2 room Focus) in classrooms
Animal Allergy	−5 Composure when a creature habitat is adjacent to their dorm
Cat Person / Bird Person / Bug Person	+10 morale with that creature family present, −3 otherwise
Show-off	+1 Rift Fortune in a team of 3, −1 alone
Sleepwalker	1% per bell to wander into a random room (harmless; small Composure loss to roommates)

Troublesome

Trait	Effect
Prankster	Misbehavior ×1.5; pranks are funnier (event text) and slightly messier
Firebug	Alchemy Lab and Kitchen fire hazard ×2 when present
Lazy	Essence yield −15%; Rest decays slower
Glutton	Eats 2 Provisions per meal
Butterfingers	3% per bell to break a fragile furniture item (it becomes "Damaged", −50% stats until repaired for 25% of its cost; never destroyed)
Nervous	Composure decays ×1.4; −1 Rift Fortune
Loudmouth	−5 Composure to roommates when morale < 40
Rule-Bender	Detention Hall effects ×0.5; misbehavior ×1.3
Summoner-at-Heart	0.3% per bell to open an unauthorized minor portal (a hazard)
Daredevil	+1 Rift Fortune, but recovers from Rift injuries 25% slower

Traits may be gained or lost through events, Detention Hall rehabilitation, the Sunlit Sanctum, certain furniture and spells (each source states which).

14.4 Needs

All characters except the wizard have five needs, each 0–100.

Need	Served by	Target value when served	Target when not served
Hunger	Refectory seats + 1 Provision per character per meal (every 5 bells)	100 (110 with Kitchen "Hearty" meals, capped 100 but grants Well-Fed)	25 (no food) / 50 (food but no seat — eating standing)
Rest	A bed (Dormitory, Grand Dormitory, Faculty Quarters)	`55 + 45 × min(1, (room_efficiency − 0.5))`	25 (no bed)
Leisure	Common Room, Faculty Lounge, Bathhouse, Orchard, decor Charm	`30 + 70 × min(1, leisure_capacity / headcount)`	30
Composure	Sanctum, Bathhouse, Charm, creatures, a strong Veil	`60 + sanctum_bonus (≤ +15) + charm_bonus (≤ +20) + creature_bonus (≤ +10) + veil_modifier (0 / −10 / −20 / −35)`	—
Health	Baseline 100; reduced by injuries (Section 23)	100	Capped by injury severity

Update rule each bell: `need += (target − need) × 0.25 − decay × personality_multiplier × trait_multipliers`, where base decay is Hunger 4, Rest 3, Leisure 3, Composure 1 per bell (Health has no decay). Values are clamped 0–100.

Meals: every 5th bell. Refectory meal capacity = seats × 2 (two sittings). Provisions are consumed first-come (faculty and students alike). If Provisions run short, the shortfall characters get the "no food" target until the next meal. Nobody ever starves to death or leaves; hunger only lowers morale.

Creatures have their own food rules (Section 22).

14.5 Need-state labels (for UI)

Range	Hunger	Rest	Leisure	Composure
80–100	Well-fed	Rested	Cheerful	Serene
50–79	Fine	Fine	Fine	Steady
25–49	Peckish	Tired	Bored	Jittery
0–24	Hungry	Exhausted	Restless	Rattled

14.6 Morale

`morale = 0.25 Hunger + 0.25 Rest + 0.20 Leisure + 0.20 Composure + 0.10 Health + personality_loves_bonus + trait_modifiers` (clamped 0–100).

Morale	Essence yield factor	Misbehavior ×	Career output ×	Rift modifier
0–29	0.6–0.8 (linear)	×2.0	×0.85	−5
30–59	0.8–1.0	×1.25	×1.0	0
60–79	1.0–1.1	×1.0	×1.0	0
80–100	1.1–1.2	×0.6	×1.05	+5

14.7 Away behaviour

Needs **do not tick** on days without reviews (no bells ring). When the user returns, needs are exactly as they were. This is the “the school sleeps under the Veil” rule (BR-04).

15. The School Map, Construction and Room Quality

15.1 Role and player goal

Construction is where mana becomes something you can see. The Academy is a side-on cross-section castle (rooms stacked on floors) plus a strip of outdoor grounds. The player's goal is a connected, compact Academy where every room is reachable, services sit near the people who use them, special rooms get the sky or ground they need, and furniture lifts rooms from Bare to Splendid, all without letting the Veil's daily cost run away.

15.2 The grid

Area	Start size	Expansions	Maximum
Castle (cross-section)	8 columns × 5 floors (floor 0 = ground)	+2 columns and +2 floors at wizard total level 8, 25 and 40 (Castle Expansion plans I–III; each costs mana and Stone, see 33.5)	14 columns × 11 floors
Grounds (outdoor strip, left and right of the castle)	6 plots	+4 plots at total level 3 and 20; +2 per "Land Grant" draft card (max 2 cards)	18 plots

The starting Academy is pre-built: **Entrance Hall** (floor 0, columns 3–4), **Wizard's Study** (column 5, floors 0–1, with an internal stair), one **Spiral Stair** (column 2, floor 0), a **Storage Vault** (column 1, floor 0), and on the grounds a **Vegetable Patch** and a **Woodlot**.

15.3 Placement rules

ID	Rule	Detail
P1	Bounds	All tiles inside the current grid. Grounds rooms only on plots; castle rooms only in the castle.
P2	Free tiles	No overlap with rooms or construction sites.
P3	Support	Every tile of the room's bottom row is on floor 0 or directly above an occupied tile (a room, construction site or **Support Pillar**).
P4	Connection	A castle room must share a vertical wall edge with a reachable room on the same floor, **or** be a Spiral Stair stacked on/under a reachable stair. Rooms that fail P4 can still be built but show **Unreachable** and run at 0% until connected.

ID	Rule	Detail
P5	Sky access	Rooms flagged `needs_sky` (Greenhouse, Observatory, Aviary) must have nothing above any of their tiles, now or later. The build tool blocks placing anything over them.
P6	Floor limits	Some rooms are restricted: Tide Tank floor 0 only; Ember Pit floors 0–1; Observatory must be on the highest built floor of its column when placed.
P7	Grounds	Plots are always reachable (the grounds path connects to the Entrance Hall). Multi-plot rooms need adjacent plots.

Reachability is computed by BFS from the Entrance Hall: same-floor rooms that share a wall edge are linked; vertical travel happens only through stacked Spiral Stairs (the Wizard's Study's internal stair links floors 0–1 of its column). Distance in BFS steps is used by some adjacency and service rules (Section 15.9).

15.4 Construction flow (BR-28: blueprint + materials + mana + build time in reviews)

1. **Choose a blueprint** from the Build menu: a **standard plan** (always available once unlocked, unlimited uses) or a **blueprint draft card** from the player's hand (consumed on placement).
2. **Place a ghost** on the grid; the tool shows rule violations in red and adjacency effects in green/orange.
3. **Confirm**: pay mana and materials immediately (Alchemy buff reduces materials by 3% per step, rounded down per material, minimum 1).
4. A **construction site** appears and joins the **construction queue**.
5. **Build time** is measured in honest reviews. Each active construction slot advances by 1 point per honest review (1.25 for a builder's slot). Alchemy reduces the required points by 3% per step.
6. On completion the room appears with its required furniture **empty**; a prompt offers to place required furniture (Section 17). Rooms with missing required furniture are **Unfinished** and run at 0%.

Construction slots: 1 (the wizard) + 1 per Builder (max 4 total). Sites beyond the slot count wait in order; the player can reorder.

Cancel / pause / demolish:

Action	When	Refund
Cancel	Before any progress	100% mana and materials; blueprint card returned to hand
Pause	Any time	Nothing lost; progress kept forever
Demolish a finished room	Any time (not during an active hazard in that room)	50% materials, 0% mana; furniture returned to inventory; occupants reassigned. A drafted blueprint is not returned.

Demolition is always a deliberate player action with a confirmation dialog. No system ever demolishes or damages a room on its own (BR-04, BR-25).

15.5 Room levels

Rooms have levels 1–3.

Level	How	Effect
1	Built	—
2	Play a **duplicate blueprint card** of the same room on it, **or** (standard plans only) **Renovate** for 1.5 × the base mana and materials and 50% of the base build time	+10% efficiency, +capacity (per room), +1 optional furniture slot, improved art
3	As above, from level 2; requires wizard level 6 in the room's school (Foundation rooms: total level 20)	Another +10% efficiency, +capacity, +1 slot, best art; some rooms gain a **level-3 perk** (listed in Section 16)

15.6 Room stats and quality

Every furniture item adds points to five **room stats**:

Stat	Represents	Typical sources
Comfort	Softness, warmth, rest	Beds, rugs, couches, hearths
Focus	Light to work by, quiet, tools	Desks, lecterns, candles, bookshelves
Order	Tidiness, discipline, storage	Shelves, wardrobes, wards, clocks
Aura	Ambient magic	Focus items, crystals, altars, cauldrons
Charm	Beauty and personality	Plants, tapestries, portraits, statues

Each room type has **two primary stats** and **one secondary stat** (Section 16). Its **quality score**:

```
Q = Σ over furniture of (points in primary stats) + 0.5 × (points in secondary stat)
```

Other stats on the furniture still count toward the Academy-wide Charm total (Section 14.4) but not toward that room's Q.

Q	Quality tier	Art	Quality factor
0–9	**Bare**	Stone walls, single torch	`1 + Q/160`
10–24	**Plain**	Plastered walls	(same formula)
25–44	**Cozy**	Warm lighting, rugs	
45–69	**Fine**	Panelled walls, banners	
70+	**Splendid**	Gilded trim, floating lights, sparkle particles	capped at Q = 80 → ×1.5

15.7 Room efficiency

```
efficiency = level_factor × quality_factor × cleanliness_factor × staff_factor × adjacency_factor
```

Factor	Values
level_factor	1.0 / 1.1 / 1.2
quality_factor	`1 + min(Q, 80) / 160` (1.0–1.5)
cleanliness_factor	`0.7 + 0.3 × cleanliness / 100` (0.7–1.0)
staff_factor	1.0 if the room's workplace is staffed (or needs no staff); 0.5 if a staffable room has no staff
adjacency_factor	Product of the adjacency bonuses/penalties in Section 15.9 (clamped 0.8–1.3)
Overrides	0 if Unfinished (missing required furniture), Unreachable, or Locked by a hazard

Efficiency multiplies the room's outputs (yields, healing, meal sittings counted as capacity × efficiency rounded down, etc.) as described for each room.

15.8 Cleanliness

Each room has cleanliness 0–100 (starts at 100).

- **Dirt per bell:** `0.5 + 0.3 × occupants_this_bell` (×0.7 if any occupant is Tidy; ×1.5 for Alchemy Lab, Kitchen and habitats).
- **Cleaning:** each idle Builder cleans 4 points per bell (to the dirtiest rooms first); Mopslime creatures 3 per bell; an Enchanted Broom 1 per bell in its room; the spell *Tidy Wind* restores a floor to 100; the Polishing Spirit potion restores one room to 100.
- Misbehavior and hazards can dirty a room by 10–40 at once.
- Cleanliness never affects anything except efficiency (minimum ×0.7). A dirty room is never damaged.

15.9 Adjacency and layout effects

"Adjacent" means sharing a wall edge on the same floor (castle) or neighbouring plots (grounds). "Within N" means BFS distance ≤ N rooms.

Effect	Rule
Kitchen ↔ Refectory	A Kitchen adjacent to a Refectory serves **Hearty meals** (Well-Fed: +5 morale for 5 bells) and cuts Provisions use by 10%. A Kitchen not adjacent to a Refectory works at 50%.
Greenhouse ↔ Kitchen	Greenhouse +10%.
Library ↔ Classroom	Each adjacent classroom +5% (max +10% from two libraries).
Classroom wings	Each same-school classroom adjacent to another +5% (max +10%).
Alchemy Lab / Workshop ↔ Dorm or Faculty Quarters	The dorm/quarters loses 10 Comfort-equivalent Q (noise and fumes). A Fume Hood in the Lab removes the Lab's penalty.
Habitat ↔ Classroom	Classroom −5% (noise) unless the habitat contains only Gentle-favoured creatures (Section 22).
Menagerie ↔ Common Room	Common Room +10% leisure capacity.

Effect	Rule
Infirmary ↔ Sunlit Sanctum	Both +10%.
Detention Hall ↔ Classroom	That classroom's misbehavior −10%.
Warding Chamber radius	Rooms within 3 BFS steps: hazards −20%.
Observatory on the highest floor of the castle	+10%.
Dormitory within 4 of a Refectory	Residents' Hunger target +5 (short walk to meals).
Dormitory more than 8 from any Common Room	Residents' Leisure target −10 (long walk).

15.10 Academy size (for the Veil)

Veil upkeep (Section 18) uses **Academy size**, never headcount (BR-24):

```
size_points = Σ over castle rooms of room.veil + Σ over grounds rooms of room.veil
```

where each room's `veil` value is listed in Section 16 and scales with level (+25% per level above 1). Construction sites count at 50%.

15.11 Design purpose and tensions

The grid creates the central spatial puzzle of Spellcaster-style games, kept simple enough for an AI to implement: support, edge connection, stairs, sky access and a handful of adjacency rules. The tension is height and width versus daily Veil cost: every room adds upkeep, so a well-furnished, levelled room is often better than two Bare ones. Furniture quality lets the player improve without growing the Veil bill, which is the intended mid-game optimization.

16. Room Catalog

Cost key: m = mana; T = Timber; S = Stone; G = Glass; C = Cloth; In = Insight; He = Herbs; Ra = Radiance; Um = Umbra; Re = Reagents; Cr = Crystal; Hw = Heartwood; Ss = Sunstone; Vp = Void Pearl; Es = Ember Salt; Ms = Moonsilk; Sm = Starmetal. **Build** is in honest reviews. **Veil** is the room's contribution to daily Veil upkeep at level 1 (+25% per level above 1). **Size** is width × height in tiles (castle) or plots (grounds). **Stats** lists primary stats, then the secondary stat in brackets. **Unlock:** *Start* = pre-built or available from day one; *Std (X Ln)* = standard plan at wizard level n in school X; *Draft (deck, rarity)* = blueprint draft card; *Grant* = given by a milestone.

There are **56 room types**: 6 core and circulation, 8 living, 10 teaching, 3 knowledge, 4 nature, 3 light, 3 shadow, 6 craft and habitat, 3 Rift and display, and 10 grounds plots.

16.1 Core and circulation

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`entrance_hall`	Entrance Hall	Start (pre-built)	2×1	—	—	—	4
`wizard_study`	Wizard's Study	Start (pre-built)	1×2	Wizard	L2: 600m·40S·20G; L3: 1,500m·60S·30G·5Cr	L2 60 / L3 120	6
`spiral_stair`	Spiral Stair	Start	1×1	—	40m·10S	5	1
`corridor`	Corridor	Start	1×1	—	25m·8T	4	1
`support_pillar`	Support Pillar	Start	1×1 (structural, no interior)	—	15m·6S	3	0.5
`storage_vault`	Storage Vault	Start (one pre-built)	1×1	+200 material cap (+150/level)	120m·20T·10S	20	2

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Entrance Hall	Grand Doors (built-in)	4	Charm, Order [Comfort]	Root of reachability. Ceremonies (mastery, retirement) play here. Charm placed here counts double toward Renown from decor. Cannot be demolished.

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Wizard’s Study	Orb Stand, Lectern (built-in)	3	Focus, Aura [Charm]	Training sessions (Section 7.3). Level = Study level (queue 1/2/4); L2 needs total wizard level 15, L3 needs 30. L3 perk: the wizard’s spells cost 10% less. Has an internal stair linking its two floors. Cannot be demolished.
Spiral Stair	—	0	—	The only vertical connector. Stack stairs to reach higher floors.
Corridor	—	1 (decor only)	Charm [—]	Horizontal connector; lets you space noisy rooms apart.
Support Pillar	—	0	—	Satisfies support (P3) for rooms above; not walkable, not a connector.
Storage Vault	Shelving	2	Order [Aura]	Raises the cap on every common material (Timber, Stone, Glass, Cloth, Provisions, Greens, Fish, Nectar) by 200 each (+150 per level). Base cap without vaults: 300 each. Essences and rare materials are uncapped. L3 perk: the vault auto-sorts, so every builder cleans 1 extra point per bell.

16.2 Living

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`dormitory`	Dormitory	Start	2×1	4 beds (+1/level)	250m·30T·10S	40	4
`grand_dormitory`	Grand Dormitory	Draft (Foundation, uncommon)	3×1	7 beds (+1/level)	650m·60T·25S·10C	80	6
`faculty_quarters`	Faculty Quarters	Std (appears with the first master)	2×1	2 faculty beds (+1/level)	350m·30T·15S·5C	50	4

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`faculty_lounge`	Faculty Lounge	Draft (Foundation, common)	2×1	4 faculty leisure seats (+1/level)	300m·25T·10C	40	4
`refectory`	Refectory	Start	3×1	12 seats (+3/level); meal capacity = seats × 2	400m·40T·20S	60	5
`kitchen`	Kitchen	Draft (Foundation, common); Std at Nature L4	1×1	—	250m·15T·15S	30	3
`common_room`	Common Room	Start	2×1	6 leisure seats (+2/level)	300m·30T·10C	45	4
`bathhouse`	Bathhouse	Draft (Foundation, uncommon)	2×1	4 (+1/level)	450m·20S·15G	60	5

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Dormitory	Beds (any bed item), up to capacity; at least 1	4	Comfort, Charm [Order]	Student beds. Capacity = beds placed (max per level). Rest target from efficiency (Section 14.4). L3 perk: roommates share Kind Heart / Sunny effects at double strength.
Grand Dormitory	Beds, at least 1	5	Comfort, Charm [Order]	As Dormitory; −15% Veil per bed compared with small dorms. L3 perk: +5 Leisure target for residents (a little common nook).
Faculty Quarters	2 Faculty Beds, Writing Desk	4	Comfort, Focus [Order]	Faculty beds (BR-22 housing term). Faculty Rest target +10. L3 perk: residents’ career proficiency +5 (flat).
Faculty Lounge	2 Armchairs	4	Comfort, Charm [Focus]	Faculty-only Leisure and Composure. At Fine quality or better: all career outputs +3%.

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Refectory	Long Table, Hearth	5	Comfort, Order [Charm]	Serves meals every 5 bells. Meals use 1 Provision per diner. L3 perk: meals restore +5 Composure.
Kitchen	Cooking Range	2	Order, Aura [Comfort]	Adjacent to a Refectory: Hearty meals (Well-Fed) and −10% Provisions per meal. Otherwise 50% effect. Fire hazard ×1.5 in this room.
Common Room	Hearth **or** Couch	5	Comfort, Charm [Order]	Leisure capacity. Social and Mischievous students love it. L3 perk: misbehavior Academy-wide −5%.
Bathhouse	Soaking Tub	4	Comfort, Aura [Charm]	Users' Rest target +10, Composure +5; injured users heal 10% faster.

16.3 Teaching (BR-08)

Five basic classrooms (standard plan at wizard level 5 in their school) and five Advanced Halls (school draft deck, rare, wizard level 7).

ID	Room	School	Unlock	Size	Seats	Cost	Build	Veil
`class_arcane`	Glyph Classroom	Arcane	Std (Arcane L5)	2×1	5 (+1/level)	300m·25T·15S	45	5
`class_nature`	Grove Classroom	Nature	Std (Nature L5)	2×1	5 (+1/level)	300m·25T·15S	45	5
`class_light`	Dawn Classroom	Light	Std (Light L5)	2×1	5 (+1/level)	300m·25T·15S	45	5
`class_shadow`	Hush Classroom	Shadow	Std (Shadow L5)	2×1	5 (+1/level)	300m·25T·15S	45	5
`class_alchemy`	Bench Classroom	Alchemy	Std (Alchemy L5)	2×1	5 (+1/level)	300m·25T·15S	45	5
`hall_arcane`	Hall of Stars	Arcane	Draft (Arcane, rare; L7)	3×1	8 (+1/level)	900m·50T·40S·20In·3Cr	100	9
`hall_nature`	Grove Hall	Nature	Draft (Nature, rare; L7)	3×1	8 (+1/level)	900m·50T·40S·20He·3Hw	100	9

ID	Room	School	Unlock	Size	Seats	Cost	Build	Veil
`hall_light`	Dawn Hall	Light	Draft (Light, rare; L7)	3×1	8 (+1/level)	900m·50T·40S·20Ra·3Ss	100	9
`hall_shadow`	Hall of Whispers	Shadow	Draft (Shadow, rare; L7)	3×1	8 (+1/level)	900m·50T·40S·20Um·3Vp	100	9
`hall_alchemy`	Crucible Hall	Alchemy	Draft (Alchemy, rare; L7)	3×1	8 (+1/level)	900m·50T·40S·20Re·3Es	100	9

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Basic classrooms	Teacher's Lectern + the school's focus item (Arcane: Orrery Globe; Nature: Potting Bench; Light: Sun Altar; Shadow: Veiled Mirror; Alchemy: Bench Cauldron)	4	Focus, Aura [Order]	Seats students of that school; teacher workplace (1). Each seated student yields 0.20 essence/bell × efficiency × morale factor (Section 8.3). 1% chance per bell per staffed classroom to also yield 1 of the school's rare material (Cr / Hw / Ss / Vp / Es). L3 perk: +1 tutor slot (resident masters) and +5% essence.
Advanced Halls	Master Lectern + the school's grand focus (Arcane: Star Orrery; Nature: Living Tree; Light: Dawn Organ; Shadow: Obsidian Mirror; Alchemy: Great Athanor)	6	Focus, Aura [Charm]	As basic, yield ×1.5, rare-material chance 2%. The teacher here gets +10 career proficiency points. L3 perk: +10% essence yield and the Hall's teacher gains a further +5 career proficiency points. (Nothing in Ankicaster alters the teacher-delay formula except the Light buff, BR-16.)

16.4 Knowledge (Arcane)

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`library`	Library	Std (Arcane L2)	2×1	4 readers (+1/level); 2 scholar slots	500m·40T·10G	70	5

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`observatory`	Observatory	Draft (Arcane, uncommon; L4)	1×2, needs sky	1 scholar slot	650m·20S·25G·3Cr	80	7
`scriptorium`	Scriptorium	Draft (Arcane, uncommon; L3)	2×1	1 scholar slot	450m·30T·10C·10In	60	5

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Library	2 Bookshelves, Reading Desk	4	Focus, Order [Charm]	Readers count as Leisure capacity. +1 Insight/bell per scholar. Sells known grimoires (Section 29) and lets the wizard research recipes and *Veil Theory* (Section 18.2) (Section 24.5). Adjacent classrooms +5%. L3 perk: grimoire study costs −10% mana.
Observatory	Great Telescope	2	Aura, Focus [Charm]	**Foresight:** the next day's random event is shown in advance and can be warded or prepared for. With a scholar: +1 Rift Fortune (max 2 scholars across observatories) and +1 Insight/bell. +10% on the highest castle floor.
Scriptorium	Scribe's Desk	3	Focus, Order [Aura]	Crafts scrolls with a scholar (Section 25.4): Rift Map, Ward Scroll, Recipe Notes, Calming Verse.

16.5 Nature

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`greenhouse`	Greenhouse	Std (Nature L2)	2×1, needs sky	2 crop beds (+1/level); 1 gardener	450m·30T·25G	60	5
`meadow_paddock`	Meadow Paddock	Std (Nature L3)	2 plots (grounds)	3 creatures (+1/level), Meadow habitat	200m·30T	30	3

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`tide_tank`	Tide Tank	Std (Nature L3)	2×1, floor 0 only	2 creatures (+1/level), Tide habitat	400m·20S·30G	50	4
`menagerie`	Menagerie	Draft (Nature, rare; L4)	3×2	6 creatures (+2/level), any habitat with modules	1,400m·60T·40S·40G·5Hw	140	12

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Greenhouse	2 Planter Beds	3	Aura, Comfort [Charm]	Each crop bed grows one crop (Section 21.3). Base output per bed per bell: 0.75 Provisions + 0.25 Herbs (or the crop's listed output). Gardener ×1.5. L3 perk: a third crop type can be planted in each bed (mixed planting).
Meadow Paddock	Feeding Trough	2	Comfort, Charm [Order]	Meadow creatures. Grounds rooms have no cleanliness penalty from creatures.
Tide Tank	Glass Tank	2	Aura, Comfort [Order]	Tide creatures. Must sit on floor 0.
Menagerie	2 Habitat Modules (max 4; each gives one habitat type)	4	Comfort, Charm [Aura]	Creatures here give +10% buff strength. Visitors count as 6 Leisure capacity. Adjacent Common Room +10%. L3 perk: wary (untamed) creatures here escape 50% less.

16.6 Light

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`infirmary`	Infirmary	Std (Light L2)	2×1	3 patient beds (+1/level); 1 healer	450m·25T·10C·10G	60	5
`aviary`	Aviary	Std (Light L3)	1×2, needs sky	3 creatures (+1/level), Sky habitat	350m·25T·15G	45	4

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`sunlit_sanctum`	Sunlit Sanctum	Draft (Light, uncommon; L4)	2×1	6 seats (+2/level); 1 healer	600m·30S·15G·15Ra	70	6

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Infirmary	2 Infirmary Cots, Remedy Cabinet	3	Comfort, Order [Aura]	Injured characters are admitted automatically. Healing per review = efficiency × (1 + healer) × Light recovery buff (Section 23.3). Hexed injuries need a healer or a potion. Adjacent Sanctum +10%. L3 perk: also treats injured creatures (otherwise they rest in their habitat at half speed).
Aviary	Perch Tree	2	Charm, Aura [Comfort]	Sky creatures. Needs sky.
Sunlit Sanctum	Sun Window, 2 Meditation Cushions	4	Aura, Charm [Comfort]	Composure target +15 × min(1, seats / headcount). With a healer: +1 Composure per bell to users. **Rite of Renewal** (once per 7 days, 50 Radiance): remove one troublesome trait from a chosen student.

16.7 Shadow

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`warding_chamber`	Warding Chamber	Std (Shadow L2)	1×1	1 warder	400m·25S·5G	50	4
`gloom_hutch`	Gloom Hutch	Std (Shadow L3)	1×1	2 creatures (+1/level), Gloom habitat	250m·15T·15S	35	3
`detention_hall`	Detention Hall	Draft (Shadow, uncommon; L4)	1×1	3 detainees; 1 warder	350m·25S	45	3

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Warding Chamber	Ward Circle	2	Order, Aura [Focus]	−5% daily Veil cost per chamber (max 3 chambers count). Rooms within 3 BFS steps: hazards −20%. With a warder: +5 Veil strength per active day; −10% hazards Academy-wide. L3 perk: breaches are auto-contained (Section 19.4) in this chamber's radius.
Gloom Hutch	Lightless Burrow	2	Comfort, Order [Aura]	Gloom creatures. Must not be adjacent to an Aviary (both rooms −10%).
Detention Hall	Penitent Benches	2	Order, Focus [—]	Misbehaving students are sent here for 3 bells (no lesson yield) and are 50% less likely to misbehave for the next 30 bells. With a warder: 5% per detention to remove Prankster or Rule-Bender. Adjacent classroom misbehavior −10%.

16.8 Craft and habitats (Alchemy and Arcane)

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`alchemy_lab`	Alchemy Lab	Std (Alchemy L2)	2×1	1 alchemist; 2 brewing stations	450m·20S·20G	60	6
`ember_pit`	Ember Pit	Std (Alchemy L3)	1×1, floors 0–1	2 creatures (+1/level), Ember habitat	300m·25S·5G	40	4
`workshop`	Workshop	Draft (Alchemy, uncommon; L4); the first Alchemy draft after L4 always offers it	2×1	2 builders	500m·40T·20S	60	5

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`glassworks`	Glassworks	Draft (Alchemy, common; L3)	1×1	—	350m·30S	40	3
`weaving_room`	Weaving Room	Draft (Foundation, common)	1×1	—	250m·20T	30	2
`crystal_vivarium`	Crystal Vivarium	Std (Arcane L3)	1×1	2 creatures (+1/level), Crystal habitat	350m·10S·20G	40	4

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Alchemy Lab	Brewing Cauldron, Ingredient Rack	3	Aura, Order [Focus]	Brewing (Section 24). Without an alchemist only basic recipes can be brewed, at normal speed. Explosion hazard (Section 19). Adjacency noise penalty to dorms unless a Fume Hood is installed. L3 perk: a third brewing station.
Ember Pit	Brazier Nest	2	Aura, Comfort [Order]	Ember creatures. Floors 0–1 only.
Workshop	Workbench	3	Order, Focus [Aura]	Builder workplace (2). Each builder adds a construction slot. Furniture costs −10% materials while a Workshop exists. Repairs Damaged furniture for 25% of its cost. L3 perk: builder slots advance 1.4 per review instead of 1.25.
Glassworks	Kiln	1	Aura, Order [—]	Converts 2 Stone → 1 Glass per bell × efficiency (toggle on/off).
Weaving Room	Loom	2	Order, Charm [Comfort]	0.5 Cloth per bell × efficiency; ×2 with a tamed Moonsheep. With Moonsilk input (1 per 20 bells) also makes Moonsilk Cloth for rare furniture.
Crystal Vivarium	Geode Bed	2	Aura, Charm [Focus]	Crystal creatures.

16.9 Rift and display

ID	Room	Unlock	Size	Capacity	Cost	Build	Veil
`rift_gate`	Rift Gate	Grant (when the Academy has 3 masters, BR-27)	2 plots (grounds)	1 expedition at a time (+1 at level 3)	1,000m·60S·20G·5Cr	120	10
`training_yard`	Training Yard	Std (with the Rift Gate)	2 plots (grounds)	3 adventurers (+1/level)	300m·30T·10S	40	2
`trophy_hall`	Trophy Hall	Draft (Foundation, rare)	2×1	4 artifact plinths (+1/level)	700m·40S·20G	80	5

Room	Required furniture	Opt. slots	Stats	Effect, placement and level perks
Rift Gate	Gate Arch, Anchor Stones (both granted with the blueprint)	2	Aura, Order [—]	Launches expeditions (Section 27). Level 2: +1 Rift Fortune; level 3: +2 Rift Fortune and a second simultaneous expedition (the level-3 renovation also needs 2 Golem Cores). Gate levels never change the odds. Levels via Rift Gate blueprint duplicates (Rift loot) or Renovation.
Training Yard	Practice Dummies	3	Order, Focus [Comfort]	Adventurers between expeditions rest here; each yard level gives drilled teams +1 Rift Fortune (max +3).
Trophy Hall	Artifact Plinths (any number, up to capacity)	2	Charm, Aura [Order]	**Artifacts only work while displayed** (Section 25.5). +2 Renown per displayed artifact. Hosts the Alumni Gallery wall.

16.10 Grounds plots

ID	Plot	Unlock	Size	Output per bell (× efficiency)	Cost	Build	Veil
`vegetable_patch`	Vegetable Patch	Start (one pre-built)	1 plot	1.0 Provisions + 0.2 Greens	60m·10T	15	2

ID	Plot	Unlock	Size	Output per bell (× efficiency)	Cost	Build	Veil
herb_garden	Herb Garden	Std (Nature L1)	1 plot	0.6 Herbs	80m·10T	20	2
woodlot	Woodlot	Start (one pre-built)	1 plot	2 Timber; 0.5% Heartwood	50m·5S	15	2
quarry	Quarry	Start	1 plot	1 Stone; 2% Crystal	80m·10T	20	2
pond	Pond	Std (Nature L2)	2 plots	0.8 Fish; Tide creatures may visit	150m·20S	30	3
orchard	Orchard	Draft (Nature, common)	2 plots	0.8 Provisions + 0.3 Nectar; +4 Leisure capacity	250m·30T	50	3
apiary	Apiary	Draft (Nature, uncommon)	1 plot	0.5 Nectar; adjacent plots +5%	200m·20T	30	2
courtyard_garden	Courtyard Garden	Draft (Foundation, common)	1 plot	+3 Leisure capacity; +3 Academy Charm	120m·15S	20	1
standing_stones	Standing Stones	Draft (Shadow, uncommon)	1 plot	Grounds hazards −50%; breach chance −25%; +2 Veil strength per active day	400m·40S	50	2
moonwell	Moonwell	Draft (Light, rare)	1 plot	Injury healing +20% Academy-wide; Composure target +5	500m·30S·10Ra	60	3

Plots have no furniture slots, except that each plot accepts **one** outdoor decor item (Section 17.10). Gardeners can be assigned to up to 3 food/herb plots (×1.5 output). Plots are always reachable and never dirty.

17. Furniture and Decor Catalog

17.1 Rules

- **Placing furniture is instant**: pay mana and materials, choose a free slot, done. (The Alchemy buff and a Workshop reduce material costs; nothing reduces mana cost except the Library L3 perk for grimoires and spellbooks that say so.)
- **Slots.** Each room has **required slots** (one per required item listed in Section 16, plus bed slots in dorms and plinths in the Trophy Hall) and **optional slots** (number listed per room, +1 per room level). Most items take 1 optional slot; items marked **(2)** take 2.
- **Required items** must be present for the room to function. A required item can be replaced by an upgraded item of the same role (e.g., any bed in a bed slot; the Master Lectern can replace a Teacher's Lectern).
- **Stacking.** Stat points from every item always add up. **Special effects** of the *same item* do not stack within one room unless the entry says so; across rooms they stack, subject to global caps (Section 33.6).
- **Moving and selling.** Items can be moved between rooms freely (instant). Removing an item returns it to inventory; selling it to the "Curio Cart" returns 25% of its mana cost.
- **Damage.** Some events and the Butterfingers trait can mark an item **Damaged** (−50% stats and effects). Repair at a Workshop for 25% of the item's cost, or with the *Mend Object* spell. Items are never destroyed.
- **Unlock sources:** *Std* = always buyable (optionally after a level gate); *Draft (deck, rarity)* = a furniture draft card grants one copy (more copies via more cards or by spending its "pattern" after the first copy — see 26.6); *Rift (destination)* = expedition loot; *Grant* = milestone or achievement.

Stat abbreviations: **Cmf** Comfort, **Foc** Focus, **Ord** Order, **Aur** Aura, **Chm** Charm.

There are **114 catalog entries** (the grand-focus entry comes in 5 school variants and the Habitat Module in 6 habitat variants, so there are 123 distinct placeable pieces): 16 general decor, 9 beds and bedroom pieces, 9 faculty and leisure, 7 refectory and kitchen, 13 classroom, 7 knowledge, 14 nature and habitat, 7 light, 6 shadow, 12 craft, storage and Rift, 8 outdoor decor, and 6 Rift-legendary pieces.

17.2 General decor (any room with a free optional slot)

ID	Item	Cost	Stats	Special effect	Unlock
`potted_fern`	Potted Fern	30m·2T	Chm 2	—	Std
`candle_sconce`	Candle Sconce	40m·2S	Foc 1, Chm 1	—	Std
`woven_rug`	Woven Rug	60m·3C	Cmf 3	—	Std
`tapestry`	Tapestry	80m·4C	Chm 4, Cmf 1	—	Std
`bookshelf`	Bookshelf	100m·10T	Foc 3, Ord 1	Counts for Well-Read trait	Std
`academy_banner`	Academy Banner	100m·5C	Chm 4	+1 Renown each (max 5 banners Academy-wide)	Std

ID	Item	Cost	Stats	Special effect	Unlock
grandfather_clock	Grandfather Clock	180m·10T·2G	Ord 4	—	Std (total level 10)
founders_portrait	Portrait of the Founder	120m·3T·2C	Chm 5	+1 Composure target to occupants (it's you!)	Grant (first master)
glow_lantern	Glow Lantern	90m·2G·2In	Foc 2, Aur 1	—	Draft (Arcane, common)
window_box	Window Box of Herbs	70m·3T·2He	Chm 2, Cmf 1	+0.05 Herbs/bell	Draft (Nature, common)
fish_bowl	Fish Bowl	90m·3G·2 Fish	Chm 3	+1 Composure target to occupants	Draft (Nature, common)
hush_chimes	Wind Chimes of Hush	110m·3G·3Um	Ord 3	Misbehavior in room −10%	Draft (Shadow, common)
radiant_lamp	Radiant Lamp	160m·2G·5Ra	Chm 3, Aur 2	—	Draft (Light, common)
enchanted_broom	Enchanted Broom	150m·5T·5Re	Ord 2	Cleans 1/bell in its room	Draft (Alchemy, common)
familiar_statue	Statue of a Familiar	300m·20S	Chm 6	—	Draft (Foundation, uncommon)
floating_candles	Floating Candles	150m·5In	Foc 2, Chm 2	—	Draft (Arcane, common)

17.3 Beds and bedroom pieces (Dormitory, Grand Dormitory, Faculty Quarters)

ID	Item	Role	Cost	Stats	Special effect	Unlock
simple_cot	Simple Cot	Bed	30m·5T	Cmf 1	—	Std
feather_bed	Feather Bed	Bed	90m·6T·3C	Cmf 3	—	Std
faculty_bed	Faculty Bed	Faculty bed	100m·8T·3C	Cmf 3	Faculty Quarters only	Std
canopy_bed	Canopy Bed	Bed	220m·8T·6C·1Ms	Cmf 5, Chm 2	Occupant Rest target +5	Draft (Foundation, uncommon)
dreamcatcher_bed	Dreamcatcher Bed	Bed	300m·8T·4C·5In	Cmf 4	Occupant Composure target +5	Draft (Arcane, uncommon)
mossbed	Mossbed	Bed	180m·6T·10He	Cmf 4	Occupant heals 20% faster	Draft (Nature, uncommon)
wardrobe	Wardrobe	Optional	70m·8T	Ord 2, Cmf 1	—	Std
footlocker	Footlocker	Optional	40m·4T	Ord 1	—	Std

ID	Item	Role	Cost	Stats	Special effect	Unlock
moon_curtains	Moon-Shroud Curtains	Optional	140m·6C·4Um	Ord 2	Dorm Rest target +5 (dark sleep)	Draft (Shadow, common)

17.4 Faculty and leisure (Faculty Quarters, Faculty Lounge, Common Room, Bathhouse)

ID	Item	Role	Cost	Stats	Special effect	Unlock
writing_desk	Writing Desk	Req (Faculty Quarters)	90m·8T	Foc 3	—	Std
armchair	Armchair	Req (Faculty Lounge)	80m·6T·3C	Cmf 3	—	Std
couch	Couch	Req-alt (Common Room)	90m·8T·4C	Cmf 3	—	Std
hammock	Hammock	Optional	60m·4C	Cmf 2	—	Std
tea_service	Tea Service	Optional	120m·2G·2He	Cmf 2, Chm 2	Lounge: career outputs +1% (stacks to 2)	Draft (Light, common)
wizard_chess	Wizard’s Chess Table	Optional	140m·10T·2In	Chm 2	+2 Leisure capacity	Draft (Arcane, common)
music_box	Music Box of Calm	Optional	160m·3G·3Ra	Chm 2	Occupants +3 Composure target	Draft (Light, common)
hex_darts	Dartboard of Harmless Hexes	Optional	120m·6T·3Um	Chm 1	Misbehavior Academy-wide −5% (max 2 boards)	Draft (Shadow, common)
warm_hearthstone	Warm Hearthstone	Optional (dorms, Common Room, Lounge)	160m·10S·3Re	Cmf 3	Halves Cold Snap effects in its room	Draft (Alchemy, common)

17.5 Refectory and kitchen

ID	Item	Role	Cost	Stats	Special effect	Unlock
long_table	Long Table	Req (Refectory)	120m·15T	Ord 2	—	Std
hearth	Hearth	Req (Refectory), Req-alt (Common Room)	150m·20S	Cmf 3	—	Std

ID	Item	Role	Cost	Stats	Special effect	Unlock
`cooking_range`	Cooking Range	Req (Kitchen)	180m·15S·2G	Ord 2, Aur 1	—	Std
`spice_rack`	Spice Rack	Optional	70m·5T·3He	Chm 1	Hearty meals give +2 extra morale	Draft (Nature, common)
`frostbox`	Frostbox	Optional (Kitchen)	200m·5S·5G·3In	Ord 2	+100 Provisions cap	Draft (Arcane, common)
`stewpot`	Bottomless Stewpot	Optional (Kitchen)	400m·10S·10Re·2Es	Aur 2	Meals use 20% fewer Provisions (stacks with Kitchen’s 10%)	Draft (Alchemy, rare)
`banquet_bell`	Banquet Bell	Optional (Refectory)	250m·10S·3G	Chm 3	Once per 7 days, ring for a Feast: next meal gives Well-Fed to all and +5 Leisure	Draft (Foundation, uncommon)

17.6 Classroom items

ID	Item	Role	Cost	Stats	Special effect	Unlock
`teachers_lectern`	Teacher’s Lectern	Req (basic classrooms)	80m·8T	Foc 2	—	Std
`orrery_globe`	Orrery Globe	Req (Glyph Classroom)	160m·5S·5G	Aur 3, Foc 1	—	Std (Arcane L5)
`potting_bench`	Potting Bench	Req (Grove Classroom)	120m·10T·3He	Aur 3, Cmf 1	—	Std (Nature L5)
`sun_altar`	Sun Altar	Req (Dawn Classroom)	150m·10S·2G	Aur 3, Chm 1	—	Std (Light L5)
`veiled_mirror`	Veiled Mirror	Req (Hush Classroom)	150m·3G·3C	Aur 3, Ord 1	—	Std (Shadow L5)
`bench_cauldron`	Bench Cauldron	Req (Bench Classroom)	150m·10S·2G	Aur 3, Ord 1	—	Std (Alchemy L5)
`student_desks`	Student Desks (set)	Optional	100m·12T	Foc 3	—	Std
`chalkboard`	Chalkboard	Optional	60m·5T·2S	Foc 2, Ord 1	—	Std
`star_chart`	Star Chart	Optional	120m·3C·5In	Foc 3 (Arcane rooms ×2)	—	Draft (Arcane, common)
`specimen_jars`	Specimen Jars	Optional	110m·4G·3He	Aur 2, Foc 1	Nature and Alchemy classrooms +3% yield	Draft (Nature, common)

ID	Item	Role	Cost	Stats	Special effect	Unlock
master_lectern	Master Lectern	Req (Advanced Halls)	400m·15T·10 (school essence)·2 (school rare)	Foc 4, Chm 2	—	Grant (with any Advanced Hall blueprint)
grand_focus	Grand focus (Star Orrery / Living Tree / Dawn Organ / Obsidian Mirror / Great Athanor) **(2)**	Req (matching Advanced Hall)	600m·20S·25 (school essence)·3 (school rare)	Aur 6, Chm 2	—	Grant (with the Hall blueprint)
inspiring_quote	Framed Inspiring Quote	Optional	50m·2T·1C	Foc 1, Chm 1	—	Std

17.7 Knowledge

ID	Item	Role	Cost	Stats	Special effect	Unlock
reading_desk	Reading Desk	Req (Library)	90m·8T	Foc 3	—	Std
rolling_ladder	Rolling Ladder	Optional (Library)	80m·8T	Ord 1	Library capacity +1	Std
great_telescope	Great Telescope **(2)**	Req (Observatory)	400m·10S·15G·2Cr	Aur 5	—	Grant (with the blueprint)
scribes_desk	Scribe’s Desk	Req (Scriptorium)	150m·10T·3C	Foc 3	—	Grant (with the blueprint)
globe_elsewhere	Globe of Elsewhere	Optional	350m·5G·10In·1Cr	Aur 3	+0.2 Insight/bell	Draft (Arcane, rare)
silent_quill	Self-Writing Quill	Optional	220m·2Ms·5In	Foc 3	Scriptorium crafts 25% faster	Draft (Arcane, uncommon)
map_table	Expedition Map Table **(2)**	Optional (Library or Training Yard)	300m·10T·5C·5In	Foc 2, Ord 2	Rift Fortune +1 (max 1)	Draft (Arcane, uncommon)

17.8 Nature and habitats

ID	Item	Role	Cost	Stats	Special effect	Unlock
planter_bed	Planter Bed	Req (Greenhouse crop bed)	60m·8T	Aur 1	Holds one crop	Std
rain_cistern	Rain Cistern	Optional (Greenhouse)	140m·10S·3G	Cmf 1	Greenhouse +10%	Std (Nature L4)

ID	Item	Role	Cost	Stats	Special effect	Unlock
`compost_bin`	Compost Bin	Optional (Greenhouse)	50m·6T	Ord 1	Greenhouse +5%; turns Murky Sludge into Herbs (1:2)	Std
`glowcap_lamp`	Glowcap Grow-Lamp	Optional (Greenhouse)	180m·3G·5He·2In	Aur 2	Greenhouse +15%	Draft (Nature, uncommon)
`feeding_trough`	Feeding Trough	Req (Meadow Paddock)	60m·8T	Ord 1	—	Std
`glass_tank`	Glass Tank **(2)**	Req (Tide Tank)	200m·20G	Aur 2	—	Std (Nature L3)
`perch_tree`	Perch Tree	Req (Aviary)	120m·10T	Chm 2	—	Std (Light L3)
`lightless_burrow`	Lightless Burrow	Req (Gloom Hutch)	100m·10S·2C	Cmf 2	—	Std (Shadow L3)
`brazier_nest`	Brazier Nest	Req (Ember Pit)	140m·15S	Aur 2	—	Std (Alchemy L3)
`geode_bed`	Geode Bed	Req (Crystal Vivarium)	160m·10S·1Cr	Aur 2	—	Std (Arcane L3)
`habitat_module`	Habitat Module (choose: Meadow, Tide, Sky, Gloom, Ember, Crystal) **(2)**	Req (Menagerie, 2–4)	250m + that habitat's required item cost	Cmf 2	Adds one habitat type to the Menagerie	Grant (with the blueprint)
`creature_toybox`	Creature Toybox	Optional (any habitat)	90m·6T·3C	Chm 2	Escape chance in room −20%	Draft (Nature, common)
`treat_dispenser`	Treat Dispenser	Optional (any habitat)	120m·5S·2Re	Ord 1	Creatures here eat 25% less (rounded down, min 1)	Draft (Alchemy, uncommon)
`warm_rock`	Sunning Rock	Optional (any habitat)	80m·10S	Cmf 2	Taming in this room +10%	Draft (Nature, common)

17.9 Light

ID	Item	Role	Cost	Stats	Special effect	Unlock
`infirmary_cot`	Infirmary Cot	Req (Infirmary bed)	90m·6T·3C	Cmf 2	—	Std
`remedy_cabinet`	Remedy Cabinet	Req (Infirmary)	150m·8T·4G	Ord 2	Healing +10%	Std
`healing_basin`	Healing Spring Basin **(2)**	Optional (Infirmary, Bathhouse)	350m·15S·10Ra	Aur 3	Healing +25%	Draft (Light, rare)

ID	Item	Role	Cost	Stats	Special effect	Unlock
sun_window	Sun Window	Req (Sunlit Sanctum)	200m·15G·5Ra	Chm 3, Aur 3	—	Grant (with the blueprint)
meditation_cushions	Meditation Cushions	Req ×2 (Sanctum); optional elsewhere	60m·4C	Cmf 2	Sanctum capacity +1	Std
soaking_tub	Soaking Tub **(2)**	Req (Bathhouse)	220m·10S·5G	Cmf 4	—	Grant (with the blueprint)
halo_candles	Halo Candles	Optional	130m·2G·4Ra	Aur 2, Chm 1	Composure target +2 in room	Draft (Light, common)

17.10 Shadow

ID	Item	Role	Cost	Stats	Special effect	Unlock
ward_circle	Ward Circle	Req (Warding Chamber)	200m·10S·2G	Ord 3, Aur 2	—	Std (Shadow L2)
ward_lantern	Ward Lantern	Optional	160m·5G·5Um	Ord 2	Hazards in room −30%, adjacent rooms −10%	Draft (Shadow, common)
penitent_benches	Penitent Benches	Req (Detention Hall)	100m·10T	Ord 2	—	Grant (with the blueprint)
silence_bell	Silence Bell	Optional	250m·5S·8Um	Ord 3	Misbehavior in adjacent rooms −15%	Draft (Shadow, uncommon)
shadow_ledger	Shadow Ledger	Optional (Warding Chamber)	300m·10C·10Um	Ord 2	Shows the next hazard roll’s target room on the map	Draft (Shadow, uncommon)
nightglass	Nightglass Pane	Optional	180m·10G·4Um	Aur 2, Chm 1	Gloom creatures in adjacent habitats +10% buff	Draft (Shadow, common)

17.11 Craft, storage, Rift and outdoor decor

ID	Item	Role	Cost	Stats	Special effect	Unlock
brewing_cauldron	Brewing Cauldron	Req (Alchemy Lab)	220m·15S·2G	Aur 3	—	Std (Alchemy L2)

ID	Item	Role	Cost	Stats	Special effect	Unlock
`ingredient_rack`	Ingredient Rack	Req (Alchemy Lab)	100m·8T·2G	Ord 2	—	Std (Alchemy L2)
`fume_hood`	Fume Hood	Optional (Alchemy Lab)	250m·10S·10G·3Re	Ord 2	Explosion chance −40%; removes the Lab's noise penalty	Draft (Alchemy, uncommon)
`alembic_array`	Alembic Array **(2)**	Optional (Alchemy Lab)	450m·20G·10Re·2Es	Aur 4	Potion yield +10%, brew time −20%	Draft (Alchemy, rare)
`workbench`	Workbench	Req (Workshop)	150m·15T·5S	Ord 2, Foc 1	—	Grant (with the blueprint)
`tool_wall`	Tool Wall	Optional (Workshop)	90m·8T·3S	Ord 2	Builder slots +5% speed	Std (Alchemy L4)
`kiln`	Kiln	Req (Glassworks)	200m·25S	Aur 2	—	Grant (with the blueprint)
`loom`	Loom	Req (Weaving Room)	150m·15T	Ord 1	—	Grant (with the blueprint)
`shelving`	Shelving	Req (Storage Vault)	50m·8T	Ord 1	—	Std
`practice_dummies`	Practice Dummies	Req (Training Yard)	80m·10T·2C	Ord 1	—	Grant (with the Rift Gate)
`supply_chest`	Supply Chest	Optional (Training Yard)	120m·10T	Ord 1	Expeditions get +1 supply slot (max 1)	Std (after the Rift opens)
`artifact_plinth`	Artifact Plinth	Req (Trophy Hall, per artifact)	150m·15S	Chm 2	Displays one artifact	Std
Outdoor decor (1 per plot):						
`garden_gnome`	Enchanted Gnome	Outdoor	60m·5S	—	+1 Academy Charm; scares away 10% of grounds hazards	Std
`bird_bath`	Bird Bath	Outdoor	80m·10S	—	Sky creatures +5% buff	Draft (Light, common)
`scarecrow`	Hexed Scarecrow	Outdoor	90m·6T·2C·2Um	—	Plot output +5%	Draft (Shadow, common)
`stone_lantern`	Stone Lantern	Outdoor	100m·12S	—	+2 Academy Charm	Std
`beehive_box`	Beehive Box	Outdoor (food plots only)	120m·8T	—	+0.1 Nectar/bell	Draft (Nature, common)

ID	Item	Role	Cost	Stats	Special effect	Unlock
`windmill_charm`	Windmill Charm	Outdoor	150m·10T·3Re	—	Plot output +8%	Draft (Alchemy, uncommon)
`fairy_ring`	Fairy Ring	Outdoor	250m·10He·5In	—	5% per day: a wild common creature visits (Section 22.3)	Draft (Nature, uncommon)
`rune_marker`	Rune Marker	Outdoor	200m·15S·5Um	—	Grounds hazards −15%	Draft (Shadow, uncommon)

17.12 Legendary furniture (Rift-only)

ID	Item	Room	Cost to place	Stats	Special effect	Source
`starfall_chandelier`	Starfall Chandelier	Any	600m	Foc 6, Chm 6	+0.3 Insight/bell	Rift: Ashen Wastes or Shattered Spire
`heartwood_bench`	Living Heartwood Bench **(2)**	Common Room, Lounge, Sanctum	600m	Cmf 6, Chm 4	Users' morale +5	Rift: Ashen Wastes or Shattered Spire
`choir_bells`	Choir of Bells	Sanctum, Entrance Hall	700m	Aur 5, Chm 5	Composure target +5 Academy-wide	Rift: Ashen Wastes or Shattered Spire
`veilweaver_loom`	Veilweaver Loom **(2)**	Warding Chamber	800m	Ord 6, Aur 4	Veil cost −5% (inside the Veil reduction cap)	Rift: Shattered Spire
`everburning_athanor`	Everburning Athanor **(2)**	Alchemy Lab	800m	Aur 8	Potion yield +15%; basic brews complete instantly	Rift: Shattered Spire
`phoenix_hearth`	Phoenix Hearth	Refectory, Common Room	700m	Cmf 6, Chm 4	Meals are always Hearty; replaces the Hearth requirement	Rift: Shattered Spire

18. The Veil and Upkeep

18.1 Role and player goal

The Veil is the shimmering enchantment that hides and protects the Academy. It is Ankicaster's one recurring fixed cost and its gentle replacement for Spellcaster University's looming Dark Lord: instead of a clock that ends the game, a daily bill that grows with the size of what you have built. The player's goal is to keep the Veil Strong at a cost that leaves room to grow, using Shadow levels, wards and warders to bring the bill down.

18.2 Daily upkeep (BR-24)

```
veil_cost_per_day = round((40 + 3 × size_points) × (1 − veil_reduction))
```

- `size_points` is the sum of room Veil values (Section 15.10). It depends only on **buildings, rooms and grounds**, never on how many characters or creatures live there.
- `veil_reduction` is the sum of every Veil-cost reduction, **capped at 50%**. Draft 8 names five kinds, and all five are present:

Kind (Draft 8)	Sources in this GDD	Reduction
Shadow wizard levels	Shadow buff, levels 6–10 (BR-09)	3% per level, max 15%
Warders	Each warder on duty in a Warding Chamber (Section 11)	3% each, max 3 warders (9%); Warder alumni (Section 31.3) up to 10%
Some creatures	Umbral Wyrmling (5%), Velvet Bat Swarm (2%)	As listed in Section 22.7
Arcane research	*Veil Theory* I / II / III, researched at a Library (below)	3% per rank, max 9%
Draft cards	*Veil Thread* (Shadow, Uncommon boon: −10% for 7 Anki days); *Warding Primer* (Shadow, Rare: a permanent −3%, once); Veilweaver Loom pattern (5%)	As listed
Other buffs	Warding Chambers (5% each, max 3 counted); Ledger of Thin Places artifact (5%); spellbooks (Section 31)	As listed

Veil Theory (Arcane research). At a Library, the wizard or a scholar researches one rank at a time. The research advances by honest reviews, like a training session, and does not block orb training.

Rank	Requires	Cost	Research time	Effect
I	Arcane L3, a Library	40 Insight + 150 mana	60 reviews	Veil cost −3%
II	Arcane L6	80 Insight + 300 mana	120 reviews	−3% more
III	Arcane L9	150 Insight + 600 mana	200 reviews	−3% more

Example Academy	size_points	Cost before reductions	With Shadow L10 + 3 chambers (−30%)
Day 1 starting layout	17	91	64
Week 2 (dorm, refectory, 2 classrooms, greenhouse, a few plots)	55	205	144
Week 6 (≈20 rooms, 8 plots)	110	370	259
Month 4 (≈35 rooms, 14 plots)	190	610	427
Late game (≈55 rooms, 18 plots, Menagerie, Rift Gate)	300	940	658

At 150 reviews a day (≈1,500 mana) the Veil takes roughly 6% of income on day 1, 15–25% in the mid game and 30–45% late, which is the intended pressure curve.

18.3 Veil strength

The Veil has a **strength** 0–100 (starts at 100).

Band	Strength	Hazard chance ×	Breaches	Composure modifier	Visual
Strong	80–100	×1.0	None	0	Faint shimmering dome
Thinning	50–79	×1.5	None	−10	Dome flickers
Thin	20–49	×2.5	1% per bell	−20	Patchy dome, wisps outside
Torn	0–19	×4.0	3% per bell	−35	Visible tears, shadowy shapes

Strength changes:

Cause	Change
Active day, upkeep paid in full	+5 (up to 100)
Each Warder on duty (active day)	+5
Standing Stones (active day)	+2 each
Reinforce the Veil (player action, pay 1× today's upkeep, any number of times)	+15 each
Veil Tonic potion	+15
Active day, upkeep **not** paid	−20 × unpaid fraction
Breach "Veil Rent", Umbral Wyrmling escape	−10
Away day (no reviews), paid or not	No change: the Veil never thins while the user is away (see 18.5)

18.4 Charging and non-payment (BR-25)

At each day tick the game processes every elapsed Anki day since the last processed day, in order:

```
for each day d not yet processed:
    cost = veil_cost_per_day(d)
    if day_had_honest_reviews(d):                # active day
        paid = min(bank, cost)
        bank -= paid
        if paid == cost: strength += 5 + warder_regen + stones_regen
        else:            strength -= 20 * (1 - paid / cost)
    else:                                         # away day (BR-26)
        floor = 0.10 * bank_at_last_review
        paid = clamp(bank - floor, 0, cost)
        bank -= paid
        # strength never changes on an away day: the Veil holds
    strength = clamp(strength, 0, 100)
    record ledger entry (day, cost, paid, strength_after, active/away)
```

What non-payment does: **only** lowers strength. A thin Veil makes hazards and breaches more likely during *future bells*. It never deletes, damages or removes anything (BR-25). The worst outcome of a Torn Veil is a busy few sessions of locked rooms, minor injuries and loose creatures, all of which resolve.

The payment policy can be set to **Auto-pay** (default) or **Ask me** (the arrival summary asks whether to pay; unpaid days thin the Veil as above). There is no way to accidentally drain the bank: Auto-pay on active days never takes more than one day's cost per day.

18.5 Breaks and the 10% floor (BR-26)

- `bank_at_last_review` is the mana bank value immediately after the most recent honest review.
- On away days (no honest reviews that Anki day), the Veil auto-pays from the bank **until the bank reaches 10% of `bank_at_last_review`**; after that, payments stop and the **Veil holds where it is**. It does not thin, even though upkeep is no longer paid: strength is frozen, no bells ring, no hazards or breaches can happen, needs do not tick, and nothing else changes.
- When the user returns and studies, the next active day resumes normal charging.
- Example: the bank was 12,000 after the last review; daily cost is 400. The user is away for 40 days. Days 1–27 are paid (10,800 total), leaving 1,200 (10%); on day 28 the remaining 0 is paid and from then the Veil holds. On return the bank shows 1,200 and the Veil is exactly as strong as when they left.

The arrival summary after a break says: "Welcome back! The Veil kept the Academy safe for 40 days (27 paid from your bank, then it held on its own). Everyone is exactly as you left them."

18.6 Design purpose and tensions

The Veil turns Academy size into an ongoing cost so that the player weighs breadth against quality, and gives the Shadow school and warders a clear job. The away rules make sure that a holiday costs at most 90% of the bank and never anything else, so a big bank is a comfortable cushion, not a score that can be wiped out.

19. Hazards and Wards

19.1 Role and player goal

Hazards are small magical problems that make the Academy feel alive and give wards, warders and the Shadow school a purpose. They are frequent enough to notice, rarely severe, and always temporary. The player's goal is to keep them rare (strong Veil, wards) and short (warders, spells, potions).

19.2 Hazard rolls

Once per bell the Academy rolls for a hazard:

```
p_hazard = 0.02 × band_multiplier × (1 + size_points / 200) × (1 − hazard_reduction)
```

- `band_multiplier` from the Veil band (Section 18.3).
- `hazard_reduction` = Shadow buff (3%/step) + warders (10% each) + creature buffs + artifacts + spellbooks, **capped at 75%**. Local reductions (Warding Chamber radius −20%, Ward Lantern −30%/−10%, Standing Stones and Rune Markers on grounds) apply when choosing and resolving the target room: a room's local reduction is the chance that a hazard aimed at it "fizzles" instead.

If a hazard occurs, pick its type by weight among types whose target rooms exist, then pick a target room (rooms weighted by their Veil value). Typical frequency: a mid-game Academy with a Strong Veil and some wards sees one hazard every 40–80 reviews.

19.3 Hazard catalog

Durations are in honest reviews. "Locked" rooms run at 0% for the duration; occupants are reassigned temporarily; nothing in them is lost.

ID	Hazard	Targets	Weight	Effect	Duration	Faster resolution
`mana_static`	Mana Static	Any room	10	Furniture special effects in the room switch off	20	Warder: instant; *Dispel* spell
`poltergeist`	Poltergeist	Castle rooms	10	Cleanliness −30; occupants Composure −5	30	Warder 10; Calming Verse scroll
`wild_growth`	Wild Growth	Greenhouse, plots, Nature rooms	8	Room locked by vines	30	Gardener 10. Silver lining: +5 Herbs when cleared

ID	Hazard	Targets	Weight	Effect	Duration	Faster resolution
`hex_fog`	Hex Fog	A whole floor	6	Floor Composure −10; Focus −3 in its rooms	40	*Clear Skies* spell; Wardwater
`gravity_hiccup`	Gravity Hiccup	A whole floor	5	Rooms on the floor ×0.5 efficiency (everything floats)	20	None needed; very funny animation
`imp_infestation`	Imp Infestation	Storage, Kitchen	6	Imps **hide** 10% of Provisions (returned in full when cleared)	40	Warder 10; Shade Hound instant
`echo_loop`	Echo Loop	Library, Arcane rooms	6	Room yields 0 (everyone repeats the same sentence)	20	Scholar 5
`wisp_swarm`	Will-o'-Wisp Swarm	Grounds	6	Plot outputs 0 on 2 plots	30	Silver lining: 25% chance a Lantern Moth stays as a visitor
`cold_snap`	Cold Snap	Greenhouse, dorms	5	Greenhouse −50%; dorm Rest target −10	40	Hearth/Warm Hearthstone in the room halves it
`night_ink`	Spill of Night	A whole floor	5	Focus −5 in rooms on the floor	20	*Glow* spell
`bubbling_cauldron`	Bubbling Cauldron	Alchemy Lab (while brewing)	special	Cleanliness −40; 30% chance one occupant is Bruised. The brew continues.	—	Fume Hood, Steady Hand, alchemist reduce chance
`kitchen_fire`	Kitchen Fire	Kitchen, Refectory	4	Room locked; next meal at 50% capacity	20	Fireproof Salve prevents; Void-like *Douse* spell
`drafty_window`	Mischievous Draft	3 random castle rooms	6	Cleanliness −15 each	—	—

ID	Hazard	Targets	Weight	Effect	Duration	Faster resolution
`rift_tremor`	Rift Tremor	Rift Gate (after the Rift opens)	4	Gate locked; expeditions launched while it lasts get −1 Rift Fortune (running expeditions are unaffected)	30	Warder 10

19.4 Rift breaches (Thin and Torn Veil only)

ID	Breach	Effect	Resolution
`shade_breach`	Shade Breach	A shade drifts into a room: room locked; each occupant 40% Bruised or 20% Hexed	A warder or the *Banish* spell resolves instantly; otherwise it leaves after 60 reviews
`gloom_pack`	Gloomhound Pack	Prowls the grounds: every habitat rolls escape ×5 this bell	Leaves after one bell; Shade Hound halves
`veil_rent`	Veil Rent	Strength −10; 3 random rooms locked	30 reviews; Reinforce the Veil ends it immediately
`wanderer`	A Wanderer from Beyond	A strange, harmless-looking creature sits in the Entrance Hall: Composure −10 Academy-wide	Feed it 20 Provisions **or** a tamer can tame it (160 reviews) into a random rare creature of any school (silver lining)

Breaches are the only events that can injure several characters at once. Injuries always heal (Section 23).

19.5 Wards summary

Ward source	Effect
Shadow wizard levels 6–10	−3% hazards per level (and −3% Veil cost, −3% misbehavior)
Warding Chamber	−5% Veil cost (max 3); −20% hazards within 3 BFS steps
Warder (career)	−10% hazards Academy-wide each; +5 Veil strength/day; instant resolution of most hazards; auto-resolves breaches
Ward Lantern	−30% in its room, −10% adjacent
Standing Stones, Rune Marker, Enchanted Gnome	Grounds hazards −50% / −15% / −10%
Ward Scroll	Prevents the next hazard on one floor
Wardwater potion	Prevents the next hazard on one floor; ends Hex Fog
Creatures (Glyph Cat, Velvet Bats, Night Lynx, Hush Raven)	See bestiary
Spells (*Ward Pulse*, *Dispel*, *Banish*)	See Section 29

20. Misbehavior

20.1 Role and player goal

Students are young wizards; they get up to things. Misbehavior is the student-driven counterpart to Veil hazards: personality-flavoured, often funny, occasionally useful, never cruel. The player's goal is to keep morale up, place Order furniture, and build a Detention Hall (and hire warders) when the Academy gets big.

20.2 Rolls

Each bell, each **student** (faculty do not misbehave) rolls:

```
p_misbehave = 0.004 × personality_mult × trait_mults × morale_mult × (1 − misbehavior_reduction) × room_order_factor
```

- `morale_mult` from Section 14.6 (×2.0 at low morale down to ×0.6 at high morale).
- `misbehavior_reduction` = Shadow buff (3%/step) + warders (10% each) + Common Room L3 (5%) + Dartboards (5% each, max 2) + Duskmouse (5%) + artifacts/spellbooks, **capped at 75%**.
- `room_order_factor` = `1 − min(0.3, room_Order_points / 100)` for the room the student is seated in (Order furniture helps), with Hush Chimes / Silence Bell / Detention adjacency applied here.

A typical 12-student mid-game Academy at morale 70 with light wards sees roughly one incident per 60–120 reviews.

20.3 Misbehavior catalog

ID	Incident	Most likely from	Effect	Silver lining / resolution
`graffiti_glyphs`	Graffiti Glyphs	Mischievous, Prankster	Room cleanliness −20; room Charm −3 until cleaned to 80+	Glyphs occasionally (10%) reveal one ingredient of an undiscovered recipe
`food_fight`	Food Fight	Social, Mischievous	Refectory cleanliness −40; next meal uses +5 Provisions	Leisure +10 for everyone involved
`unauthorized_summoning`	Unauthorized Summoning	Curious, Summoner-at-Heart	Triggers an Echo Loop or Imp Infestation hazard	15% chance a common creature visitor appears
`sneaking_out`	Sneaking Out to the Woods	Bold, Curious	Student absent 20 reviews (no lesson yield)	Returns with 3–8 of a random common material

ID	Incident	Most likely from	Effect	Silver lining / resolution
`hex_duel`	Hallway Hex Duel	Bold, Daredevil	Two students: 20% Bruised each	Both gain +1 Rift Fortune on their first expedition if they ever become adventurers (they "trained")
`creature_release`	Creature Release	Gentle (out of sympathy), Mischievous	One creature becomes Loose	—
`potion_prank`	Potion Prank	Curious, Prankster	A random cosmetic status for 1 day (green hair, floating, tiny voice)	Composure +5 for onlookers (it's hilarious)
`midnight_feast`	Midnight Feast	Social, Glutton	Provisions −10 (max 5% of stock)	Dorm Leisure +10
`bell_prank`	The Bell Tower Prank	Mischievous	All students' Rest −5	—
`library_fort`	Library Fort	Bookish (!), Dreamy	Library readers displaced for 20 reviews	+2 Insight
`portrait_mustache`	The Founder's Mustache	Prankster	Founder's Portrait Charm −2 until the room is cleaned	The wizard's sprite briefly wears the mustache. Achievement.
`broom_joyride`	Broom Joyride	Bold, Daredevil	30% Bruised	Leisure +5
`alchemy_dare`	Alchemy Dare	Curious, Firebug	Lab explosion chance ×3 next bell	—
`rift_peek`	Peeking Through the Rift	Curious, Bold (after the Rift opens)	Student wanders to the Gate; Gate locked 10 reviews	10% chance they bring back a Mossy Hollow trinket (3–8 Timber or Herbs)
`secret_club`	A Secret Club	Social, Mischievous	3 students: Leisure +10, misbehavior ×1.2 for 30 bells	The club gets a funny name in the chronicle
`dorm_swap`	Midnight Dorm Swap	Social	Bed assignments shuffled for 1 day	Roommates' Composure +3 (new friends)

20.4 Detention and consequences

If a Detention Hall with a free spot exists, the offending student is sent there automatically for 3 bells (no lesson yield; Composure −5). Afterwards their misbehavior chance is halved for 30 bells. With a warder present, each detention has a 5% chance to remove Prankster or Rule-Bender. There is **no expulsion** and no permanent penalty; misbehavior can never affect card links, skill, mastery or anything else in Anki.

21. Food, Gardens and the Greenhouse

21.1 Role and player goal

Food is the Academy's first logistics chain: plots and greenhouses grow it, storage holds it, the refectory serves it, people and creatures eat it. The player's goal is to keep production a little ahead of headcount and to choose crops that also feed the potion and creature economies.

21.2 Food resources

Resource	Eaten by	Main sources	Default cap
Provisions	All characters (1 per meal), omnivore creatures	Vegetable Patch, Greenhouse, Orchard, events, alumni	300 (+200 per Storage Vault, +100 Frostbox)
Greens	Herbivore creatures; potions	Vegetable Patch (side output), Greenhouse crops, Featherfern	300 (+200 per vault)
Fish	Fish-eating creatures; potions	Pond, Tide Tank side output (Puddlewyrm)	300 (+200 per vault)
Nectar	Insects, birds, some Light creatures; potions	Orchard, Apiary, Beehive Box, Glimmerberry crop	300 (+200 per vault)
Herbs (essence of Nature, also a crop)	Potions, some furniture	Herb Garden, Greenhouse, Nature lessons	Uncapped

Overflow above the cap is not produced (production pauses); nothing is destroyed.

21.3 Greenhouse crops

Each Greenhouse crop bed grows one crop chosen by the player (changing crops is free and instant). Outputs are per bed per bell before efficiency, gardener (×1.5) and the Nature buff (+3%/step).

Crop	Unlock	Output per bed per bell	Notes
Hearthroot	Start (with the Greenhouse)	1.0 Provisions	The staple
Moonbeans	Start	0.75 Provisions + 0.25 Greens	Default crop
Featherfern	Nature L3	0.8 Greens	For herbivore-heavy menageries
Silverleaf	Start	0.6 Herbs	Potion staple
Glimmerberry	Draft (Nature, common)	0.5 Provisions + 0.3 Nectar	Feeds moths and finches
Dreamthistle	Draft (Arcane, uncommon)	0.4 Herbs + 0.05 Insight	Arcane cross-crop
Emberpepper	Draft (Alchemy, uncommon)	0.6 Provisions + 0.05 Reagents	Kitchen-friendly
Shadecap Mushrooms	Draft (Shadow, uncommon)	0.3 Herbs + 0.05 Umbra	Must be in a Greenhouse adjacent to a Gloom Hutch or on floor 0

Crop	Unlock	Output per bed per bell	Notes
Sunwheat	Draft (Light, uncommon)	1.2 Provisions	Needs a Rain Cistern in the Greenhouse
Starfruit	Rift (Crystal Caverns) — Rare Seed	0.6 Provisions + 0.1 Insight + 0.02 Crystal	Gardener required
Heartwood Sapling	Rift (Whispering Fen, “rare herbs”) — Rare Seed	0.5 Herbs + 0.01 Heartwood	Gardener required
Lumen Lily	Rift (Whispering Fen, “rare herbs”) — Rare Seed	0.4 Herbs + 0.06 Radiance	Gardener required

21.4 Plot outputs (recap)

Plot	Output per bell
Vegetable Patch	1.0 Provisions + 0.2 Greens
Herb Garden	0.6 Herbs
Pond	0.8 Fish
Orchard	0.8 Provisions + 0.3 Nectar
Apiary	0.5 Nectar (+5% to adjacent plots)

Gardeners assigned to plots (up to 3 plots each) multiply output ×1.5 × proficiency.

21.5 Meals

Every 5th bell is a **mealtime** (Section 14.4): each character consumes 1 Provision (Glutton 2), reduced by Kitchen (−10%) and Bottomless Stewpot (−20%), rounded per meal across the whole Academy. Refectory meal capacity is `seats × 2 × efficiency` (rounded down). Creatures eat at the same mealtimes (Section 22.5).

Food balance rule of thumb: each character needs 0.2 Provisions per bell. One Vegetable Patch (1.0/bell) feeds 5 characters; a 2-bed Greenhouse of Hearthroot with a gardener at efficiency 1.2 (≈3.6/bell) feeds about 18.

21.6 Shortages

If Provisions run out at a mealtime, characters who did not eat get the “no food” Hunger target until the next meal. Morale drops; misbehavior rises. Nobody starves, leaves or is harmed. The arrival summary flags “3 students went hungry at the last meal — the pantry is empty” with a one-click link to the Build menu filtered to food rooms.

22. Creatures and Taming (Bestiary)

22.1 Role and player goal

Creatures are Ankicaster's collection layer, in the spirit of Spellcaster University's familiars and dragons: each one is a small living buff with its own home, diet and personality, and each causes a little trouble when it gets out. The player's goal is to fill habitats with complementary creatures, keep them fed and their pens pleasant, and chase rare and legendary ones through the Rift.

22.2 Habitats

Habitat	Room(s)	Unlocked at
Meadow	Meadow Paddock (grounds), Menagerie module	Nature L3
Tide	Tide Tank (floor 0), Menagerie module	Nature L3
Sky	Aviary (needs sky), Menagerie module	Light L3
Gloom	Gloom Hutch, Menagerie module	Shadow L3
Ember	Ember Pit (floors 0–1), Menagerie module	Alchemy L3
Crystal	Crystal Vivarium, Menagerie module	Arcane L3

22.3 How creatures arrive

Source	Rarity	Rule
Visitors	Common, sometimes Uncommon	At each day tick, each habitat type with a free space has a 4% chance (+5% Fairy Ring, +Lure Musk guaranteed) that a wild creature of that habitat appears in the pen as **Wary**. Visitors that are not wanted can be released (no penalty).
Lure cards (draft)	Common–Rare	A school deck lure card summons a creature of that school (rarity on the card) into a matching free habitat.
Silver linings	Varies	Some hazards, breaches and misbehavior incidents bring a creature (e.g., Wisp Swarm → Lantern Moth; Wanderer → random rare).
The Rift	Common–Legendary	Expedition loot arrives as **eggs** that hatch in a matching habitat (Section 27.10): Common eggs from Mossy Hollow, owls and sprites from the Whispering Fen, Rares from the Crystal Caverns, fire and Epic eggs from the Ashen Wastes, Legendary eggs from the Shattered Spire. The Geode Golem is awakened from three Golem Cores.

Source	Rarity	Rule
Events	Varies	E.g., “A Basket on the Doorstep” (Section 28).

A creature with nowhere to live waits in the Entrance Hall for 1 day (“Looking for a home”), then wanders off. Rift eggs and Rift creatures never wander off: they wait in Storage until a habitat is free.

22.4 Taming

1. A newly arrived creature is **Wary**: it eats, gives no buff, and escapes more easily.
2. Assign a **tamer** (career), or the **wizard** (Nature L3+, Common creatures only, 50% speed).
3. Each honest review adds taming progress: `1.0 × tamer_proficiency × (1 + nature_buff) × modifiers` (Gentle personality +4%, Beast Whisperer +25%, Sunning Rock +10%, Beast Biscuit +30 points at once).
4. When progress reaches the creature’s taming requirement, it becomes **Tamed** and its buff turns on.
5. **Bond.** Tamed creatures gain Bond levels: Bond 2 after 150 more honest reviews, Bond 3 after 400. Buff strength ×1.25 at Bond 2, ×1.5 at Bond 3. Bond never decreases.

Rarity	Taming requirement (reviews)	Escape chance per bell (Wary / Tamed)
Common	40	1.5% / 0.20%
Uncommon	90	1.5% / 0.20%
Rare	160	2.0% / 0.25%
Epic	260	2.5% / 0.30%
Legendary	400	3.0% / 0.30%

22.5 Food

At each mealtime (every 5 bells) each creature eats its diet from stock. A creature that cannot be fed becomes **Hungry**: buff halved, escape chance ×3 until fed. Creatures never die, sicken permanently or leave because of hunger.

22.6 Escapes and “trouble”

Each bell, each creature rolls its escape chance × modifiers (habitat quality: Bare ×1.5, Plain ×1.2, Cozy ×1.0, Fine ×0.8, Splendid ×0.5; Creature Toybox −20%; tamer assigned −50%; Hungry ×3; Shadow buff −3%/step). On escape:

1. The creature’s **trouble** effect (bestiary column) happens once.
2. The creature becomes **Loose**: its sprite wanders the map; its buff is off.
3. It returns when a tamer or warder recaptures it (15 reviews of their time), when the wizard casts *Beckon* (instant), or on its own after 40 reviews if its habitat has food in stock.

Loose creatures are **never lost**. Any resource “hidden” by trouble is returned on recapture.

22.7 Bestiary

Buffs are at Bond 1 and apply only while the creature is Tamed and fed. Same-species buffs stack up to 2 copies unless stated. Diet is per meal.

#	Creature	School	Rarity	Where found	Habitat	Taming	Diet	Buff	Escape trouble
1	Mossback Tortoise	Nature	Common	Visitor; Nature lure	Meadow	40	1 Greens	Greenhouse output +5%	Naps in the Vegetable Patch: that plot −50% for 10 reviews
2	Thistle Hare	Nature	Common	Visitor; Nature lure	Meadow	40	1 Greens	+0.1 Herbs per bell	Hides 5 Herbs (returned on recapture)
3	Bramblefox	Nature	Uncommon	Visitor; Nature lure	Meadow	90	1 Provisions	Grounds hazards −15%	Hides 5 Provisions
4	Moonsheep	Nature	Uncommon	Nature lure	Meadow	90	2 Greens	Weaving Room ×2 output; 1 Moonsilk per 40 bells	Sleeps in a dorm bed: one student's Rest −15
5	Puddlewyrm	Nature	Uncommon	Pond visitor; Nature lure	Tide	90	1 Fish	Pond +15%	Floods a floor-0 room: cleanliness −30
6	Glimmerhorn Stag	Nature	Rare	Rift egg: Crystal Caverns	Meadow	160	3 Greens	Academy morale +3	Knocks the paddock fence: every creature in its pen rolls escape
7	Coral Kelpie	Nature	Rare	Rift egg: Whispering Fen	Tide	160	2 Fish	Bathhouse and Infirmary +10%	Soaks a student: Composure −10
8	Verdant Drake	Nature	Epic	Rift egg: Ashen Wastes	Meadow	260	2 Provisions + 2 Greens	Food and Herbs output +12%	Scorches the Greenhouse: locked 30 reviews
9	Elder Treant Sprout	Nature	Legendary	Rift egg: Shattered Spire	Meadow	400	3 Greens + 1 Herbs	Gardener output +20%	Roots itself in a corridor: that corridor blocks movement for 20 reviews

#	Creature	School	Rarity	Where found	Habitat	Taming	Diet	Buff	Escape trouble
10	Inkling Wisp	Arcane	Common	Visitor; Arcane lure	Crystal	40	1 Insight	Library +5%; +0.05 Insight per bell	Hides 3 books: Library −20% for 15 reviews
11	Orrery Moth	Arcane	Common	Visitor; Arcane lure	Sky	40	1 Nectar	Observatory +5%; Rift Fortune +1 (Section 27.9)	Nibbles curtains: one Cloth furniture item Damaged
12	Glyph Cat	Arcane	Uncommon	Arcane lure	Crystal	90	1 Provisions	Hazards −8%	Teleports into a classroom: its yield is 0 for 10 reviews (everyone pets it)
13	Prism Beetle	Arcane	Uncommon	Quarry visitor; Arcane lure	Crystal	90	1 Insight	1 Crystal per 25 bells	Chews glass: one Glass furniture item Damaged
14	Star Heron	Arcane	Rare	Rift egg: Whispering Fen	Sky	160	2 Fish	Foresight +1 day; Rift Fortune +1	Perches on the Observatory: locked 15 reviews
15	Phase Ferret	Arcane	Rare	Rift egg: Crystal Caverns	Crystal	160	2 Insight	Successful expeditions +1 loot roll	“Borrows” an artifact: inactive until recaptured
16	Aether Serpent	Arcane	Epic	Rift egg: Ashen Wastes	Crystal	260	3 Insight	Draft prices −8%	Tears a tiny rift: Echo Loop hazard
17	Chronoflare Owl	Arcane	Legendary	Rift egg: Shattered Spire	Sky	400	3 Insight	Once per day, a draft shows 4 cards instead of 3	Stops the clocks: Grandfather Clocks and Foresight off for 20 reviews
18	Lantern Moth	Light	Common	Visitor; Wisp Swarm silver lining; Light lure	Sky	40	1 Nectar	Academy Charm +3; Composure target +1	Swarms a lamp: a room’s Focus −5 for 10 reviews

#	Creature	School	Rarity	Where found	Habitat	Taming	Diet	Buff	Escape trouble
19	Dawn Finch	Light	Common	Visitor; Light lure	Sky	40	1 Greens	Composure target +2	Sings at night: one dorm's Rest −5
20	Halo Lamb	Light	Uncommon	Light lure	Meadow	90	1 Greens	Healing +10%	Follows a student to class: room Focus −2 for 10 reviews
21	Gilded Pangolin	Light	Uncommon	Light lure; event	Ember	90	1 Nectar	+5 Renown when tamed; Trophy Hall artifacts' Renown ×1.5	Rolls off with 10 Glass (returned on recapture)
22	Dawnhart Gryphon	Light	Rare	Rift egg: Crystal Caverns	Sky	160	3 Provisions	As a companion: expeditions return 15% sooner	Carries a student to the roof: 50% Bruised
23	Sunlit Unicorn	Light	Epic	Rift egg: Ashen Wastes	Meadow	260	2 Greens + 1 Radiance	Healing +20%; Hexed injuries heal without a healer	Hides in the Sanctum: locked 20 reviews (Composure +5 nearby)
24	Ember Phoenix	Light	Legendary	Rift egg: Ashen Wastes (rare) or Shattered Spire	Ember or Sky	400	2 Radiance + 1 Reagents	Recovery +10%; as a companion, heals the worst Rift injury on return	A small fire in its room: locked 15 reviews
25	Duskmouse	Shadow	Common	Visitor; Shadow lure	Gloom	40	1 Provisions	Misbehavior −5%	Startles the Refectory: next meal −10% capacity
26	Hush Raven	Shadow	Common	Visitor; Shadow lure	Gloom or Sky	40	1 Provisions	Hazards resolve 15% faster	Steals shinies: hides 5 Glass
27	Velvet Bat Swarm	Shadow	Uncommon	Shadow lure	Gloom	90	1 Nectar	Hazards −8%; Veil cost −2%	Roosts in a dorm: residents' Rest −10 for 10 reviews

#	Creature	School	Rarity	Where found	Habitat	Taming	Diet	Buff	Escape trouble
28	Shade Hound	Shadow	Uncommon	Shadow lure; may stay after a breach	Gloom	90	1 Umbra	Breach effects −50%; clears Imp Infestations instantly	Howls: floor Composure −10 for 15 reviews
29	Night Lynx	Shadow	Rare	Rift egg: Crystal Caverns	Gloom	160	2 Provisions	Hazards −15%	Hunts: another random creature becomes Loose
30	Umbral Wyrmling	Shadow	Epic	Rift egg: Ashen Wastes	Gloom	260	3 Umbra	Veil cost −5% (inside the cap)	Gnaws the Veil: strength −10
31	Moonless Kitsune	Shadow	Legendary	Rift egg: Shattered Spire	Gloom	400	2 Umbra + 1 Nectar	Misbehavior −20%	Plays tricks: three students get a Potion Prank cosmetic status
32	Cinder Newt	Alchemy	Common	Visitor; Alchemy lure; Rift egg: Ashen Wastes	Ember	40	1 Reagents	Potion yield +5%	Singes a room: cleanliness −15
33	Mopslime	Alchemy	Common	Visitor; Alchemy lure	Tide	40	1 Provisions	Cleans 3 points per bell Academy-wide	Absorbs 5 Provisions (returned on recapture)
34	Bellows Toad	Alchemy	Uncommon	Alchemy lure	Ember	90	1 Fish	Construction slots +5% speed	Puffs up in the Workshop: locked 10 reviews
35	Quicksilver Otter	Alchemy	Uncommon	Pond visitor; Alchemy lure	Tide	90	2 Fish	Glassworks and Weaving Room +15%	Slips into the Pond: Pond output 0 for 15 reviews
36	Alembic Salamander	Alchemy	Rare	Rift egg: Ashen Wastes	Ember	160	2 Reagents	Potion yield +10%; lab explosions −30%	Starts a lab fire: Alchemy Lab locked 20 reviews

#	Creature	School	Rarity	Where found	Habitat	Taming	Diet	Buff	Escape trouble
37	Philosopher's Chimera	Alchemy	Legendary	Rift egg: Shattered Spire	Menagerie only (needs Ember + Crystal modules)	400	2 Reagents + 2 Provisions	Room and furniture materials −8% (inside the cap)	Transmutes a stack: up to 50 Timber become the same amount of Stone
38	Fen Sprite	Nature	Uncommon	Rift egg: Whispering Fen	Meadow or Tide	90	1 Nectar	Herb Garden and Greenhouse +8%	Hides 5 Herbs in the reeds (returned on recapture)
39	Mistfeather Owl	Arcane	Rare	Rift egg: Whispering Fen	Sky	160	2 Fish	Library and Scriptorium +10%; Foresight +1 day	Hoots in the Library: locked 10 reviews
40	Geode Golem	Alchemy	Rare	Awakened from 3 Golem Cores (Crystal Caverns) at a Workshop	Crystal	160	2 Stone	Construction slots +10% speed; Storage caps +10%	Wanders into a corridor and sits down: that corridor blocks movement for 10 reviews

Totals: 40 creatures — 10 Common, 12 Uncommon, 9 Rare, 4 Epic, 5 Legendary. Per school: Nature 10, Arcane 9, Light 7, Shadow 7, Alchemy 7. Every Common species can also hatch from a Mossy Hollow egg. Every Rift destination yields at least one egg type: Common (Hollow), owl and sprite (Fen), Rare (Caverns), fire and Epic (Wastes), Legendary (Spire).

22.8 Creature caps and stacking

- Percent buffs from creatures stack additively within a category and are included in that category's global cap (Section 33.6).
- At most **2** of the same species give their buff; extra copies are cosmetic (and still eat).
- Creatures carried on Rift expeditions as companions (Section 27.11) give Rift Fortune instead of their home buff while away.

23. Injuries, Health and Healers

23.1 Role and player goal

Injuries are the “cost” side of hazards, breaches, misbehavior and the Rift, and the reason the Infirmary and healers exist. They are temporary, readable and never lethal (BR-27: no deaths). The player’s goal is to keep injured characters few and short-lived so that adventurers can go out again and careers keep running.

23.2 Injury types

Injury	Typical causes	Health cap while injured	Heal requirement (points)	Effects while injured	Needs a healer or potion?
Bruised	Hex duels, joyrides, minor hazards, minor Rift injuries (Easy–Medium)	85	30	Career output −10%	No
Sprained	Minor Rift injuries (Hard–Deadly), serious Rift injuries at Mossy Hollow, gryphon rides	70	80	Cannot go on expeditions; career output −25%	No
Hexed	Shade Breach, Rift curses (Easy–Medium), potion accidents	60	120	Composure −10; cannot go on expeditions	Heals at 25% speed without a healer, Sunlit Unicorn, or a Mending Draught
Broken Bone	Serious Rift injuries (Whispering Fen and beyond) only	50	200	Cannot work or go on expeditions; stays in an Infirmary bed if available	No, but heals at 50% speed outside an Infirmary
Rift-Touched	Rift curses (Hard–Deadly), some Rift events	80	100	Faint glow; Composure −10; immune to Hex Fog (silver lining)	No

Injury types are never shifted (the Rift’s outcome table is binding). What changes is recovery: Brave heals all injuries 15% faster; Steadfast heals Rift injuries 25% faster and Daredevil 25% slower; a Courage Draught halves the recovery requirement of injuries and curses from its expedition; the Ember Phoenix companion heals the worst Rift injury on return.

23.3 Healing

Each honest review adds healing points to every injured character:

```
heal_per_review = base × place_factor × healer_factor × (1 + light_recovery_buff) × modifiers
```

Factor	Value
base	1.0
place_factor	Infirmary bed: room efficiency; resting in own bed: 0.5; nowhere (no bed): 0.3
healer_factor	×2 × healer proficiency if a healer staffs that Infirmary; 1 otherwise
light_recovery_buff	+3% per Light step (BR-09)
modifiers	Mossbed +20% (own bed), Bathhouse +10%, Moonwell +20%, Halo Lamb +10%, Sunlit Unicorn +20%, Healing Spring Basin +25%, Remedy Cabinet +10%, Hardy trait ×1/0.7 heal requirement, artifacts

Example: a Sprained adventurer in an Infirmary at efficiency 1.1 with a healer (proficiency 1.0) and Light L8 (+9%) heals 2.4 points per review: 80 points in about 34 reviews.

Infirmary admission is automatic (most severe first). Infirmaries at level 3 also treat creatures that were hurt on expeditions.

23.4 Health

Health is 100 unless injured; while injured it is set to the injury's cap and rises toward 100 as healing points accumulate. Health feeds morale (10% weight). There is no disease system and no permanent health loss.

23.5 Design purpose

Injuries create a light, satisfying chain — something happens, a healer fixes it, the character is back — without any risk of loss. They make Light useful for every playstyle and give the Rift's "when it goes wrong" table real but recoverable weight.

24. Alchemy and Potions

24.1 Role and player goal

Alchemy turns surplus ingredients into flexible one-shot tools: heal an injury now, push a construction over the line, calm a rowdy floor, top up the Veil, or prepare an expedition. Like Spellcaster University's cauldron, recipes combine **three different ingredients** and must be discovered. The player's goal is to discover the recipes that matter for their Academy and keep a small stock of emergency potions.

Rule: no potion, recipe or ingredient ever produces mana (BR-01).

24.2 Brewing

1. Requires an **Alchemy Lab** (2 brewing stations; 3 at L3). Basic recipes can be brewed without an alchemist; Advanced, Rare and Legendary recipes need an alchemist on duty.
2. Choose three **different** ingredients and a station. Ingredients are consumed (one unit each unless the recipe lists more).
3. Brewing takes honest reviews: Basic 20, Advanced 40, Rare 80, Legendary 120 (alchemist ×0.5; Alembic Array −20%; Everburning Athanor makes Basic instant).
4. Yield: **2 potions** base × (1 + Alchemy potion-yield buff 3%/step + Cinder Newt 5% + Alembic Salamander 10% + items) with fractional potions accumulating per recipe; an alchemist adds +1 potion every 4 brews.
5. While a station is brewing, each bell rolls a lab explosion chance of 0.5% (the *Bubbling Cauldron* hazard), reduced by alchemist (−50%), Steady Hand, Fume Hood (−40%) and Alembic Salamander (−30%), increased by Firebug and Alchemy Dare. An explosion never ruins the brew.

24.3 Discovery

- Recipes are **fixed** for a save by default (setting `randomize_recipes`, default off; when on, the ingredient triples are shuffled per Timed Term like Spellcaster University's per-run recipes).
- **Experimenting:** combining an unknown triple consumes the ingredients and either discovers the recipe (and brews it) or produces **Murky Sludge** (×2), which a Compost Bin turns into Herbs. Experiments are never a total loss.
- **Research:** at a Library, spend 30 Insight to reveal one ingredient of an undiscovered recipe (scholars reduce the cost). Recipe Notes scrolls do the same.
- **Hints:** Graffiti Glyphs (misbehavior) and some events reveal ingredients.

24.4 Recipe book

#	Potion	Tier	Ingredients	Effect
1	Mending Draught	Basic	Herbs + Radiance + Greens	Fully heals one injury (including Hexed)

#	Potion	Tier	Ingredients	Effect
2	Calming Tonic	Basic	Herbs + Umbra + Nectar	Composure +30 for everyone in a room; ends a misbehavior chain there
3	Growth Elixir	Basic	Herbs + Reagents + Fish	One Greenhouse or plot ×2 output for 10 bells
4	Beast Biscuit	Basic	Greens + Fish + Reagents	+30 taming points to one creature; or instantly recaptures a Loose creature
5	Lure Musk	Basic	Nectar + Fish + Insight	Guarantees a visitor at the next day tick in a chosen habitat with space
6	Polishing Spirit	Basic	Reagents + Greens + Nectar	One room to 100 cleanliness; no dirt for 20 bells
7	Fireproof Salve	Basic	Reagents + Fish + Umbra	One room immune to fires and explosions for 50 bells
8	Dreamless Draught	Basic	Umbra + Nectar + Greens	Residents of one dorm: Rest +20
9	Veil Tonic	Advanced	Umbra + Insight + Radiance	Veil strength +15
10	Wardwater	Advanced	Umbra + Radiance + Fish	Prevents the next hazard on one floor; ends Hex Fog
11	Quickmortar	Advanced	Reagents + Crystal + Fish	+20 progress points to one construction site
12	Rift Elixir	Advanced	Insight + Herbs + Nectar	+10% safe odds on one expedition, up to the destination’s best odds (drunk at launch; the only item that changes Rift odds)
13	Courage Draught	Advanced	Radiance + Reagents + Greens	One expedition: any injury or curse it causes needs half the usual recovery
14	Glass Flux	Advanced	Reagents + Herbs + Crystal	Produces 20 Glass
15	Elixir of Fellowship	Rare	Moonsilk + Nectar + Radiance	Academy morale +10 for 30 bells
16	Starlight Ink	Rare	Insight + Starmetal + Umbra	Grants 3 draft reroll tokens

#	Potion	Tier	Ingredients	Effect
17	Transmutation Salts	Rare	Ember Salt + Reagents + Insight	Convert 50 of one common material into 40 of another
18	Phoenix Draught	Legendary	Phoenix Ash + Radiance + Reagents	Instantly heals every injury and curse of one returned team

Potions are stored in inventory (no cap) and used from the Inventory screen or directly from the relevant screen (e.g., the Rift planner offers the Rift Elixir and Courage Draught).

25. Materials, Essences and Artifacts

25.1 Common materials

Material	Abbr.	Main sources	Main uses	Cap
Timber	T	Woodlot (2/bell)	Rooms, wooden furniture	300 + vaults
Stone	S	Quarry (1/bell)	Rooms, stone furniture, Glassworks input	300 + vaults
Glass	G	Glassworks (2 Stone → 1 Glass), Glass Flux, events	Greenhouses, labs, tanks, lamps	300 + vaults
Cloth	C	Weaving Room (0.5/bell, ×2 with Moonsheep), events	Beds, rugs, banners, curtains	300 + vaults

Starting stock: 120 Timber, 60 Stone, 20 Glass, 20 Cloth, 40 Provisions, 10 Greens. (A starter kit, not a reward for past study.)

25.2 Essences (one per school)

Essence	Abbr.	School	Produced by	Used for
Insight	In	Arcane	Arcane lessons, Library scholars, Observatory, Starfruit, Dreamthistle	Recipe research, scrolls, draft rerolls (15 per reroll), Arcane furniture, Inkling Wisp food
Herbs	He	Nature	Nature lessons, Herb Garden, Greenhouse, Thistle Hare	Most potions, Nature furniture
Radiance	Ra	Light	Light lessons, Lumen Lily	Healing potions, Sanctum rites, Light furniture
Umbra	Um	Shadow	Shadow lessons, Shadecap	Wards, Shadow furniture, Shade Hound food
Reagents	Re	Alchemy	Alchemy lessons, Emberpepper	Potions, Alchemy furniture, Cinder Newt food

Essences are uncapped. Students produce them every bell (Section 8.3).

25.3 Rare materials

Material	Abbr.	Sources (non-Rift)	Rift sources	Uses
Crystal	Cr	Quarry 2%/bell; Prism Beetle; Arcane classrooms 1%/bell	Crystal Caverns	Observatory, Rift Gate, Arcane hall, potions
Heartwood	Hw	Woodlot 0.5%/bell; Nature classrooms 1%/bell	Whispering Fen	Nature hall, Menagerie
Sunstone	Ss	Light classrooms 1%/bell	Ashen Wastes, Shattered Spire	Light hall
Void Pearl	Vp	Shadow classrooms 1%/bell	Ashen Wastes, Shattered Spire	Shadow hall
Ember Salt	Es	Alchemy classrooms 1%/bell	Ashen Wastes, Shattered Spire	Alchemy hall, Stewpot, Alembic Array, Transmutation Salts
Moonsilk	Ms	Moonsheep (1 per 40 bells)	Whispering Fen	Canopy Bed, Self-Writing Quill, Elixir of Fellowship
Starmetal	Sm	—	Crystal Caverns, Ashen Wastes, Shattered Spire	Starlight Ink, Rift Gate L3 renovation, late blueprints
Golem Core	Gc	—	Crystal Caverns	Awakening a Geode Golem (3 cores, Section 27.10); Rift Gate L3 renovation (2 cores)
Phoenix Ash	Pa	—	Ashen Wastes, Shattered Spire	Phoenix Draught

25.4 Scrolls (Scriptorium)

Scroll	Cost	Craft time (reviews, with scholar)	Effect
Rift Map	20 In + 2 C	30	+2 Rift Fortune on one expedition (max 1 map per expedition; never changes the odds)
Ward Scroll	15 In + 10 Um	30	Prevents the next hazard on one floor
Recipe Notes	25 In	20	Reveals one ingredient of an undiscovered recipe
Calming Verse	10 In + 10 Ra	20	Ends a misbehavior chain; Composure +20 in one room

25.5 Artifacts

Artifacts are unique trophies found mainly in the Rift. **An artifact works only while displayed** on an Artifact Plinth in a Trophy Hall (4 plinths, +1 per level; multiple Trophy Halls allowed). Each displayed artifact also gives +2 Renown. Duplicates of the same artifact do not stack.

#	Artifact	Source	Effect while displayed
1	Founder’s Chalk	Ashen Wastes or Shattered Spire	Classroom essence yields +8%
2	Star Map of Many Doors	Ashen Wastes or Shattered Spire	Rift Fortune +1
3	Abacus of the First Bursar	Shattered Spire	Mana +2% (inside the +25% cap)
4	Gemcutter’s Loupe	Shattered Spire	Rare-material chances ×1.5
5	Astrolabe of Unwritten Hours	Shattered Spire	School-deck drafts show 4 cards
6	Seedvault Acorn	Ashen Wastes or Shattered Spire	Food output +10%
7	Beastcaller Horn	Ashen Wastes or Shattered Spire	Taming +15%
8	Everfull Pantry Jar	Ashen Wastes or Shattered Spire	Meals use 10% fewer Provisions
9	Heart of the Wildwood	Shattered Spire	Creature buffs +10%
10	Tidebell	Ashen Wastes or Shattered Spire	Fish +20%; Tide creatures +10%
11	Mender’s Satchel	Ashen Wastes or Shattered Spire	Healing +15%
12	Crown of Small Kindnesses	Ashen Wastes or Shattered Spire	Academy morale +3
13	Bell of Returning	Shattered Spire	Loose creatures return 50% faster; expeditions return 10% sooner
14	Sunken Chapel Bell	Shattered Spire	Composure target +8 Academy-wide
15	Lantern of the Long Watch	Ashen Wastes or Shattered Spire	Hazards −10%
16	Orb of Quiet Corridors	Ashen Wastes or Shattered Spire	Misbehavior −10%
17	Ledger of Thin Places	Shattered Spire	Veil cost −5% (inside the cap)
18	Mirror of Second Sight	Shattered Spire	Once per day, reroll a random event’s options
19	Umbral Keystone	Shattered Spire	Breaches cannot happen at Thin (only at Torn)
20	Crucible Shard	Ashen Wastes or Shattered Spire	Potion yield +8%
21	Builder’s Plumb of Truth	Ashen Wastes or Shattered Spire	Build time −5% (inside the build-time cap)

#	Artifact	Source	Effect while displayed
22	Phoenix Feather Quill	Shattered Spire	Once per 30 days, instantly heals one injury or curse from the Rift

26. The Card Draft

26.1 Role and player goal

The draft is Ankicaster's version of Spellcaster University's signature mechanic: spend currency to see three cards and keep one. It is how special rooms, rare furniture, creature lures, grimoires, crop seeds and one-shot boons enter the game, and it keeps every Academy different. The player's goal is to draft toward a plan (for example "a strong Shadow wing") while staying flexible.

26.2 Decks

Deck	Unlocked	Pays with	Base price	Content
Foundation	Start	Mana	120	General rooms, comfort furniture, supplies, Academy boons
Arcane	Arcane L1	Mana	150	Knowledge rooms, Arcane furniture, Arcane lures, grimoires, seeds
Nature	Nature L1	Mana	150	Gardens, habitats, Nature lures, grimoires, seeds
Light	Light L1	Mana	150	Care rooms, Light furniture and lures, grimoires
Shadow	Shadow L1	Mana	150	Wards, Shadow furniture and lures, grimoires
Alchemy	Alchemy L1	Mana	150	Craft rooms, Alchemy furniture and lures, recipes, grimoires

26.3 Drawing

1. Choose a deck and pay its current price.
2. Three **different** cards are revealed. Each card's rarity is rolled from the deck's rarity weights (26.4), then a card is picked uniformly from the eligible cards of that rarity (cards whose unlock gates are met and that are not unique-and-owned).
3. Keep **one**. The other two return to the deck.
4. **Reroll** the three (once per draw) for 15 Insight or a reroll token.
5. The kept card goes to the **hand** (limit 8). If the hand is full, the player must play or discard a card first. Discarding a card returns a small salvage (common materials worth roughly 20% of the draw price; never mana).

Price growth: `price = base × 1.15^k`, where k is the number of draws from that deck in the last 7 Anki days, capped at 2.5 × base. Prices relax on their own as days pass. Aether Serpent −8%;

spellbooks may reduce further (Section 31). This replaces Spellcaster University's ever-rising prices with a recent-use curve that cannot punish a long game.

Pity rule: at wizard level 4+ in the deck's school (Foundation: total level 20+), if 10 consecutive draws from a deck showed no Rare-or-better card, the next draw shows at least one.

Guaranteed offers: the first Alchemy draw after Alchemy L4 includes the Workshop; the first draw from a school deck after that school's L7 includes its Advanced Hall; the first Foundation draw after 6 students includes the Grand Dormitory.

26.4 Rarity weights

Wizard level in the deck's school (Foundation: total level ÷ 5)	Common	Uncommon	Rare	Epic	Legendary
1–2	85%	15%	0%	0%	0%
3	65%	30%	5%	0%	0%
4–6	55%	32%	13%	0%	0%
7–8	45%	33%	17%	5%	0%
9–10	40%	32%	18%	8%	2%

Renown shifts 1 percentage point from Common to Rare per 50 Renown (max +6).

26.5 Card types

Type	Played how	Notes
Blueprint	Place a room (consumed) or play on an existing room of the same type to level it	After a drafted room is built once, its plan becomes a **Known Plan**: further copies can be built from the Build menu at 1.5 × mana cost
Furniture pattern	Grants one copy now and unlocks the pattern	Further copies can be bought in the Build menu at the listed cost
Lure	Summons a creature of the stated rarity and school into a free matching habitat	Fizzles back into the hand if no habitat is free
Grimoire	Becomes available to study in the Wizard's Study	See Section 29
Seed	Unlocks a greenhouse crop permanently	
Recipe page	Reveals one full recipe	
Boon	One-shot effect	Listed below
Deed	Land Grant (+2 plots), Renovation Deed (level a standard room without material cost)	

26.6 Deck lists

Foundation

Rarity	Cards
Common	Kitchen (blueprint); Faculty Lounge (blueprint); Weaving Room (blueprint); Courtyard Garden (blueprint); Crate of Supplies (boon: +40 Timber, +20 Stone, +10 Glass); Bountiful Harvest (boon: +60 Provisions); Spring Cleaning (boon: every room +30 cleanliness); Builder's Afternoon (boon: +30 progress to one construction)
Uncommon	Grand Dormitory; Bathhouse; Canopy Bed (pattern); Statue of a Familiar (pattern); Banquet Bell (pattern); Feast Day (boon: next meal Well-Fed for all); Open Day (boon: +3 applicants with 2 positive traits each); Mentor's Letter (boon: add a positive trait to one student)
Rare	Trophy Hall; Land Grant (deed); Renovation Deed; Fair Weather (boon: no hazards for 50 bells)
Epic	Grand Opening (boon: +25 Renown); Architect's Insight (boon: next 3 constructions need 30% fewer reviews)

Arcane

Rarity	Cards
Common	Glow Lantern; Floating Candles; Star Chart; Wizard's Chess Table; Frostbox (patterns); Arcane Lure (common); Moment of Clarity (boon: +20 Insight)
Uncommon	Scriptorium; Observatory (L4); Dreamcatcher Bed; Self-Writing Quill; Expedition Map Table (patterns); Dreamthistle (seed); Arcane Lure (uncommon); Grimoire (Arcane tier 2)
Rare	Hall of Stars (L7); Globe of Elsewhere (pattern); Grimoire (Arcane tier 3); Starlit Night (boon: Arcane yields ×2 for 20 bells); Renovation Deed
Epic	Grimoire (Arcane tier 4); Steady Prices (boon: your next 5 draws from any deck cost base price)
Legendary	Starfall Fragment (boon: +5 Crystal, +2 Starmetal)

Nature

Rarity	Cards
Common	Window Box of Herbs; Fish Bowl; Specimen Jars; Spice Rack; Creature Toybox; Sunning Rock (patterns); Orchard (blueprint); Glimmerberry (seed); Nature Lure (common)
Uncommon	Mossbed; Glowcap Grow-Lamp; Fairy Ring; Beehive Box (patterns); Apiary (blueprint); Nature Lure (uncommon); Grimoire (Nature tier 2); Rain Dance (boon: all plots ×2 for 10 bells)
Rare	Grove Hall (L7); Menagerie (L4); Grimoire (Nature tier 3); Wild Abundance (boon: +80 Provisions, +40 Greens, +20 Herbs)

Rarity	Cards
Epic	Grimoire (Nature tier 4); Call of the Wild (boon: a Rare creature of a random school visits — the only non-Rift rare source)
Legendary	Heartwood Bough (boon: +5 Heartwood; one Heartwood Sapling seed)

Light

Rarity	Cards
Common	Radiant Lamp; Tea Service; Music Box of Calm; Halo Candles; Bird Bath (patterns); Light Lure (common); Warm Welcome (boon: all students Composure +20)
Uncommon	Sunlit Sanctum (L4); Sunwheat (seed); Light Lure (uncommon); Grimoire (Light tier 2); Day of Rest (boon: all injuries +50 healing points)
Rare	Dawn Hall (L7); Moonwell (blueprint); Healing Spring Basin (pattern); Grimoire (Light tier 3)
Epic	Grimoire (Light tier 4); Festival of Lanterns (boon: Academy morale +10 for 50 bells)
Legendary	Sunstone Reliquary (boon: +5 Sunstone, +30 Radiance)

Shadow

Rarity	Cards
Common	Wind Chimes of Hush; Moon-Shroud Curtains; Dartboard of Harmless Hexes; Ward Lantern; Nightglass Pane; Hexed Scarecrow (patterns); Shadow Lure (common); Quiet Night (boon: no misbehavior for 20 bells)
Uncommon	Detention Hall (L4); Standing Stones (blueprint); Silence Bell; Shadow Ledger; Rune Marker (patterns); Shadecap Mushrooms (seed); Shadow Lure (uncommon); Grimoire (Shadow tier 2); Veil Thread (boon: Veil cost −10% for 7 Anki days)
Rare	Hall of Whispers (L7); Grimoire (Shadow tier 3); Mending the Veil (boon: Veil strength to 100); Warding Primer (permanent Veil cost −3%, once per Academy)
Epic	Grimoire (Shadow tier 4); Long Night’s Watch (boon: hazards −50% for 100 bells)
Legendary	Pearl of the Deep Dark (boon: +5 Void Pearl, +30 Umbra)

Alchemy

Rarity	Cards
Common	Enchanted Broom; Warm Hearthstone; Treat Dispenser (patterns); Glassworks (blueprint, L3); Alchemy Lure (common); Recipe Page (Basic); Bulk Reagents (boon: +25 Reagents)
Uncommon	Workshop (L4); Fume Hood; Windmill Charm (patterns); Emberpepper (seed); Alchemy Lure (uncommon); Recipe Page (Advanced); Grimoire (Alchemy tier 2)

Rarity	Cards
Rare	Crucible Hall (L7); Bottomless Stewpot; Alembic Array (patterns); Grimoire (Alchemy tier 3); Recipe Page (Rare)
Epic	Grimoire (Alchemy tier 4); Master Builder’s Week (boon: all construction slots ×2 progress for 100 reviews)
Legendary	Salt of the Earth’s Heart (boon: +5 Ember Salt, +2 Starmetal)

27. The Rift and Expeditions

> **Source.** The destinations, skill requirements, safe and best odds, return times, main rewards, odds modifiers and “when it goes wrong” table in Sections 27.3, 27.5 and 27.6 are copied from approved Concept Draft 8 and are binding (BR-27, BR-31 to BR-33). Mana costs, supplies, loot quantities, eggs, Fortune and the other details are this GDD’s additions and are starting values to tune. The code reads every number from `content/rift.json`.

27.1 Role and player goal

The Rift is the late-game adventure: a shimmering tear on the grounds that leads to five strange places, each harder and richer than the last. Only adventurers go, in teams of one to three. It is the main source of creature eggs, golem cores, rare materials, rare draft cards, legendary furniture and artifacts, and the place where an Academy’s masters get to be heroes.

What makes the Rift Ankicaster’s own is how a team gets stronger. **A party’s skill is the average interval of its adventurers’ career cards.** An adventurer becomes able to brave the Ashen Wastes because the user has really remembered those 20 cards for a month. The player’s goal is to read the chance meter, match the party to the destination, and decide how much risk to take for how much reward. Nobody ever dies or is lost.

27.2 Opening the Rift (BR-27)

When the Academy has **3 masters** (3 characters holding at least one master title, retired alumni excluded), a scripted event plays (“The Air Over the Orchard Tears Open”) and the **Rift Gate** blueprint is granted. When the Gate is built, the **Adventurer** career becomes available to masters (any school, BR-11) and the **Training Yard** plan unlocks. If the master count later falls below 3 (for example through retirements), the Rift stays open.

All five destinations can be chosen as soon as the Gate stands. Nothing locks them: the skill requirement and the odds are the gate.

27.3 Destinations (Draft 8, binding)

Skill needed = the average interval of the party’s adventurer career cards. **Safe odds** = the chance of returning unscathed for a team of three at exactly that skill. **Best odds** = the cap, so something can always go wrong.

Destination	Tier	Skill needed	Safe odds	Best odds	Reviews to return	Main rewards
Mossy Hollow	Easy	0 days	85%	95%	50	Herbs, timber, common eggs
Whispering Fen	Mild	10 days	75%	90%	100	Rare herbs, owls and sprites

Destination	Tier	Skill needed	Safe odds	Best odds	Reviews to return	Main rewards
Crystal Caverns	Medium	20 days	65%	85%	150	Crystals, stone, golem cores
Ashen Wastes	Hard	30 days	55%	80%	250	Fire creatures, rare materials
The Shattered Spire	Deadly	45 days	45%	70%	400	Rare cards, artifacts, legendary eggs

Each destination has exactly one tier. The fixed column values above are binding; the flavour below is this GDD's.

Destination	Look and feel	Signature finds (details in 27.8)
Mossy Hollow	A green, dripping hollow under giant roots, fireflies at noon	Timber, Herbs, Greens; eggs of every Common creature; common patterns
Whispering Fen	Mist-covered reed beds where the water talks back	Herbs in quantity; rare herb seeds (Lumen Lily, Heartwood Sapling); Moonsilk; owl and sprite eggs (Mistfeather Owl, Fen Sprite, Star Heron, Coral Kelpie)
Crystal Caverns	Echoing galleries of humming crystal and sleeping stone giants	Crystal, Stone, **Golem Cores**, Starmetal, Starfruit seed; Rare eggs (Phase Ferret, Night Lynx, Glimmerhorn Stag, Dawnhart Gryphon)
Ashen Wastes	Grey dunes, cooling lava rivers and a sky the colour of embers	Fire creature eggs (Cinder Newt, Alembic Salamander; rarely Ember Phoenix); Epic eggs (Verdant Drake, Aether Serpent, Sunlit Unicorn, Umbral Wyrmling); Ember Salt, Phoenix Ash, Sunstone, Void Pearl, Starmetal; minor artifacts; legendary furniture patterns
The Shattered Spire	Floating fragments of a broken tower, staircases into nothing	Rare, Epic and Legendary draft cards; all artifacts; legendary eggs (Elder Treant Sprout, Chronoflare Owl, Ember Phoenix, Moonless Kitsune, Philosopher's Chimera); legendary furniture

27.4 Party skill, costs and logistics

Party skill. Each adventurer holds an Adventurer career card set: the 20 youngest free cards when the career was taken (BR-18, Section 11.2). An adventurer's **Rift skill** is the mean honest interval (Section 9.2) of those 20 cards, in days, *not* capped at 30. The **party skill** is the mean over every adventurer career card of every member (with 20 cards each, this is the average of the members' Rift skills). Skill is read at the day tick and when the planner opens, and it falls when those cards lapse (BR-15). Empty career-card slots count as interval 0.

Destination	Launch cost (mana)	Supplies per member (Provisions)	Loot rolls (full rewards)	Supply slots
Mossy Hollow	150	5	2	1
Whispering Fen	300	10	3	1
Crystal Caverns	500	15	4	2
Ashen Wastes	800	20	5	2
The Shattered Spire	1,200	30	6	3

- The mana cost rises with each tier, as Draft 8 requires. These five numbers are starting values to tune.
- **Supply slots** hold potions and scrolls taken along: the Rift Elixir (the only item that raises odds), Rift Map, Courage Draught and Mending Draught. The Supply Chest adds one slot.
- **One expedition at a time** (two with a level-3 Rift Gate). An expedition cannot be cancelled after launch. Characters on an expedition keep their beds, and their careers pause.
- **Who can go:** adventurers who are not injured. Bruised adventurers can go, and Bruised has no effect on odds.

27.5 The chance meter and odds (Draft 8, binding)

The planner shows a large semicircular **chance meter**: a needle and number for the chance of a **safe return**. It updates live as the player changes destination, team, supplies and companion. Every modifier is listed beneath it, for example “Party skill 27 days (needs 20): +5%”.

Calculation:

```
base       = safe_odds[destination]                      # 85 / 75 / 65 / 55 / 45
diff       = party_skill_days - skill_needed[destination]
skill_mod  = -10 * floor(-diff / 5)  if diff < 0         # each full 5 days below: -10
             +5  * floor(diff / 5)   if diff >= 0        # each full 5 days above: +5
team_mod   = 0 (3 adventurers) | -10 (2) | -25 (alone)
elixir     = +10 if a Rift Elixir is taken
veil_mod   = -5 if the home Veil is Thin or Torn (strength < 50)
chance     = base + skill_mod + team_mod + elixir + veil_mod
chance     = min(chance, best_odds[destination])         # 95 / 90 / 85 / 80 / 70
chance     = max(chance, 5)                              # floor (GDD decision)
```

Modifier (Draft 8)	Effect on safe odds
Each 5 days of party skill **below** what is needed	−10%
Each 5 days of party skill **above** what is needed	+5%, up to the destination’s best odds
Team of two	−10%
Going alone	−25%
Rift Elixir	+10%, up to the best odds
Thin Veil at home	−5%

- **These are the only things that change the odds.** Everything else that helps on expeditions (companions, personalities, traits, the Training Yard, Gate levels, scrolls, artifacts, spellbooks, alumni) adds **Rift Fortune** (Section 27.9), which improves rewards, return time or recovery

but never the odds. This keeps the meter readable and makes the user's real memory the main lever.
- **Three is best** (BR-27): only a team of three gets the full safe odds.
- The "5 days" steps use whole steps. The planner shows how far the party is from the next step ("+1.8 days to the next +5%").
- "Thin Veil" means the Thin or Torn band (strength below 50; Section 18.3).

Example. A trio with party skill 27 days targets Crystal Caverns (needs 20): 65 + 5 (one full step above) = 70%. With a Rift Elixir it is 80%, below the 85% cap. The same trio at the Ashen Wastes (needs 30) is 3 days short, less than one full 5-day step, so it keeps the full 55%. At the Shattered Spire (needs 45, 18 days short, three full steps) it is 45 − 30 = 15%.

27.6 Resolution (Draft 8, binding)

The roll is made **at launch** with the seeded RNG (Section 38.11) and stored hidden. It is revealed when the expedition's review count completes, so reopening the game cannot change it.

1. Roll `r` from 1 to 100. If `r ≤ chance`: **Safe return.** Everyone comes home unscathed with **full rewards**.
2. Otherwise **something went wrong**. Roll again on the table for the destination's tier band:

When it goes wrong	Easy–Medium (Mossy Hollow, Whispering Fen, Crystal Caverns)	Hard–Deadly (Ashen Wastes, Shattered Spire)	Rewards
Minor injury to one adventurer	60% (d100 01–60)	40% (01–40)	Full rewards
Serious injury to one adventurer	25% (61–85)	30% (41–70)	Half rewards
Curse on one adventurer	10% (86–95)	20% (71–90)	Half rewards (GDD decision)
Retreat with nothing	5% (96–100)	10% (91–100)	None (supplies used; launch mana spent)

Nobody dies or is lost. The worst case is a long infirmary recovery. The injured or cursed adventurer is chosen at random from the team.

Outcome	Injury applied (Section 23)
Minor injury	Bruised (Easy–Medium) or Sprained (Hard–Deadly)
Serious injury	Sprained (Mossy Hollow) or Broken Bone (all other destinations)
Curse	Hexed (Easy–Medium) or Rift-Touched (Hard–Deadly)
Retreat	No injury. Team morale −10 for 10 bells ("We'll get it next time")

Half rewards means half the loot rolls, rounded up. An Easy expedition (2 rolls) still brings home 1.

27.7 Expected value at a glance

At the destinations' own safe odds (a trio at exactly the needed skill, no elixir), the chances of each outcome are:

Destination	Safe return	Minor injury	Serious injury	Curse	Retreat
Mossy Hollow (85%)	85%	9.0%	3.75%	1.5%	0.75%
Whispering Fen (75%)	75%	15.0%	6.25%	2.5%	1.25%
Crystal Caverns (65%)	65%	21.0%	8.75%	3.5%	1.75%
Ashen Wastes (55%)	55%	18.0%	13.5%	9.0%	4.5%
The Shattered Spire (45%)	45%	22.0%	16.5%	11.0%	5.5%

Even at the Spire, 67% of expeditions bring home full rewards (safe return or a minor injury), and 94.5% bring home at least half.

27.8 Rewards and loot

Each loot roll picks a row from the destination's pool by weight. Quantities are scaled by Fortune (Section 27.9).

Mossy Hollow

Weight	Reward
35%	Timber 30–50
30%	Herbs 15–25
15%	Greens 15–25 or Provisions 15–25
15%	**Common egg** (any Common species, Section 22)
5%	Common furniture pattern

Whispering Fen

Weight	Reward
30%	Herbs 30–50 ("rare herbs": always includes 1 Dreamthistle or Shadecap seed)
20%	**Owl or sprite egg:** Fen Sprite (Uncommon, 50%), Mistfeather Owl (Rare, 25%), Star Heron (Rare, 15%), Coral Kelpie (Rare, 10%)
15%	Rare herb seed: Lumen Lily or Heartwood Sapling
15%	Moonsilk 1–2
10%	Heartwood 1–2
10%	Uncommon furniture pattern or blueprint

Crystal Caverns

Weight	Reward
30%	Crystal 3–6
25%	Stone 40–80
15%	**Golem Core** ×1 (three awaken a Geode Golem, Section 27.10)
10%	Starmetal 1–2
10%	**Rare egg:** Phase Ferret, Night Lynx, Glimmerhorn Stag or Dawnhart Gryphon
5%	Starfruit seed
5%	Rare furniture pattern or blueprint

Ashen Wastes

Weight	Reward
35%	Rare materials: Ember Salt 2–4, Phoenix Ash 1, Sunstone 2–3, Void Pearl 2–3 or Starmetal 2 (equal weights)
20%	**Fire creature egg:** Cinder Newt (60%), Alembic Salamander (35%), Ember Phoenix (5%)
12%	**Epic egg:** Verdant Drake, Aether Serpent, Sunlit Unicorn or Umbral Wyrmling
15%	Reagents 30–50
10%	Legendary furniture pattern (the six Rift legendaries, Section 17)
8%	Minor artifact (artifacts whose source lists the Ashen Wastes, Section 25.5)

The Shattered Spire

Weight	Reward
30%	**Rare cards:** one Rare (70%), Epic (25%) or Legendary (5%) draft card from a random school deck, added to the hand
20%	**Artifact** (any not yet owned)
15%	**Legendary egg:** Elder Treant Sprout, Chronoflare Owl, Ember Phoenix, Moonless Kitsune or Philosopher's Chimera
15%	Legendary furniture pattern
20%	Rare materials ×2 of the Ashen Wastes amounts

Duplicate artifacts and uniques are re-rolled into rare materials. The Lucky trait (+1 roll) and the Phase Ferret (+1 roll) apply whenever the expedition brings full rewards.

27.9 Rift Fortune (rewards, never odds)

Fortune is a separate score for each expedition, shown under the chance meter. Each point gives **+10% to all reward quantities** (rounded down per item) on any outcome that brings rewards. The maximum is 5 points (+50%).

Source	Fortune
Companion creature: Common/Uncommon, Rare, Epic/Legendary	+1 / +2 / +3
Team drilled at the Training Yard (every member rested ≥ 50 reviews there since their last expedition)	+1 per yard level (max +3)
Rift Map scroll	+2
Expedition Map Table; Star Map of Many Doors (artifact); Field Guide to Elsewhere (spellbook)	+1 each
Each scholar in an Observatory (max 2); Star Heron or Orrery Moth tamed at home (max 1 from these)	+1 each
Rift Gate level 2 / 3	+1 / +2
Paragon in the team; Bold personality; Brave or Daredevil trait; Show-off in a team of 3; Social in a team of 3	+1 each
Personality affinity: Dreamy at the Whispering Fen, Bookish at the Crystal Caverns, Curious at Mossy Hollow or the Ashen Wastes, Mischievous at the Shattered Spire	+1 each
Hex Duel veterans (Section 20.3)	+1 each, once
Adventurer alumni bonus; Rift Beacon spell; “Whispers from the Rift” event	Section 31; +3; +1
Team average morale ≥ 80 / < 40	+1 / −1
Nervous trait; Show-off alone; Rift Tremor hazard active at launch	−1 each

Separately from Fortune: the **Dawnhart Gryphon** companion makes the expedition return 15% sooner, and the Rift Beacon spell makes the next one return 25% sooner. A **Courage Draught** halves the recovery requirement of any injury or curse from that expedition. **Steadfast** adventurers recover from Rift injuries 25% faster; **Daredevil** adventurers 25% slower. The **Ember Phoenix** as companion heals the worst Rift injury on return.

27.10 Eggs and golem cores

- **Eggs** go to Storage. An egg is placed in a free slot of a matching habitat (Section 22.2) and **hatches after** 20 / 40 / 60 / 80 / 100 honest reviews for Common / Uncommon / Rare / Epic / Legendary. It hatches **Wary**, and taming starts as for any arrival. An egg never spoils, and it waits in Storage indefinitely.
- **Golem Cores** are a rare material. Three Golem Cores at a Workshop awaken a **Geode Golem** (Section 22.7) after 60 reviews of builder or wizard time. Golem Cores are also used in the Rift Gate level-3 renovation (2 cores).

27.11 Companions

One tamed creature may go along on each expedition. It leaves its habitat (its home buff turns off while it is away), adds Fortune, and shares the team’s fate: it comes home tired (no buff for 10 bells) on any outcome other than a safe return, and it is never hurt. It eats nothing while away.

27.12 Expected progression

Adventurer career cards start young, so a new adventurer's Rift skill begins at a few days and rises as the user really learns those 20 cards. With default scheduling, a typical set averages about 10 days after 2–3 weeks, 20 days after 4–6 weeks, 30 days after 6–9 weeks and 45 days after 10–14 weeks. Lapses pull it down.

- **Right after the Rift opens:** Mossy Hollow at 85% with a trio is dependable. The Whispering Fen is a gamble (75 − 20 = 55% with a brand-new trio at skill 0–4).
- **About a month in:** the Fen is at or above its 75%, and the Crystal Caverns are within reach.
- **Two months in:** the Ashen Wastes at 55–65%. An elixir and a well-remembered set make it a real choice.
- **Three months and more:** the Shattered Spire. Even a perfect party is capped at 70% there, so its legendary eggs and artifacts always feel earned.

A solo adventurer always pays −25%, and a pair −10%, which is why three is best. A single, very experienced adventurer can still keep materials flowing from the Hollow: with 30 days of skill they reach 85 − 25 + 30 = 90%, while a trio with the same skill reaches the 95% cap.

27.13 `rift.json`

```
{
  "opens_at_masters": 3, "team_size": [1, 3], "min_chance": 5,
  "skill_source": "adventurer_career_cards_mean_honest_ivl_uncapped",
  "destinations": [
    {"id": "mossy_hollow",    "tier": "easy",   "skill_needed": 0,  "safe": 85, "best": 95,
        "reviews": 50,
     "cost": 150,  "supplies": 5,  "loot_rolls": 2, "supply_slots": 1, "band": "low"},
    {"id": "whispering_fen",  "tier": "mild",   "skill_needed": 10, "safe": 75, "best": 90,
        "reviews": 100,
     "cost": 300,  "supplies": 10, "loot_rolls": 3, "supply_slots": 1, "band": "low"},
    {"id": "crystal_caverns", "tier": "medium", "skill_needed": 20, "safe": 65, "best": 85,
        "reviews": 150,
     "cost": 500,  "supplies": 15, "loot_rolls": 4, "supply_slots": 2, "band": "low"},
    {"id": "ashen_wastes",    "tier": "hard",   "skill_needed": 30, "safe": 55, "best": 80,
        "reviews": 250,
     "cost": 800,  "supplies": 20, "loot_rolls": 5, "supply_slots": 2, "band": "high"},
    {"id": "shattered_spire", "tier": "deadly", "skill_needed": 45, "safe": 45, "best": 70,
        "reviews": 400,
     "cost": 1200, "supplies": 30, "loot_rolls": 6, "supply_slots": 3, "band": "high"}
  ],
  "modifiers": {"per_5_days_below": -10, "per_5_days_above": 5, "team_of_two": -10, "alone": -25,
                "rift_elixir": 10, "thin_veil": -5, "thin_veil_below_strength": 50},
  "wrong_table": {
    "low":  [{"id": "minor_injury", "pct": 60, "rewards": 1.0}, {"id": "serious_injury", "pct": 25,
        "rewards": 0.5},
             {"id": "curse", "pct": 10, "rewards": 0.5}, {"id": "retreat", "pct": 5, "rewards": 0}],
    "high": [{"id": "minor_injury", "pct": 40, "rewards": 1.0}, {"id": "serious_injury", "pct": 30,
        "rewards": 0.5},
             {"id": "curse", "pct": 20, "rewards": 0.5}, {"id": "retreat", "pct": 10, "rewards": 0}]
  },
  "fortune": {"per_point_reward_pct": 10, "max_points": 5},
  "egg_hatch_reviews": {"common": 20, "uncommon": 40, "rare": 60, "epic": 80, "legendary": 100}
}
```

28. Random School Events

28.1 Role and rules

Events are short story cards with two or three choices that make the Academy feel inhabited, in the spirit of Spellcaster University's event board. They never punish and never require a fast response.

- **Daily roll:** at each day tick after an **active** day, 60% chance of one random event (Observatory foresight shows it a day early). No random events accumulate while the user is away (BR-04): after a break, at most one event is waiting.
- **Event board:** events wait on the left of the Academy screen for up to 3 days; if ignored they resolve with their **default** option, which is always harmless.
- **Eligibility:** each event has conditions (rooms, schools, Rift open, alumni existing). Weights are listed; an event cannot repeat within 10 days.
- **Scripted events** (always fire once, at the trigger): Founding Day (first launch), First Student, First Creature, First Master, The Rift Opens, First Thin Veil (explains the Veil kindly), Welcome Back (after 7+ days away), First Retirement.

28.2 Event catalog

ID	Event	Conditions	Weight	Options (default first)
`misaddressed_owl`	A Misaddressed Owl	—	10	Send it on its way: +5 Renown · Peek at the letter: +10 Insight · Offer it a perch: a Hush Raven visits if a Gloom/Sky habitat is free
`peddler`	The Curio Cart	—	8	Browse later (nothing) · Buy one of three furniture patterns at its listed cost +50% (bypasses the draft)
`doorstep_basket`	A Basket on the Doorstep	Any habitat	6	Find it a home: a Common creature of a random school arrives as Wary · Pass it to the village: +5 Renown
`rainy_week`	A Rainy Week	Grounds plots	6	Wait it out: plots −20%, Greenhouse +10% for 30 bells · Cast *Clear Skies* (if known)

ID	Event	Conditions	Weight	Options (default first)
`guild_inspector`	The Hedge-Mage Guild Inspector	8+ rooms	5	Show them around: if every room is Plain or better, +15 Renown; otherwise friendly advice naming your two lowest-quality rooms
`birthday`	A Student’s Birthday	1+ student	8	Sing: Composure +5 for the student · Bake a cake (20 Provisions): their dorm Leisure +20 and morale +5
`lost_cat`	The Lost Cat	Crystal habitat	4	Put up posters: 50% a Glyph Cat visits · Leave milk out (5 Provisions): 80%
`lantern_festival`	Lantern Festival	6+ characters	5	Watch from the windows: Composure +5 · Host it (100 mana): Academy morale +8 for 30 bells
`tutors_request`	A Master Offers to Tutor	A resident master	5	Gladly: their tutor bonus is doubled for 50 bells · Not now
`mysterious_seeds`	Mysterious Seeds	Greenhouse	4	Plant them: unlock a random crop seed you lack · Dry them: +30 Herbs
`old_map`	An Old Map in the Library	Library	4	File it: +20 Insight · Study it (Rift open): a Rift Map scroll
`ghostly_choir`	The Ghostly Choir	Veil below 80 or Shadow L3	4	Listen: Composure +10 · Bottle the echo (Shadow L5): +15 Umbra
`glass_merchant`	The Glass Merchant	—	5	Trade 40 Stone for 20 Glass · Trade 20 Timber for 10 Cloth · No thanks
`frost_fairies`	Frost Fairies	Greenhouse	3	Let them be: Cold Snap hazard · Offer 10 Nectar: they leave 5 Crystal
`collapsing_shelf`	The Collapsing Shelf	Library	3	Fix it later: Library −10% until fixed from its panel (50 Timber) · Fix now: 50 Timber

ID	Event	Conditions	Weight	Options (default first)
visiting_scholar	A Visiting Scholar	Wizard total level 10+	4	Host for tea: +5 Renown · Pay for a lecture (150 mana): reveal one recipe
founders_day	Founder’s Day	Anniversary of install (scripted yearly)	—	Celebrate: +25 Renown and a free Foundation draw
homesick	A Homesick Student	A Homesick student	6	A friend helps: 30% they gain Kind Heart · Send a letter home (5 Cloth): morale +10
strange_egg	A Strange Egg	Ember Pit or Menagerie	3	Keep it warm: hatches after 100 reviews into a random Uncommon (75%) or Rare (25%) creature · Return it to the woods: +5 Renown
crop_blight	A Touch of Blight	Greenhouse	3	Let it pass: Greenhouse −30% for 30 bells · Treat it (10 Herbs + 5 Reagents): no effect
alumni_letter	A Letter from an Alumnus	1+ alumni	4	Read it: a gift related to their career (Section 13.3)
rival_challenge	A Friendly Rival Academy	15+ rooms	3	Decline politely · Accept: choose a school; if your best room of that school is Fine or better, +20 Renown and a Rare draft card of that school, else +5 Renown for good sportsmanship
midsummer	Midsummer Night	Nature L3	3	Dance: plots ×1.5 for 20 bells and a Fairy-Ring visitor roll
veil_bell	The Bell of the Veil Rings	Veil below 70	5	Thank it (nothing) · Reinforce now: pay 1× upkeep for +20 strength
wandering_bard	A Wandering Bard	Common Room	4	Decline · Hire (80 mana): Leisure +20 Academy-wide
overdue_notice	The Library Overdue Notice	Library	3	Pay the fine (20 Insight) · Ignore it: Library −10% for 20 bells

ID	Event	Conditions	Weight	Options (default first)
`hidden_door`	A Door That Wasn’t There	Castle 80%+ full	2	Explore: +30 Timber and +30 Stone · Brick it up: a free Support Pillar
`bake_off`	The Great Bake-Off	Kitchen	3	Spectate: Leisure +5 · Enter (30 Provisions + 10 Nectar): morale +10, +5 Renown
`shooting_star`	A Shooting Star	Observatory	3	Catch it: +2 Crystal · Make a wish: a random Boon card from any unlocked deck
`rift_whispers`	Whispers from the Rift	Rift open	4	Listen: next expedition +1 Rift Fortune · Seal the gate for the night: Rift Gate locked 10 reviews, Veil +10

29. Wizard Spells and Abilities

29.1 Rules

- Spells are learned by **Study a Grimoire** sessions (Section 7.3) and cast with mana from the Wizard screen or from context menus (e.g., right-click a hazard → *Dispel*).
- Grimoire tiers per school: **T1** (school L1, in the Library shop or starting kit for the first school), **T2** (L3), **T3** (L5), **T4** (L8), **Capstone** (L10). T2–T4 grimoires also appear as draft cards; once owned they can be studied any time.
- Study cost and time by tier: T1 150 mana / 40 reviews; T2 300 / 80; T3 500 / 120; T4 800 / 180; Capstone 1,200 / 250.
- Cooldowns are measured in honest reviews unless they say “per day”. Cooldowns do not advance while away (no reviews) and nothing is lost.
- Spells never affect Anki, card links, skill or mastery, and never produce mana.

29.2 Spell list

School	Tier	Spell	Cast cost	Cooldown	Effect
Arcane	T1	Glow	40	20	Ends Spill of Night or Echo Loop on a floor
Arcane	T2	Dispel	120	60	Ends any one hazard (not a breach)
Arcane	T3	Foresight	200	1 per day	Shows the next 3 days’ events; skip one of them
Arcane	T4	Rift Beacon	300	200	Next expedition: +3 Rift Fortune and returns 25% sooner (never changes the odds)
Arcane	Cap	Starlit Conjunction	400	1 per 3 days	All essence yields ×1.5 for 50 bells
Nature	T1	Beckon	30	10	A Loose creature returns home
Nature	T2	Rain Dance	120	60	All plots ×1.5 for 10 bells
Nature	T3	Clear Skies	150	40	Ends Cold Snap, Hex Fog and Rainy Week effects
Nature	T4	Wild Kinship	250	100	+50 taming points, split among all Wary creatures
Nature	Cap	Verdant Surge	400	1 per 3 days	Food and Herbs ×2 for 30 bells

School	Tier	Spell	Cast cost	Cooldown	Effect
Light	T1	Mend	40	15	+30 healing points to one injured character
Light	T2	Soothing Light	100	40	Composure +20 for everyone on a floor
Light	T3	Mend Object	80	20	Repairs one Damaged furniture item
Light	T4	Sunburst	300	150	+60 healing points to every injured character
Light	Cap	Dawn Blessing	400	1 per 3 days	Academy morale +15 for 50 bells; lifts every Rift curse (Hexed and Rift-Touched characters heal at once)
Shadow	T1	Hush	30	10	Ends a misbehavior chain in a room; Composure +5 there
Shadow	T2	Ward Pulse	150	80	No hazards for 10 bells
Shadow	T3	Banish	200	60	Resolves a breach instantly
Shadow	T4	Veil Stitch	0.8 × today’s upkeep	1 per day	Veil strength +20
Shadow	Cap	Moonless Night	400	1 per 3 days	Hazards and misbehavior −60% for 50 bells (outside the 75% cap)
Alchemy	T1	Tidy Wind	40	20	Every room on a floor to 100 cleanliness
Alchemy	T2	Douse	60	20	Ends a fire or explosion; that room is fireproof for 20 bells
Alchemy	T3	Quickening	250	60	+25 progress points to one construction site
Alchemy	T4	Transmute	150	40	Convert 60 of one common material into 50 of another

School	Tier	Spell	Cast cost	Cooldown	Effect
Alchemy	Cap	Philosopher’s Hour	400	1 per 3 days	The next 3 brews complete instantly with ×2 yield

29.3 Passive wizard abilities

Ability	Unlock	Effect
Headmaster’s Presence	Start	The wizard’s sprite visits a random room each bell; that room gets +5% efficiency that bell (purely cosmetic-feeling, small real effect)
Stand-in Teacher	Any school at L5	Section 7.7
Practised Builder	Alchemy L5	The wizard’s construction slot advances 1.1 per review
Kindly Tamer	Nature L3	Wizard can tame Commons at 50% speed
Steady Gaze	Shadow L5	Hazards in the Wizard’s Study’s column −25%
Warm Welcome	Light L5	New students start with +10 morale for 7 days
Scholar’s Eye	Arcane L5	Draft rerolls cost 10 Insight instead of 15

30. Game Modes: Timed Term and Endless School; Reset Rules

30.1 Overview (BR-29)

	Endless School (default)	Timed Term
Fantasy	Build one Academy forever	Run a term with goals, graduate a class, start again stronger
End	Never	When the exam timer ends (if on) or when the player calls Commencement
Goals	Milestones and achievements	A three-star **term charter**
Reset	Never	At term end (Section 30.4)
Meta rewards	Spellbooks at Renown milestones	Spellbook choice by stars earned; alumni from the term's masters

The mode is chosen at creation and can be changed from Settings: Endless → Timed starts a new term *after* the player confirms the reset rules; Timed → Endless converts the current term into an Endless School with no reset.

30.2 Timed Term setup

Setting	Options	Default
Term length (minimum before Commencement)	60 / 90 / 120 / 180 / 365 real days	120
Exam timer	Off / On	**Off**
Charter	Pick 3 goals from 6 offered	—
Randomize recipes	Off / On	Off

Exam timer off (default): the term can end at any time after the minimum length; the player presses **Commencement** in the Headmaster panel. There is no deadline pressure at all.

Exam timer on: the term ends automatically at the end of the chosen length (real Anki days). The countdown is shown on the Academy screen. Because the timer is real-calendar, a break uses term time; this is the only place where time away matters, and the player opts into it. The timer can be switched off at any time (it cannot be switched back on in the same term).

30.3 Term charter goals

At the start of a term, six goals are offered (drawn from this pool, scaled to term length); the player picks three. Each completed goal is one star.

Goal	Scaling (120-day term)
Graduate N masters	4
Have masters in N different schools	3

Goal	Scaling (120-day term)
Reach N Renown	150
Tame N creature species	8
Have N rooms at Fine quality or better	6
Bring N expeditions home safely	3 (offered only if the Rift opened in a previous term or the term is ≥ 120 days)
Discover N potion recipes	8
Build an Advanced Hall	1
Keep the Veil Strong on N% of active days	90%
Have N students enrolled at once	12

30.4 Reset rules (BR-30)

At the end of a Timed Term (Draft 8, BR-30 and BR-34):

Carries over	Resets
Wizard levels in all schools (with spells known and the Study's level, which belong to the wizard)	**Buildings** (rooms, furniture, grounds), back to the starting layout
Alumni bonuses (from retirements during the term **and** from every master at Commencement, who graduates as an alumnus)	**Materials** (and essences, potions, eggs, inventory), back to the starting stock (Section 25.1)
Spellbooks	**Unspent mana**: the bank goes to 0. Mana is earned again only from honest reviews (BR-01)
Achievements, the Alumni Gallery and the term record (meta records, not game state)	**Students and faculty** (all characters). Their card links are released, so **their Anki cards become free again** for the next term's students
	Creatures
	Known Plans and patterns (unless a spellbook says otherwise), recipes (when randomized), Renown, the Veil (back to Strong), the draft hand and price history, research ranks

Anki is never touched. A reset only deletes rows in Ankicaster's own save (`card_links`, `characters`, `rooms` and so on). No Anki card, note, deck or scheduling value changes. The term record keeps a read-only summary of every reset character.

Each new term starts from the starting layout and starting stock with a 0 mana bank, but with the wizard already strong, so the early game moves faster and the interesting choices arrive sooner.

30.5 Endless School

Endless School never resets. Long-term goals come from milestones (Section 32), spellbook milestones (Section 31.1) and the bestiary/achievement collections. Breaks are handled exactly as in Sections 14.7 and 18.5.

31. Spellbooks and Alumni Bonuses (Meta Progression)

31.1 Earning spellbooks

Mode	When	Choice
Timed Term	At Commencement, if at least one star was earned	Pick 1 of 3 random spellbooks from the tier equal to the stars (1–3)
Endless School	At Renown 100 (T1), 250 (T1), 400 (T2), 600 (T2), 850 (T3), 1,200 (T3), then every +400 Renown (T3)	Pick 1 of 3 random spellbooks of that tier not yet owned

Spellbooks are permanent, carry over between terms (BR-30) and sit on a shelf in the Wizard's Study (each one visibly added). Owned books are never offered again; when a tier is exhausted, the next higher tier is offered, and when all are owned, a cosmetic "Gilded Edition" is offered.

31.2 Spellbook catalog

Tier	Spellbook	Effect
1	Almanac of Kind Weather	Plot outputs +10%; Rainy Week and Cold Snap effects halved
1	Primer of Patience	Draft price growth ×1.12 per recent draw instead of ×1.15
1	Little Book of Wards	Hazards −10%
1	Housekeeper's Hymnal	All dirt −25%
1	The Tamer's Pocketbook	Taming +15%
1	Cookery for Fifty	Meals use 10% fewer Provisions
1	Notes on Furnishing	Furniture materials −10%
2	Treatise on Thin Places	Veil cost −10% (inside the 50% cap)
2	Codex of Careful Hands	Lab explosions −50%; potion yield +10%
2	The Open Door	Applicant board +1 applicant per school; applicant quality one Renown band higher
2	Field Guide to Elsewhere	+1 Rift Fortune and +1 supply slot
2	Architect's Folio	Known Plans and furniture patterns persist across Timed Terms; build time −5%
2	Healer's Compendium	Healing +20%
2	The Quiet Corridor	Misbehavior −15%
3	The Eternal Index	Recent draws count half toward draft price growth
3	Atlas of the Unmapped	+1 loot roll on every expedition that brings full rewards

Tier	Spellbook	Effect
3	Book of Bright Mornings	Mana +3% (inside the +25% cap)
3	Covenant of the Veil	Veil cost −10% (inside the cap); breaches only at Torn
3	Menagerie Royale	Creature buffs +15%; up to 3 of a species give buffs
3	The Founder’s Testament	Timed Terms start with ×3 starting materials and one Known Plan of choice
3	Grimoire of Swift Hands	Spell cooldowns −30%

31.3 Alumni bonuses

Each alumnus grants one bonus based on their final career, plus a small bonus per extra mastered school. Bonuses are scaled by `clamp(retirement_age / 90, 0.67, 2.0)` and capped per category.

Final career	Bonus per alumnus	Category cap
Teacher	Classroom essence yields +1%	+15%
Scholar	Mana +0.5%	+5% (and inside the global +25%)
Gardener	Food and Herbs +2%	+20%
Tamer	Taming +2% and creature buffs +1%	+20% / +10%
Healer	Healing +2%	+20%
Warder	Veil cost −1%	−10% (inside the 50% cap)
Alchemist	Potion yield +2%	+20%
Builder	Build time −1%	−10% (inside the build-time cap)
Adventurer	Rift Fortune +0.2 (summed, rounded down)	+1
None (resident master)	Essence yield of their school +1%	+10% per school
Each extra school mastered	+5 Renown at the start of every term (Timed) or once (Endless)	—

The Alumni Gallery lists every alumnus with portrait, titles, careers, retirement date and the bonus they gave.

32. Achievements and Milestones

32.1 Rules

- Achievements celebrate **play choices and the honest passage of learning**, never review totals, speed, streaks, accuracy or new cards (BR-03).
- Each achievement grants Renown (listed) and sometimes a cosmetic (robe colour, crest frame, title).
- Achievements persist across Timed Terms.

32.2 Achievement list

ID	Achievement	Condition	Reward
`first_light`	First Light	Complete your first training session	+5 Renown; candle crest frame
`five_orbs`	The Five Orbs	Level 1 in every school	+10
`archmagus`	Archmagus	Level 10 in any school	+15; school aura
`grand_magus`	Grand Magus	Level 10 in all five schools	+40; golden robe
`doors_open`	Doors Open	Enroll your first student	+5
`full_house`	Full House	Every bed filled with at least 8 students	+10
`master_class`	Master Class	Your first master	+15
`triple_crown`	Triple Crown	Three masters (the Rift opens)	+20
`paragon`	Paragon	One character masters all five schools	+40; “Paragon” banner
`long_game`	The Long Game	A character with 2+ career sets retires	+20
`first_farewell`	First Farewell	Your first retirement	+10
`hall_of_fame`	Hall of Fame	10 alumni	+25
`splendid`	Splendid!	Any room reaches Splendid quality	+10
`every_room_a_joy`	Every Room a Joy	10 rooms at Fine or better	+20
`wing_commander`	Wing Commander	Three adjacent classrooms of the same school	+10
`skyline`	Sky’s the Limit	Build on the top floor of a fully expanded castle	+15
`green_fingers`	Green Fingers	A Greenhouse with every bed planted and a gardener	+5
`feast_for_all`	A Feast for All	Serve a meal to 20+ characters with nobody hungry	+10
`calm_waters`	Calm Waters	An active day with 10+ students and no misbehavior	+10

ID	Achievement	Condition	Reward
`veilkeeper`	Veilkeeper	Veil Strong for 30 consecutive days	+15
`welcome_back`	Welcome Back	Return after 7+ days away (the Veil held!)	+5; “Seasoned Traveller” title
`first_friend`	First Friend	Tame your first creature	+5
`menagerie_keeper`	Menagerie Keeper	10 creatures tamed at once	+15
`bestiary_25`	Naturalist	25 different species tamed (lifetime)	+25
`legend_keeper`	Keeper of Legends	Tame a Legendary creature	+30
`bond_three`	Best of Friends	A creature reaches Bond 3	+10
`first_brew`	First Brew	Brew your first potion	+5
`potion_master`	Potion Master	Discover all Basic recipes	+10
`grand_alchemist`	Grand Alchemist	Discover all 18 recipes	+30
`founders_mustache`	The Founder’s Mustache	Suffer the portrait prank	+5 (and the mustache cosmetic)
`through_the_rift`	Through the Rift	Your first expedition returns	+10
`hero_hollow` / `hero_fen` / `hero_caverns` / `hero_wastes` / `hero_spire`	Hero of [Destination]	A safe return from that destination	+10 each
`against_all_odds`	Against All Odds	A safe return from the Shattered Spire launched at 30% odds or less	+30
`curator`	Curator	10 artifacts displayed at once	+20
`term_scholar`	Cum Laude / Magna / Summa	Finish a Timed Term with 1 / 2 / 3 stars	+10 / +20 / +30

32.3 Renown

Renown is the Academy’s standing. It never affects mana.

Source	Renown
Each new master title	+10
Each retirement	+5
Each displayed artifact	+2 (while displayed)
Academy Banners	+1 each (max 5)
Academy Charm	+1 per 50 Charm (Entrance Hall Charm counts double)
Achievements, events	As listed

Renown	Academy title	Unlocks
0	Hedge School	—
100	Respected Academy	Applicant band 2; first Endless spellbook
300	Renowned College	Applicant band 3; crest frame
600	Grand University	Gilded gate art

Renown	Academy title	Unlocks
1,000	Legendary Athenaeum	Floating-lantern sky over the Academy

32.4 Milestone track (shown in the Headmaster panel)

A list of the next three meaningful milestones with progress bars: e.g., “Nature 5 — teaching unlocks (61/125 reviews)”, “First master — Tamsin: 41/50 cards over 30 days”, “The Rift — 1/3 masters”. Milestones are informational and never phrased as review targets (“Tamsin’s cards are maturing” rather than “do more reviews”).

33. Economy Balance Tables

> **All numbers in this document are starting values to tune.** They were chosen to be internally consistent for a reference user doing 100–200 honest reviews per day and adding roughly 10–25 new cards per day. Every number lives in `content/*.json` or `config.json`, never in code, so a balancing pass needs no code changes. The simulation harness (Section 40.4) should be used to retune them.

33.1 Income

Honest reviews per day	Mana per day (no Arcane)	With Arcane L10 (+10%)	Bells per day	Mealtimes per day
30	300	330	3	0–1
50	500	550	5	1
100	1,000	1,100	10	2
150	1,500	1,650	15	3
200	2,000	2,200	20	4
300	3,000	3,300	30	6

Because production, needs and meals are all per bell, the *ratio* of production to consumption is the same at every study volume; heavier studying simply moves the whole Academy faster. The only thing that differs by volume is how fast fixed daily costs (the Veil) are covered.

33.2 Spending reference

Category	Item	Mana	Other
Training	Level n	80 × n	25 × n reviews
Spells	Study T1/T2/T3/T4/Cap	150 / 300 / 500 / 800 / 1,200	40 / 80 / 120 / 180 / 250 reviews
Rooms	Corridor … Menagerie	25 … 1,400	See Section 16
Furniture	Fern … Legendary	30 … 800	See Section 17
Draft	Foundation / school deck base	120 / 150	×1.15 per draw in the last 7 days, max ×2.5
Enrollment	Per student	60	50 free young cards, bed, seat
Applicant reroll	Per school	100	or 15 Insight
Rift	Mossy Hollow … Shattered Spire launch	150 / 300 / 500 / 800 / 1,200	Provisions 5–30 per member
Veil	Daily	`(40 + 3 × size) × (1 − reductions)`	—
Reinforce the Veil	Per use	1 × today’s upkeep	+15 strength
Expansions	Grounds I / II	400 / 1,200	40 T / 80 T + 40 S

Category	Item	Mana	Other
Expansions	Castle I / II / III	1,500 / 3,000 / 5,000	100 S + 40 T / 200 S + 80 T + 10 G / 300 S + 120 T + 20 G + 10 Cr

33.3 Where mana goes, by phase (target shares at 150 reviews/day)

Phase	Days	Training	Construction & furniture	Veil	Draft	Rift, spells, misc.
Founding	1–7	45%	40%	7%	8%	0%
Growing	8–30	35%	35%	15%	12%	3%
Masters	31–90	15%	30%	25%	15%	15%
Mature	90+	5% (grimoires)	25%	35%	15%	20%

33.4 Production reference (per bell, before efficiency and buffs)

Source	Output
Student in a staffed classroom	0.20 essence of their school
Vegetable Patch	1.0 Provisions + 0.2 Greens
Greenhouse bed (Hearthroot)	1.0 Provisions
Herb Garden	0.6 Herbs
Woodlot / Quarry	2 Timber / 1 Stone
Pond / Orchard / Apiary	0.8 Fish / 0.8 Provisions + 0.3 Nectar / 0.5 Nectar
Glassworks / Weaving Room	2 Stone → 1 Glass / 0.5 Cloth
Library scholar	+1 Insight
Consumption per character	0.2 Provisions (1 per mealtime)

33.5 Typical Academy at three points (150 reviews/day, illustrative)

	Day 7	Day 30	Day 90
Wizard total level	9	27	50
Rooms / plots	9 / 4	22 / 8	40 / 14
Students / faculty	3 / 0	9 / 0	10 / 6
Masters	0	0	4
Creatures	1	5	12
Daily Veil	~140	~300	~420 (after reductions)
Bank (if the player spends steadily)	~500	~1,500	~4,000

33.6 Global caps

Category	Cap	Notes
Mana multiplier (Arcane + artifacts + spellbooks + alumni)	**+25%**	Only multiplies honest-review mana

Category	Cap	Notes
Veil cost reduction	**50%**	
Hazard reduction	**75%**	Moonless Night is outside the cap (temporary)
Misbehavior reduction	**75%**	Moonless Night is outside the cap
Build-time reduction	**40%**	Boons (Architect's Insight, Master Builder's Week) are outside the cap
Material-cost reduction	**40%**	Minimum 1 of each listed material
Rift safe odds	Each destination's **best odds** (95 / 90 / 85 / 80 / 70%, Draft 8)	Floor 5%. Rift Fortune max 5 points (+50% reward quantities)
Food, herbs, essence, potion-yield percentage bonuses	**+100%** each	Gardener ×1.5 and efficiency are separate multipliers
Creature buff bonuses	**+50%**	Bond multipliers are separate

33.7 Balance intentions (what to protect when tuning)

1. The first level-up must happen within the first ~35 reviews of a user's first session.
2. Teaching the first school must unlock within the first 2–4 days at 100–200 reviews/day.
3. Veil upkeep must never exceed ~45% of income for an average-size Academy at 150 reviews/day with reasonable wards.
4. A user at 30 reviews/day must be able to sustain a small Academy (≈10 rooms) indefinitely.
5. No setting or combination may make mana come from anything other than honest reviews.
6. The Shattered Spire is capped at 70% by Draft 8, and only the six Draft 8 modifiers change Rift odds. Never add an odds modifier when tuning; tune Fortune and rewards instead.

34. Progression Pacing

34.1 Assumptions

Default Anki scheduling (or FSRS at ~90% retention), 10–25 new cards per day, honest answering with ~10% lapses, the player training continuously and spending sensibly. “Free young cards at install” depends on the collection: an active learner typically has 200–800 cards with intervals ≤ 30 days at install. (Linking these existing young cards grants no mana or resources; it only allows enrollment, so BR-01 holds.)

34.2 Milestone timetable

Milestone	What gates it	100 reviews/day	150 reviews/day	200 reviews/day
First level-up (L1)	8 reviews of mana + 25 reviews	Session 1	Session 1	Session 1
First school at L5 (teaching)	375 reviews	Day 4	Day 3	Day 2
First student	L5 + Dormitory + Classroom + 50 free young cards	Day 4–5	Day 3–4	Day 2–3
First creature	Habitat (L3 of a school) + visitor roll	Days 3–8	Days 2–6	Days 2–5
All five schools teachable	1,875 reviews of training (+ mana)	Days 19–22	Days 13–16	Days 10–12
10 students	Beds, seats, and ~500 free young cards (often card-limited)	Weeks 3–6	Weeks 2–5	Weeks 2–4
First mastery-ready student	All 50 linked cards > 30 days	Days 35–60	Days 35–60	Days 35–60
First master	+ teacher delay (wizard L5–L10: 20–10 days, Light-reduced)	Days 45–75	Days 42–70	Days 40–68
The Rift opens (3 masters)	Three masters	Days 55–90	Days 50–85	Days 48–80
Wizard at L10 in all schools	6,875 reviews (+22,000 mana)	~Day 75–90	~Day 50–65	~Day 38–50
First Ashen Wastes trip at full safe odds	Party skill 30 days (adventurer cards’ real intervals; 6–9 weeks after the adventurers start)	Days 100–150	Days 95–145	Days 95–140
First Legendary egg	Shattered Spire (party skill 45 days for full safe odds; earlier at lower odds)	Months 5–7	Months 4–7	Months 4–6

Milestone	What gates it	100 reviews/day	150 reviews/day	200 reviews/day
Castle fully expanded	Total level 40 + materials	Months 2–3	Months 2–3	Month 2
First retirement	Every linked card > 90 days	Months 5–8	Months 5–8	Months 5–8

Note how card-driven milestones (mastery, retirement) are the same at every review volume: they follow the user's real memory, not their effort level. Review volume changes only mana-and-review-driven milestones (wizard, construction, taming, expedition return times). Rift *odds* are card-driven too: party skill is the real average interval of the adventurers' cards.

34.3 The shape of each phase

Phase	Player's main toys	Typical session
Founding (days 1–7)	Orbs, first rooms, first plot, first student and creature	Queue the next orb, place a room, admire the new student's constellation
Growing (weeks 2–5)	Five schools, dorm/refectory logistics, furniture quality, draft, first events, Veil bill	Fix the lowest-morale need, furnish a room to Cozy, draft once
Masters (weeks 6–12)	Careers, faculty limit, teacher delay, Rift opening, alchemy	Assign a new master, brew, plan the first expedition
Mature (month 3+)	Rift tiers, legendary hunts, Splendid rooms, retirements, spellbooks, terms	Launch an expedition, display an artifact, welcome an alumnus

34.4 Avoiding dead zones

The riskiest gap is between "all five schools teachable" (~week 3) and "first master" (~week 7–10), when the wizard is nearly maxed and students are still maturing. Content placed there on purpose: grimoire study (25 spells), the draft's uncommon and rare tiers, the Menagerie and Advanced Halls (L7), creature bonds, furniture quality tiers, events and achievements, and the visible "constellations" filling with gold as cards mature. The milestone track (Section 32.4) always shows the next mastery-ready student and how many of their cards are past 30 days.

35. Tutorial and Onboarding

35.1 Principles

- Never interrupt reviewing. All tutorial content lives in the game window or in tiny toasts.
- One hint at a time, always dismissible; “Skip tutorial” is available on the first screen; every hint is also in the Codex.
- Make the honesty promise first and plainly.

35.2 Flow

Step 0 — Install and first open (Tools → Ankicaster).

A storybook welcome page:

> “Every honest review you do from today earns your Academy a little mana. Again, Hard, Good and Easy are worth exactly the same — answer however is true. Your past reviews don’t count, nothing here ever changes your Anki scheduling, and taking a break never costs you anything: the Veil keeps your Academy safe while you’re away.”

Buttons: **Begin** / Skip tutorial. Then wizard creation (Section 7.2), then a 4-panel guided tour of the Academy (Study, Entrance Hall, grounds, top bar). The tutorial grants the starting kit and one **free Foundation draft** (taught right away with a fixed first draw: Courtyard Garden / Crate of Supplies / Bountiful Harvest).

First goal shown: **“Study in Anki as usual. Your first 8 reviews pay for your first orb.”**

Step 1 — First reviews. After the first honest review, a toast: “+10 mana ✨”. At 80 mana the toast says “You can afford to ponder an orb — open Ankicaster when you like.” The game remembers; it does not nag again.

Step 2 — First orb. In the Study, the player chooses a school (Nature is suggested, with a one-line reason for each school). The session progress ring is explained: “This finishes after 25 more honest reviews.”

Step 3 — First build. The Build menu opens on standard plans; the hint points at a Herb Garden or Quarry (cheap, useful). Construction progress is explained the same way.

Step 4 — First level-up (≈ review 33). Celebration, and the hint “Level 5 lets you teach this school. Until then, your Academy is just you, your tower and your gardens — enjoy the quiet.”

Step 5 — Level 3 of the first school. Habitat plan unlocked; the tutorial guarantees a Common visitor of that school at the next day tick if a habitat is built (Mossback Tortoise for Nature).

Step 6 — First day tick. The arrival summary introduces the **Veil** with the first bill (small, ~60–90 mana): what it is, that it scales with buildings not people, that Shadow and wards reduce it, and that while you’re away it pays itself down to 10% of your bank and then simply holds.

Step 7 — Level 5 (teaching). The Classroom plan and the **Recruitment** screen unlock. The enrollment tracker is explained: “Each student is linked to 50 of your youngest cards that no one else has. Right now you have 214 free young cards.” The player enrolls their first student; the character card shows the constellation with the explanation: “Each dot is one of your cards. As your intervals grow, Tamsin grows. When every one of her cards is past 30 days, and her teacher’s delay has passed, she becomes a Master.”

Step 8 — Needs and meals. When the second student arrives, a hint introduces Hunger and Rest and points at the Refectory and Dormitory capacity chips.

Step 9 — The draft. After the second school reaches L1, a hint introduces school decks and the 3-keep-1 draw.

Contextual hints later (fire once each): first hazard; first misbehavior; first room reaching Cozy; first event; first mastery-ready student (explains teacher delay with that student’s actual numbers); first master (careers, faculty limit); the Rift opening; first expedition planner (chance meter); first retirement; first time back after 7+ days (Welcome Back).

35.3 Codex (in-game help)

An illustrated book with one page per concept: Honest Mana, Bells, The Wizard, Schools, Students and Cards, Skill and Mastery, Teacher Delay, Careers, Faculty Limit, Enrollment Tracker, Retirement, Needs and Morale, Rooms and Quality, Furniture, The Veil, Hazards, Misbehavior, Food, Creatures, Injuries, Potions, Materials, The Draft, The Rift, Events, Spells, Modes, Spellbooks, Alumni, Settings, and “Our Promises” (the guardrails in plain language).

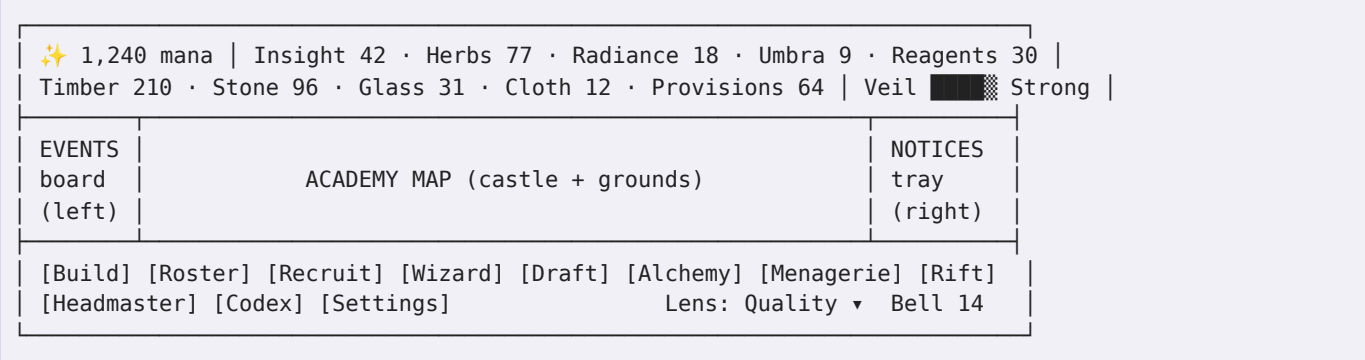
36. UI/UX Specification

36.1 How Ankicaster appears inside Anki

Surface	Implementation	Behaviour
Tools menu item	`QAction("Ankicaster", mw)` added to `mw.form.menuTools`	Opens (or raises) the game window
Top toolbar link	`gui_hooks.top_toolbar_did_init_links`	A small "🔮 1,240" link (mana bank) that opens the window; hidden if the setting `show_toolbar_link` is off
Keyboard shortcut	Configurable, default `Ctrl+Shift+M` (checked against existing Anki shortcuts at startup; if taken, no shortcut is bound and Settings says so)	Opens the window
Game window	Non-modal `QDialog` (or `QMainWindow`) containing an `AnkiWebView`; default 1280×800, resizable, remembers geometry per profile	The full game. Closing it never affects reviewing
Dock panel (optional)	`QDockWidget` added to `mw` (right side), containing a compact `AnkiWebView`	"Tower card": mana bank, active session ring, next bell, construction progress, notices count, Veil gauge. Toggle in Settings (default off)
Post-review toast	A small frameless, click-through overlay widget anchored to the top-right of the main window (not `tooltip()`, which can cover the answer buttons), fades in 120 ms, holds 1.2 s, fades out	Frequency setting: `every_review` ("+10 ✨"), **`every_bell`** (default: "+100 mana · Bell 14 · Nature 6: 61/150"), `events_only` (level-ups, finished builds, masters, hazards), `off`
Profile switch	Saves and closes the game window; loads the next profile's save on open	—

The game window is never opened automatically except once, after install, via a one-time "Open Ankicaster?" prompt on the Deck Browser.

36.2 Global layout of the game window



Screens open as large panels over the map (the map stays visible behind a translucent veil). Esc closes the top panel.

36.3 Screens

1. Academy map (home). Side-on castle grid (64 px tiles, zoom 50–150%) with grounds to the left and right and sky above. Rooms are drawn with their quality-tier art, furniture icons, occupants (small sprites idling) and status badges (Unfinished, Unreachable, Locked, Dirty, Hazard). Creatures wander in their habitats; Loose creatures wander the map and can be clicked. Construction sites show scaffolding and a progress ring. **Lenses:** Quality (tier colour), Cleanliness, Hazards and wards (Warding Chamber radii), Reachability (BFS distance), Needs (per-room satisfaction), Veil (per-room contribution). Clicking a room opens the **Room panel**: name, level, efficiency with a factor breakdown (level × quality × cleanliness × staff × adjacency), room stats and Q, quality tier, required and optional furniture slots (drag from inventory or click "+"), occupants and staff, outputs per bell, level-up/renovate, demolish.

2. Wizard. Large portrait in the Study; five school dials (level, progress ring, current buff, next unlock); the training queue with drag-to-reorder and pause; spells (cast buttons with cooldown rings); spellbook shelf; titles; appearance editor.

3. Roster. Filterable grid (students, masters, faculty, adventurers, injured, alumni). Character card: portrait, personality, traits, the **constellation** (dots for each linked card coloured by interval; hover shows the card's front text and interval; dots for suspended or missing cards show a moon or empty ring), skill and seasoning bars, mastery-ready countdown with teacher delay explanation ("Ready for 4 of 12 days. Teacher: Prof. Alder — teaching cards average 22 days → delay 8 days, −9% Light → 8 days"), needs bars, morale, assignments (bed, seat, workplace), chronicle, actions (assign career, enroll second school, swap a suspended/missing card, move bed/seat, rename).

4. Recruitment. Enrollment tracker at the top (Section 12.3); one tab per teachable school with three applicant cards each; capacity chips (beds, seats); Enroll buttons (disabled with a reason when blocked); reroll.

5. Build menu. Category tabs (Core, Living, Teaching, Knowledge, Nature, Light, Shadow, Craft, Rift, Grounds, Expansions, Furniture). Each entry: art, size, cost (unaffordable parts in red), build time, Veil per day ("+12 Veil/day"), effects, placement notes. Hand blueprints appear in their categories with a card frame. Selecting enters placement mode (ghost, rule feedback, adjacency preview, Veil delta).

6. Draft. Six deck "books" on a shelf, each with price and rarity weights; the draw animation reveals three cards; click to keep; reroll button; the hand (8 slots) below with play/discard.

7. Alchemy & Inventory. Tabs: Materials (with caps), Essences, Potions (use buttons), Recipe Book (known recipes, partially revealed ones, experiment slots), Brewing stations (choose three ingredients, see whether the triple is known), Scrolls, Artifacts, Curios.

8. Menagerie. Habitat list with occupancy; creature cards (portrait, rarity, status Wary/Tamed/Hungry/Loose, taming progress, bond, diet, buff, trouble); assign tamer; release; bestiary collection grid (silhouettes for undiscovered species, with "found in" hints).

9. Rift. Five portal cards (Mossy Hollow to the Shattered Spire), each showing its tier, skill needed, safe and best odds, reviews to return, launch cost and main rewards; a **party skill** readout (each adventurer's Rift skill in days, the party average, and "+1.8 days to the next +5%"); the **team picker** (drag up to three adventurers; injured ones greyed with reason); companion slot; supply

slots; the **chance meter** (needle and number for a safe return) with the six Draft 8 modifiers listed, the best-odds cap marked on the dial, and the Fortune score shown separately; launch cost; Launch button; active expedition card (progress in reviews, team portraits); expedition log with outcomes and loot.

10. Headmaster. Veil ledger (last 30 days: cost, paid, strength, active/away), faculty limit breakdown (Section 11.6), milestone track, term charter and timer (Timed Term), achievements, Alumni Gallery, statistics (mana earned per day, mana spent by category, rooms by quality tier, creatures tamed). There is deliberately **no** review-count leaderboard or streak display.

11. Settings. Every `config.json` option (Section 38.14) with plain-language descriptions, grouped: Honesty & counting, Students & cards, Veil, Notifications, Display & accessibility, Mode, Data (export save, backups, reset with double confirmation).

12. Codex. Section 35.3.

36.4 Arrival summary

Shown when the game window opens if anything happened since the last open: reviews counted (desktop and other devices), mana earned and spent automatically (Veil), bells rung, sessions/constructions/tamings/expeditions finished, masters, retirements, events waiting, hazards that happened and how they resolved. One “Got it” button; each line links to the relevant screen.

36.5 Accessibility and comfort

- Colour-blind-safe palette with icons and patterns on every colour-coded element (school colours always paired with school glyphs).
- Text scaling 80–150%; minimum 13 px.
- **Reduce motion** setting (disables wandering, particles, fades).
- Full keyboard navigation (Tab order, Enter/Space, arrow keys on the map); ARIA labels on all controls.
- Sound off by default.
- Dark and light themes following Anki’s night mode (`theme_manager.night_mode`).

37. Art Direction and Audio

37.1 Direction

Cozy storybook cross-section. A warm, hand-drawn-looking castle cut open like a doll's house, against a sky that shifts with the Veil (clear twilight blue when Strong, flickering aurora when Thinning, inky with stars and drifting shapes when Thin or Torn). Think illustrated children's book rather than dark fantasy: rounded shapes, soft outlines, lamplight, plants everywhere.

37.2 Practical approach for v1 (AI-buildable)

- **Rendering:** HTML/CSS/SVG inside the `AnkiWebView`. The map is an absolutely positioned grid of room `div`s with layered backgrounds (wall tier, floor, furniture icons) or a single `<canvas>` if performance requires (target 60 rooms, 80 sprites).
- **Assets:** an SVG sprite sheet generated as part of the build (simple shapes: rooms as rounded rectangles with tier-specific wall patterns; furniture as 24–32 px icons; characters as layered "paper doll" SVGs: body, robe in school colour, hair, hat). Emoji may be used as placeholder icons in the MVP only.
- **Tile size:** 64 × 64 px at 100% zoom; rooms 1×1 to 3×2 tiles.
- **Quality tiers** change wall and floor art: Bare (grey stone), Plain (plaster), Cozy (warm wood, rug), Fine (panelling, banner), Splendid (gilding, floating candles, sparkle).
- **Animation:** CSS transitions only; idle bobbing for characters, wandering creatures (simple paths within their room), scaffolding progress, level-up sparkles. All disabled by Reduce motion.

37.3 Palette

Element	Colour
Arcane	Violet #6B4FBB, silver #C9CED6
Nature	Moss #4E8B3A, amber #E0A33A
Light	Gold #E8C547, ivory #FFF6DC
Shadow	Indigo #2F3A73, teal #2E8B8B
Alchemy	Copper #B5651D, vermilion #D9472B
UI parchment	#F4EBD9 (light theme), #2A2533 (dark theme)
Veil	Strong #9BD4FF, Thinning #B9A6FF, Thin #7D5BA6, Torn #3D2856

37.4 Typography

A readable humanist sans-serif for UI (system UI font stack) and a single display face for headings (bundle an open-licence storybook serif such as a SIL OFL font; include its licence file).

37.5 Audio (optional, off by default)

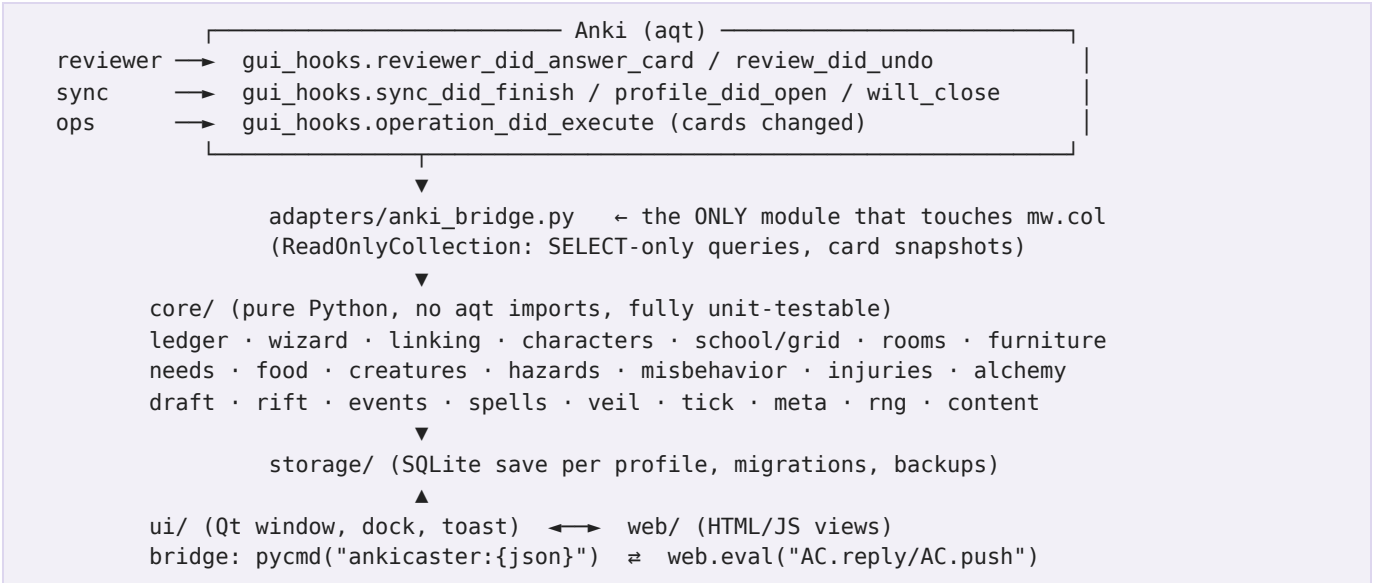
Soft bell chime per bell (only while the game window is open), page-turn for the draft, gentle “ding” on level-up, purr/chirp on taming. Never play sound in the reviewer.

38. Technical Implementation Specification

38.1 Target environment

Item	Requirement
Anki	Desktop 2.1.66 or newer, including 23.x, 24.x and 25.x (Qt6 builds only)
Language	Python 3.9-compatible syntax (no `match`, no 3.10+ typing syntax such as `X | Y`; use `from __future__ import annotations` + `typing.Optional`)
UI toolkit	`aqt.qt` (PyQt6 via Anki's wrapper; never import `PyQt6` directly)
Web UI	`aqt.webview.AnkiWebView` with local HTML/CSS/JS served through `mw.addonManager.setWebExports`
JS	Vanilla ES2019 modules, no CDN, no build step required (a small bundled micro-framework of your own is fine)
Save storage	Python's standard `sqlite3`, one file per Anki profile in the add-on's `user_files/` folder
Dependencies	None beyond the standard library and Anki

38.2 Architecture



Rules:

1. `core/` never imports `aqt` or `anki`. It receives plain data (card snapshots, revlog rows, today's day number) and returns plain data (state changes, notices).
2. `adapters/anki_bridge.py` is the only code allowed to read the collection, and it can only read (Section 38.4).
3. The engine (Python) is authoritative. The web UI renders state and sends *intents*; every intent is validated in Python.

38.3 Suggested file structure

```
ankicaster/
├── __init__.py                # entry point: registers hooks, menu action, web exports
├── manifest.json              # package metadata for .ankiaddon
├── config.json                # global user preferences (Section 38.14)
├── config.md                  # help text for Anki's config dialog
├── adapters/
│   ├── anki_bridge.py         # ReadOnlyCollection, card/revlog snapshot queries
│   └── hooks.py               # hook handlers (thin; defer heavy work)
├── core/
│   ├── constants.py           # binding constants (25×n, 30 days, 20 career cards, 10% floor…)
│   ├── content.py             # loads and validates content/*.json
│   ├── rng.py                 # seeded, named RNG streams
│   ├── ledger.py              # honest-review checks, mana, undo, catch-up
│   ├── progress.py            # per-review progress dispatch (sessions, builds, taming…)
│   ├── bells.py               # bell processing order (Section 38.12)
│   ├── tick.py                # day tick, break handling (Section 38.10)
│   ├── wizard.py              # levels, sessions, buffs, spells
│   ├── linking.py             # youngest-free-card selection, maintenance
│   ├── characters.py          # students, skill, mastery, careers, retirement
│   ├── enrollment.py          # tracker, applicants
│   ├── grid.py                # placement rules, reachability BFS, adjacency
│   ├── rooms.py               # construction, levels, efficiency, quality
│   ├── furniture.py
│   ├── needs.py               # needs, meals, morale
│   ├── food.py                # production, crops, storage caps
│   ├── creatures.py           # arrival, taming, feeding, escapes
│   ├── veil.py                # upkeep, strength, away floor
│   ├── hazards.py
│   ├── misbehavior.py
│   ├── injuries.py
│   ├── alchemy.py
│   ├── draft.py
│   ├── rift.py
│   ├── events.py
│   ├── meta.py                # modes, terms, spellbooks, alumni, achievements, renown
│   ├── modifiers.py           # collects buffs and applies global caps (Section 33.6)
│   └── state_view.py          # builds the JSON snapshot sent to the UI
├── storage/
│   ├── db.py                  # connection, transactions, backups
│   └── migrations/
│       ├── 001_initial.sql
│       └── …
├── ui/
│   ├── window.py              # game QDialog + AnkiWebView
│   ├── dock.py                # optional QDockWidget
│   ├── toast.py               # click-through overlay toast
│   └── bridge.py              # pycmd router, reply/push helpers
├── content/                   # all tunable game data (Section 38.15)
│   ├── economy.json  rooms.json  furniture.json  creatures.json  crops.json
│   ├── potions.json  materials.json  artifacts.json  scrolls.json  hazards.json
│   ├── misbehavior.json  traits.json  personalities.json  events.json  spells.json
│   ├── spellbooks.json  alumni.json  achievements.json  draft_decks.json  rift.json
│   ├── wizard.json  names.json  tutorial.json  strings_en.json
├── web/
│   ├── index.html  app.js  views/*.js  styles.css  sprites.svg  fonts/
├── tests/
```

```
│   ├── test_ledger.py  test_linking.py  test_mastery.py  test_veil.py  test_grid.py
│   ├── test_rift.py  test_tick_breaks.py  test_readonly_guard.py  sim/simulate.py
└── user_files/                # created at runtime; preserved by Anki on updates
    └── <profile name>/ankicaster.sqlite, backups/, logs/
```

38.4 Read-only collection access (BR-02)

All collection access goes through one class:

```python
class ReadOnlyCollection:
    """The only gateway to Anki data. SELECT statements only."""
    def __init__(self, col):
        self._col = col

    def _select(self, sql: str, *args):
        s = sql.lstrip().lower()
        if not s.startswith("select") or ";" in sql:
            raise RuntimeError("Ankicaster: non-SELECT query refused")
        return self._col.db.all(sql, *args)

    def today(self) -> int:            return self._col.sched.today
    def day_cutoff(self) -> int:       return self._col.sched.day_cutoff  # epoch seconds

    def card_rows(self, ids):          # chunked by 500
        ...  # SELECT id, nid, did, odid, type, queue, ivl FROM cards WHERE id IN (...)

    def last_answer_ivls(self, ids):
        ...  # SELECT cid, ivl FROM revlog WHERE cid IN (...) AND type BETWEEN 0 AND 3
             #   AND ease BETWEEN 1 AND 4 ORDER BY id   (keep the last per cid)

    def revlog_since(self, after_id):
        ...  # SELECT id, cid, ease, ivl, time, type FROM revlog WHERE id > ? ORDER BY id

    def revlog_entry(self, revlog_id): ...
    def latest_revlog_for_card(self, cid): ...
    def eligible_young_cards(self, deck_ids, young_days, exclude_ids): ...  # Section 38.9
    def deck_names(self): return self._col.decks.all_names_and_ids()
```

Forbidden anywhere in the add-on (enforce with a unit test that scans the source tree): `col.update_card`, `col.update_cards`, `card.flush`, `col.sched.answerCard`, `sched.answer_card`, `sched.suspend_cards`, `sched.bury_cards`, `sched.reschedule_cards`, `sched.set_due_date`, `sched.forget_cards`, `col.remove_cards`, `col.add_note`, `col.update_note`, `note.flush`, `col.decks.save`, `col.set_config`, `col.db.execute`, `col.db.executemany`, `col.save`, `mw.col.mod`, `checkpoint`, `col.modSchema`, any `UPDATE/INSERT/DELETE/REPLACE` string sent to `col.db`. The reviewer's answer flow must never be altered (no wrapping of `Reviewer._answerCard`, no monkey-patching).

Heavier reads (refreshing thousands of links) use `aqt.operations.QueryOp(parent=mw, op=lambda col: ..., success=...)` so they run off the main thread with proper collection locking and never freeze the UI.

38.5 Hooks used

Hook	Handler	Work done on the hook thread (≤ 5 ms)
`gui_hooks.main_window_did_init`	Add Tools-menu action and shortcut; `setWebExports`	Trivial
`gui_hooks.profile_did_open`	Open the save for `mw.pm.name`; first-run install timestamp; schedule catch-up scan and day tick	Open DB only; defer the rest with `mw.progress.single_shot(500, …)`
`gui_hooks.profile_will_close`	Flush and close the save; close windows	Short
`gui_hooks.reviewer_did_answer_card(reviewer, card, ease)`	Record `card.id` and `time.time()` in a pending queue; schedule `process_pending()` in 50 ms	Append to a list
`gui_hooks.review_did_undo(card_id)`	Schedule ledger reconciliation for that card	Append
`gui_hooks.state_did_undo(changes)`	Schedule a broader ledger reconciliation (Edit → Undo)	Append
`gui_hooks.operation_did_execute(changes, handler)`	If `changes.card` or `changes.note` is true, mark links dirty for refresh at the next window open / day tick	Set a flag
`gui_hooks.sync_did_finish`	Schedule catch-up scan of revlog (other devices) and link refresh	Schedule only
`gui_hooks.top_toolbar_did_init_links(links, toolbar)`	Add the mana link	String append

A `QTimer` (every 10 minutes while a profile is open) checks whether `col.sched.today` has advanced and runs the day tick if so, so a user who leaves Anki open overnight still gets their tick.

38.6 Review processing pipeline

```
def process_pending():                       # runs ~50 ms after an answer, on the main thread
    for (cid, t) in drain(pending):
        row = ro.latest_revlog_for_card(cid)        # id, cid, ease, ivl, time, type
        if row is None: continue
        if ledger.has(row.id): continue             # H4 idempotence
        verdict = ledger.check_honest(row, install_ms, per_card_day_counts)   # H1–H7
        ledger.record(row, verdict, source="live")
        if verdict.ok:
            effects = engine.apply_honest_review(row)   # mana, progress, bells (38.12)
            ledger.attach_effects(row.id, effects)      # for undo
            toast.maybe_show(effects)
    engine.maybe_day_tick()

def catch_up_scan():                         # profile open, after sync
```

```
    for row in ro.revlog_since(max(ledger.watermark(), install_ms)):
        if ledger.has(row.id): continue
        verdict = ledger.check_honest(row, ...)
        ledger.record(row, verdict, source="catchup")
        if verdict.ok: engine.apply_honest_review(row, quiet=True)
    notices.add_summary("reviews on other devices")

def reconcile_undo(card_id=None):
    for entry in ledger.recent_counted(card_id, within_ms=3_600_000):
        if not ro.revlog_entry(entry.revlog_id):        # Anki removed it
            engine.reverse(entry.effects)               # mana (→ debt if spent), progress
            ledger.mark_reversed(entry.revlog_id)
```

Notes:

- The live path and the catch-up path share the same ledger, so a review is never counted twice.
- `apply_honest_review` is fast (O(active items)); bells and the day tick are deferred further if they exceed 5 ms.
- **Undo edge case (accepted):** if the undone review had *completed* something (a session, a build), the completion stands; only the review's own progress and mana are reversed (mana via debt). The internal `review_index` used for durations is monotonic and is never decremented.

38.7 Data model (SQLite schema)

One save file per Anki profile: `user_files/<profile>/ankicaster.sqlite` (WAL journal mode, `PRAGMA foreign_keys=ON`). JSON columns hold small structured values. All times are epoch milliseconds; all `day` values are Anki day numbers (`col.sched.today`).

```
-- 001_initial.sql
CREATE TABLE meta (key TEXT PRIMARY KEY, value TEXT NOT NULL);
-- keys: schema_version, install_ms, rng_seed, last_processed_day, last_review_ms,
--       bank_at_last_review, review_index, honest_review_count, last_bell_index,
--       mode ('endless'|'timed'), current_term_id, renown, tutorial_step

CREATE TABLE wallet (
  id INTEGER PRIMARY KEY CHECK (id = 1),
  mana REAL NOT NULL DEFAULT 0, mana_debt REAL NOT NULL DEFAULT 0,
  lifetime_earned REAL NOT NULL DEFAULT 0, lifetime_spent REAL NOT NULL DEFAULT 0);

CREATE TABLE mana_ledger (
  revlog_id INTEGER PRIMARY KEY,          -- Anki revlog.id; guarantees no double counting
  card_id INTEGER NOT NULL, day INTEGER NOT NULL,
  source TEXT NOT NULL,                    -- 'live' | 'catchup'
  status TEXT NOT NULL,                    -- 'counted' | 'too_fast' | 'card_cap' | 'pre_install'
                                           -- | 'not_study' | 'reversed'
  mana REAL NOT NULL DEFAULT 0,
  effects_json TEXT,                       -- what this review advanced (for undo)
  counted_ms INTEGER NOT NULL);
CREATE INDEX ix_ledger_card_day ON mana_ledger(card_id, day);

CREATE TABLE day_log (
  day INTEGER PRIMARY KEY, honest_reviews INTEGER NOT NULL DEFAULT 0,
  mana_earned REAL NOT NULL DEFAULT 0, veil_cost INTEGER, veil_paid INTEGER,
  veil_strength_after INTEGER, was_active INTEGER, processed INTEGER NOT NULL DEFAULT 0);

CREATE TABLE wizard (
```

```
  id INTEGER PRIMARY KEY CHECK (id = 1), name TEXT, title TEXT, pronouns TEXT,
  appearance_json TEXT, sigil TEXT, academy_name TEXT, created_ms INTEGER,
  study_level INTEGER NOT NULL DEFAULT 1);
CREATE TABLE wizard_levels (school TEXT PRIMARY KEY, level INTEGER NOT NULL DEFAULT 0);

CREATE TABLE sessions (
  id INTEGER PRIMARY KEY, kind TEXT NOT NULL,             -- 'orb' | 'grimoire'
  school TEXT NOT NULL, target_level INTEGER, spell_id TEXT,
  cost_paid REAL NOT NULL, required INTEGER NOT NULL, progress REAL NOT NULL DEFAULT 0,
  state TEXT NOT NULL,                                     -- 'active'|'queued'|'paused'|'done'
  queue_pos INTEGER, created_ms INTEGER, completed_ms INTEGER);

CREATE TABLE spells_known (spell_id TEXT PRIMARY KEY, learned_day INTEGER,
  cooldown_until_index INTEGER DEFAULT 0, cooldown_until_day INTEGER DEFAULT 0);

CREATE TABLE characters (
  id INTEGER PRIMARY KEY, name TEXT NOT NULL, pronouns TEXT, appearance_json TEXT,
  personality TEXT NOT NULL, traits_json TEXT NOT NULL,
  status TEXT NOT NULL,             -- 'student'|'resident'|'faculty'|'away'|'alumnus'
  current_school TEXT,              -- school currently being studied (NULL if none)
  mastered_json TEXT NOT NULL DEFAULT '[]',
  career TEXT, career_state TEXT,   -- 'active'|'sabbatical'
  needs_json TEXT NOT NULL,         --
      {"hunger":80,"rest":70,"leisure":60,"composure":75,"health":100}
  morale REAL NOT NULL DEFAULT 70,
  injury_json TEXT,                 -- {"type":"sprained","points":12,"required":80}
  bed_room_id INTEGER, seat_room_id INTEGER, workplace_room_id INTEGER,
  ready_days_json TEXT NOT NULL DEFAULT '{}',   -- per school: real days ready
  enrolled_day INTEGER, retired_day INTEGER,
  chronicle_json TEXT NOT NULL DEFAULT '[]', flags_json TEXT NOT NULL DEFAULT '{}');

CREATE TABLE card_sets (
  id INTEGER PRIMARY KEY, character_id INTEGER NOT NULL REFERENCES characters(id),
  kind TEXT NOT NULL,               -- 'study' | 'career'
  school TEXT, career TEXT, size INTEGER NOT NULL, created_day INTEGER NOT NULL,
  released INTEGER NOT NULL DEFAULT 0);

CREATE TABLE card_links (
  anki_card_id INTEGER PRIMARY KEY,           -- PRIMARY KEY enforces BR-13 (never shared)
  set_id INTEGER NOT NULL REFERENCES card_sets(id),
  slot INTEGER NOT NULL, linked_day INTEGER NOT NULL,
  cached_ivl INTEGER NOT NULL DEFAULT 0,      -- cards.ivl
  last_answer_ivl INTEGER NOT NULL DEFAULT 0, -- from revlog (type 0–3)
  honest_ivl INTEGER NOT NULL DEFAULT 0,      -- max(0, min(cached_ivl, last_answer_ivl))
  card_type INTEGER, card_queue INTEGER, note_id INTEGER,
  status TEXT NOT NULL DEFAULT 'ok',          -- 'ok'|'suspended'|'new'|'missing'
  status_since_day INTEGER, refreshed_ms INTEGER,
  UNIQUE (set_id, slot));
CREATE INDEX ix_links_set ON card_links(set_id);

CREATE TABLE rooms (
  id INTEGER PRIMARY KEY, type TEXT NOT NULL, area TEXT NOT NULL,   -- 'castle'|'grounds'
  x INTEGER NOT NULL, y INTEGER NOT NULL, w INTEGER NOT NULL, h INTEGER NOT NULL,
  level INTEGER NOT NULL DEFAULT 1, state TEXT NOT NULL,           -- 'site'|'built'
  build_required REAL, build_progress REAL DEFAULT 0, build_paused INTEGER DEFAULT 0,
  build_slot INTEGER, cleanliness REAL NOT NULL DEFAULT 100,
  locked_until_index INTEGER DEFAULT 0, lock_reason TEXT,
  settings_json TEXT NOT NULL DEFAULT '{}',   -- crops, habitat modules, toggles
  built_day INTEGER);
```

```
CREATE TABLE furniture (
  id INTEGER PRIMARY KEY, item_id TEXT NOT NULL, room_id INTEGER REFERENCES rooms(id),
  slot_kind TEXT,                    -- 'required:<role>' | 'optional' | 'bed' | 'plinth'
  slot_index INTEGER, damaged INTEGER NOT NULL DEFAULT 0, acquired_day INTEGER);

CREATE TABLE known_patterns (item_id TEXT PRIMARY KEY);
CREATE TABLE known_plans (room_type TEXT PRIMARY KEY);
CREATE TABLE inventory (item_id TEXT PRIMARY KEY, qty REAL NOT NULL DEFAULT 0);
CREATE TABLE hand (id INTEGER PRIMARY KEY, card_def_id TEXT NOT NULL, drawn_day INTEGER);
CREATE TABLE draft_history (id INTEGER PRIMARY KEY, deck TEXT, day INTEGER, price REAL,
  shown_json TEXT, kept TEXT);
CREATE TABLE draft_pity (deck TEXT PRIMARY KEY, draws_without_rare INTEGER DEFAULT 0);

CREATE TABLE creatures (
  id INTEGER PRIMARY KEY, species_id TEXT NOT NULL, nickname TEXT,
  room_id INTEGER REFERENCES rooms(id),
  status TEXT NOT NULL,              --
        'egg'|'wary'|'tamed'|'loose'|'away'|'crated'|'visitor_waiting'
  hatch_progress REAL DEFAULT 0,      -- eggs only (Section 27.10)
  hungry INTEGER NOT NULL DEFAULT 0, taming_progress REAL DEFAULT 0, tamer_id INTEGER,
  bond_level INTEGER DEFAULT 1, bond_progress REAL DEFAULT 0,
  loose_since_index INTEGER, hidden_json TEXT, arrived_day INTEGER);

CREATE TABLE expeditions (
  id INTEGER PRIMARY KEY, destination TEXT NOT NULL, tier TEXT NOT NULL,
  team_json TEXT NOT NULL, companion_id INTEGER, supplies_json TEXT,
  party_skill REAL NOT NULL,                              -- mean ivl of adventurer cards at launch
  chance INTEGER NOT NULL, fortune INTEGER NOT NULL DEFAULT 0,
  roll INTEGER NOT NULL, wrong_roll INTEGER NOT NULL,  -- both hidden from the UI
  breakdown_json TEXT NOT NULL, required INTEGER NOT NULL, progress REAL DEFAULT 0,
  state TEXT NOT NULL,                                    -- 'active'|'returned'
  outcome TEXT,               -- 'safe'|'minor_injury'|'serious_injury'|'curse'|'retreat'
  loot_json TEXT, injury_json TEXT, launched_day INTEGER, returned_day INTEGER);

CREATE TABLE events (id INTEGER PRIMARY KEY, event_id TEXT NOT NULL, created_day INTEGER,
  expires_day INTEGER, state TEXT NOT NULL, choice TEXT, result_json TEXT);
CREATE TABLE active_effects (id INTEGER PRIMARY KEY, kind TEXT, target TEXT,
  until_index INTEGER, until_day INTEGER, data_json TEXT);      -- hazards, boons, spells
CREATE TABLE applicants (id INTEGER PRIMARY KEY, school TEXT, day INTEGER, data_json TEXT);
CREATE TABLE notices (id INTEGER PRIMARY KEY, created_ms INTEGER, kind TEXT, text TEXT,
  data_json TEXT, seen INTEGER DEFAULT 0);
CREATE TABLE achievements (id TEXT PRIMARY KEY, unlocked_day INTEGER);
CREATE TABLE spellbooks (id TEXT PRIMARY KEY, acquired_day INTEGER, source TEXT);
CREATE TABLE alumni_bonuses (character_id INTEGER, category TEXT, value REAL);
CREATE TABLE recipes_known (recipe_id TEXT PRIMARY KEY, revealed_mask INTEGER NOT NULL);
CREATE TABLE research (id TEXT PRIMARY KEY, rank INTEGER NOT NULL DEFAULT 0,      -- e.g.
        'veil_theory'
  active_rank INTEGER, progress REAL DEFAULT 0, required INTEGER, researcher_id INTEGER);
CREATE TABLE rng_streams (name TEXT PRIMARY KEY, counter INTEGER NOT NULL DEFAULT 0);
CREATE TABLE terms (id INTEGER PRIMARY KEY, started_day INTEGER, min_length_days INTEGER,
  exam_timer INTEGER, charter_json TEXT, ended_day INTEGER, stars INTEGER, record_json TEXT);
```

38.8 UI state snapshot (JSON)

`state_view.build()` returns one JSON object (≈ 20–150 KB) that the web UI renders. Deltas may be pushed after bells; a full snapshot is sent on window open.

```
{
  "version": 1,
  "day": 20123, "reviewIndex": 4182, "bell": 418, "nextBellIn": 8,
  "wallet": {"mana": 1240.0, "debt": 0},
  "resources": {"timber": 210, "stone": 96, "glass": 31, "cloth": 12, "provisions": 64,
                "greens": 20, "fish": 0, "nectar": 4,
                "insight": 42, "herbs": 77, "radiance": 18, "umbra": 9, "reagents": 30,
                "crystal": 3}, "caps": {"timber": 500},
  "veil": {"strength": 86, "band": "strong", "costToday": 302, "reductions": [["Shadow L7", 0.06]]},
  "wizard": {"name": "Hazel", "levels": {"arcane": 3, "nature": 7, "light": 5, "shadow": 2,
        "alchemy": 5},
             "sessions": [{"id": 9, "kind": "orb", "school": "nature", "target": 8,
                           "progress": 61, "required": 200, "state": "active"}]},
  "grid": {"castle": {"w": 10, "h": 7}, "grounds": {"plots": 10}},
  "rooms": [{"id": 1, "type": "entrance_hall", "x": 3, "y": 0, "w": 2, "h": 1, "level": 1,
             "state": "built", "efficiency": 1.08, "quality": {"q": 18, "tier": "plain"},
             "clean": 92, "badges": []}],
  "characters": [{"id": 3, "name": "Tamsin Reed", "status": "student", "school": "nature",
                  "skill": 64, "seasoning": 22, "ready": false, "cardsOver30": 31,
                  "needs": {"hunger": 80, "rest": 72, "leisure": 55, "composure": 70, "health":
        100},
                  "morale": 72}],
  "tracker": {"freeYoung": 63, "needed": 50, "bedsFree": 2, "seatsFree": {"nature": 3}},
  "creatures": [], "expeditions": [], "events": [], "notices": [], "hand": [],
  "settings": {"reduceMotion": false, "textScale": 1.0}
}
```

Card front text for constellation tooltips is fetched lazily (`character.card_preview`), stripped of HTML and truncated to 60 characters, and never stored in the save.

38.9 Card-linking algorithm (implementation)

```
def pick_youngest_free(ro, store, n, require_young, deck_ids):
    linked = store.all_linked_card_ids()            # set[int]
    rows = ro.eligible_cards(deck_ids)              # see SQL below
    cands = []
    for r in rows:
        if r.id in linked: continue
        hon = honest_ivl(r.ivl, store.last_answer_ivl_cache.get(r.id, r.ivl))
        if require_young and hon > YOUNG_THRESHOLD_DAYS: continue
        cands.append((hon, -r.first_review_ms, r.id, r.nid))
    cands.sort()                                    # youngest, newest-started, stable id
    chosen, notes = [], set()
    for hon, _, cid, nid in cands:
        if settings.avoid_siblings and nid in notes: continue
        chosen.append(cid); notes.add(nid)
        if len(chosen) == n: return chosen
    if settings.avoid_siblings:                     # relax sibling rule if short
        for hon, _, cid, nid in cands:
            if cid not in chosen:
                chosen.append(cid)
                if len(chosen) == n: return chosen
    return None                                     # not enough: caller blocks (BR-23)
```

```
-- eligible cards (read-only). Deck scope uses the home deck for cards in filtered decks.
SELECT c.id, c.nid, c.ivl, c.type, c.queue,
```

```
       (SELECT MIN(r.id) FROM revlog r WHERE r.cid = c.id) AS first_review_ms
FROM cards c
WHERE c.queue != -1                          -- not suspended
  AND c.type IN (1, 2, 3)                    -- studied at least once (unless link_new_cards)
  AND (CASE WHEN c.odid != 0 THEN c.odid ELSE c.did END) IN (/* deck ids in scope */)
```

For large collections (100k+ cards) the query is run inside a `QueryOp`, and the `first_review_ms` subquery may be replaced by `c.id` ordering if it exceeds 300 ms (documented fallback).

Link maintenance at each day tick:

1. Refresh `cached_ivl`, `card_type`, `card_queue` for all links; refresh `last_answer_ivl` for links whose card was answered since the last refresh (from the ledger) or every 7 days for all.
2. Missing ids → status `missing` (`status_since_day` = today).
3. `queue == -1` → `suspended` (keep `status_since_day`); `type == 0` → `new`.
4. If more than 25% of all links became `missing` in one tick, stop and raise the "Many linked cards are missing" dialog (Section 9.6).
5. Release and refill: links `missing` (any age), `suspended` longer than `suspended_release_days`, `new` longer than 30 days → delete the link row and, for study sets, fill the slot with `pick_youngest_free(n=1, require_young=True)`; for career sets, `require_young=False`. If no card is available the slot stays empty (counts as interval 0).

38.10 Day tick and break handling

```
def maybe_day_tick():
    today = ro.today()
    last = meta.last_processed_day
    if last is None: meta.last_processed_day = today; return
    if today <= last: return                        # same day, or clock moved backwards: ignore
    for d in range(last, today):                    # process every completed day [last, today)
        veil.charge_day(d, active=day_log.honest_reviews(d) > 0)   # Section 18.4
        if day_log.honest_reviews(d) > 0:
            events.daily_roll(d)                     # only after active days
    elapsed = today - last
    links.refresh_and_maintain(today)                # QueryOp; then continue in success()
    characters.recompute_skill()
    characters.advance_mastery(elapsed_days=elapsed) # ready_days += elapsed if ready now
    characters.check_retirements(today)
    enrollment.refresh_applicants(today)
    creatures.daily_visitors(today, active=day_log.honest_reviews(today - 1) > 0)
    meta.term_timer(today); meta.check_achievements()
    meta.last_processed_day = today
```

- **Long breaks:** the Veil loop is O(days) with trivial arithmetic; for breaks over 400 days use the closed form (pay `min(cost × days, bank − floor)` and stop). Nothing else iterates per day.
- **Away days** are days with zero honest reviews in `day_log`. Reviews from other devices arriving later are credited to their own day for mana, but a day already processed as away is not re-processed (no retroactive Veil changes).
- **Mapping a revlog id to an Anki day:** `day = today − floor((day_cutoff × 1000 − 1 − revlog_id) / 86_400_000)`.
- **Clock moved forward** (e.g., the system clock changes): treated like an away period. Card intervals cannot change without answers, so this cannot speed mastery-readiness; it only

advances day-based countdowns, which is accepted for a single-player game.

38.11 Randomness and determinism

- A 64-bit `rng_seed` is created at save creation. Every random decision draws from a **named stream** (`"hazard"`, `"misbehavior"`, `"draft:arcane"`, `"rift"`, `"events"`, `"applicants"`, `"visitors"`, `"escape"`, `"loot"`, …): `value = PCG/xorshift(seed, hash(stream), counter++)`. Counters are persisted.
- Consequences: reopening the window or restarting Anki never rerolls anything; the Rift roll is fixed at launch; bug reports can include the seed and counters to reproduce behaviour.
- Tests can inject a fixed seed.

38.12 Per-review and per-bell processing order

Every honest review (in `apply_honest_review`):

1. Mana: `+mana_per_review × (1 + capped multiplier)`; pay off `mana_debt` first.
2. `review_index += 1; day_log[today].honest_reviews += 1; bank_at_last_review = mana`.
3. Progress +1 (with each item's own multiplier) to: the active training session; active research; each active construction slot; each taming; each egg; each healing; each brewing station; each scroll craft; each active expedition. Completions are queued as notices.
4. Expire `active_effects` whose `until_index` ≤ `review_index`.
5. If `review_index` crosses a multiple of 10 → **ring a bell**.

Each bell:

1. Gather modifiers (`modifiers.py`) once for the bell.
2. Needs update for every character (Section 14.4).
3. Production: plots, greenhouses, lessons (essence + rare chance), scholars, Glassworks, Weaving Room, creatures with production buffs; apply storage caps.
4. If `bell_index % 5 == 0`: mealtime — characters, then creatures.
5. Cleanliness: dirt, then cleaning (builders, Mopslime, brooms).
6. Hazard roll (and breach roll if Thin/Torn).
7. Misbehavior rolls.
8. Creature escape rolls; Loose creature auto-return checks; bond progress.
9. Morale recompute.
10. Achievement checks; push a UI delta if the window is open.

Target: one bell ≤ 30 ms for 60 rooms, 40 characters, 40 creatures.

38.13 Anti-exploit rules (summary)

Exploit	Rule
Counting old history	Only `revlog.id > install_ms` (H1); install time recorded once per profile and never reset by reinstalling (stored in the save; if the save is deleted, the new install time is "now")
Mashing buttons	`revlog.time ≥ min_answer_ms` (H5)
Looping one card	Per-card per-day cap (H6)

Exploit	Rule
Manual reschedule / “set due date” to fake maturity	Honest interval = min(current ivl, last answered ivl) (Section 9.2)
Suspending hard cards	Suspended cards stay linked at their interval; released after 30 days and replaced with a young card
Deleting young cards	Missing slots count as 0 until refilled with another young card
Forget/reset	Interval 0, honest
Undo after counting	Reversal via ledger; spent mana becomes debt
Double counting (hook + sync scan)	Ledger primary key on `revlog_id`
Rerolling randomness by restarting	Persisted RNG stream counters; Rift roll fixed at launch
Switching profiles	Saves and revlogs are per profile
Editing `config.json` to inflate mana	`mana_per_review` is bounded 1–20 and changes apply only to future reviews; a changed value shows a “custom economy” badge in the Headmaster panel
Importing another collection’s recent revlog	Accepted residual risk; H1–H7 still apply

38.14 `config.json` (global preferences)

Anki stores this per add-on (shared across profiles). Game state that belongs to a save (mode, term) lives in the save, not here.

```
{
  "mana_per_review": 10,
  "min_answer_ms": 800,
  "max_counted_answers_per_card_per_day": 6,
  "count_reviews_from_other_devices": true,
  "cards_per_student": 50,
  "retirement_age_days": 90,
  "deck_scope": "all",
  "link_new_cards": false,
  "avoid_siblings": true,
  "honest_intervals_only": true,
  "suspended_release_days": 30,
  "veil_payment": "auto",
  "toast_mode": "every_bell",
  "show_toolbar_link": true,
  "dock_panel": false,
  "shortcut": "Ctrl+Shift+M",
  "reduce_motion": false,
  "sound": false,
  "text_scale": 1.0,
  "debug_logging": false
}
```

Key	Allowed values	Notes
`mana_per_review`	1–20	Economy is balanced for 10
`min_answer_ms`	0–3000	0 disables H5
`max_counted_answers_per_card_per_day`	1–20	

Key	Allowed values	Notes
`cards_per_student`	10–200	BR-12 setting, default 50; applies to **all** students, existing and new (resize rule in Section 9.4)
`retirement_age_days`	31–365	BR-21 setting; must exceed 30
`deck_scope`	`"all"` or list of deck ids	
`link_new_cards`	bool	Off by default
`veil_payment`	`"auto"` / `"ask"`	
`toast_mode`	`"every_review"`, `"every_bell"`, `"events_only"`, `"off"`	

Fixed constants (not configurable, in `core/constants.py`, because they are binding): reviews per level `25 × n`; max level 10; teaching at level 5; buff percentages; mastery threshold 30 days; teacher delay formula; 20 teaching/career cards; away floor 10%; 3 masters to open the Rift; team size 1–3.

38.15 Content files

All game content is data. `core/content.py` loads every file at startup, validates it against a schema (required keys, referenced ids exist, numbers in range) and refuses to start the game (with a clear in-game message, never affecting reviewing) if validation fails.

File	Contents
`economy.json`	Mana per review default, caps (Section 33.6), Veil formula constants, draft price constants, meal interval, bell size, starting kit
`wizard.json`	Level unlock table, total-level unlocks, Study levels, passive abilities
`rooms.json`	All 56 rooms (Section 16)
`furniture.json`	All furniture (Section 17)
`crops.json`	Greenhouse crops (Section 21.3)
`creatures.json`	Bestiary (Section 22.7) plus habitat definitions
`materials.json`	Materials and essences with caps and display info
`potions.json`	Recipes and effects (Section 24.4)
`scrolls.json`, `artifacts.json`	Sections 25.4–25.5
`hazards.json`, `misbehavior.json`	Sections 19.3–19.4, 20.3
`traits.json`, `personalities.json`, `names.json`	Section 14
`events.json`	Section 28
`spells.json`	Section 29
`spellbooks.json`, `alumni.json`, `achievements.json`	Sections 31–32
`draft_decks.json`	Section 26
`research.json`	Arcane research projects (Veil Theory ranks: requirements, costs, review times)
`rift.json`	Destinations with tier, skill needed, safe/best odds, return reviews, costs; modifiers; “when it goes wrong” table; loot pools; Fortune; egg hatch times (Section 27.13)

File	Contents
`tutorial.json`, `strings_en.json`	Tutorial steps and all UI strings (localizable)

Effect vocabulary. Effects are expressed with a small set of typed operations so the engine stays generic:

```
{"op": "pct", "target": "food_output", "value": 0.05, "scope": "academy"}
{"op": "pct", "target": "room_efficiency", "value": 0.10, "scope": "room", "room_types":
        ["greenhouse"]}
{"op": "flat", "target": "composure_target", "value": 5, "scope": "room_occupants"}
{"op": "produce", "resource": "herbs", "per_bell": 0.1}
{"op": "chance_mult", "target": "hazard", "value": -0.08}
{"op": "fortune", "value": 1, "condition": {"destination": "crystal_caverns"}}
{"op": "special", "id": "cleans_room", "value": 1}
```

`modifiers.py` aggregates all active effects into category totals, applies the global caps, and records a human-readable source list for tooltips.

Example entries

```
// rooms.json
{
  "id": "greenhouse", "name": "Greenhouse", "category": "nature", "area": "castle",
  "size": [2, 1], "needs_sky": true, "floor_rule": null,
  "unlock": {"type": "standard", "school": "nature", "level": 2},
  "cost": {"mana": 450, "timber": 30, "glass": 25}, "build_reviews": 60, "veil": 5,
  "capacity": {"crop_beds": 2, "per_level": 1}, "workplace": {"career": "gardener", "slots": 1},
  "required": [{"role": "planter_bed", "count": 2}], "optional_slots": 3,
  "stats": {"primary": ["aura", "comfort"], "secondary": "charm"},
  "outputs": [{"per_crop_bed": true}],
  "level3_perk": "mixed_planting"
}
```

```
// furniture.json
{
  "id": "ward_lantern", "name": "Ward Lantern", "rooms": "any", "role": "optional", "slots": 1,
  "cost": {"mana": 160, "glass": 5, "umbra": 5},
  "stats": {"order": 2},
  "effects": [{"op": "chance_mult", "target": "hazard", "value": -0.30, "scope": "room"},
              {"op": "chance_mult", "target": "hazard", "value": -0.10, "scope": "adjacent"}],
  "unlock": {"type": "draft", "deck": "shadow", "rarity": "common"}
}
```

```
// creatures.json
{
  "id": "mopslime", "name": "Mopslime", "school": "alchemy", "rarity": "common",
  "habitat": ["tide"], "taming": 40, "diet": {"provisions": 1},
  "sources": ["visitor", "lure:alchemy"],
  "buff": [{"op": "special", "id": "academy_clean_per_bell", "value": 3}],
  "trouble": {"id": "hide_resource", "resource": "provisions", "amount": 5},
  "companion_bonus": 3
}
```

The complete `rift.json` is given in Section 27.13.

38.16 Web UI bridge protocol

- **JS → Python:** `pycmd("ankicaster:" + JSON.stringify({id: 17, cmd: "build.place", args: {...}}))`. Register with `web.set_bridge_command(handler, context)`.
- **Python → JS (reply):** `web.eval("AC.reply(17, " + json + ")")` with `{"ok": true, "delta": {...}}` or `{"ok": false, "error": "not_enough_mana", "message": "You need 40 more mana."}`.
- **Python → JS (push):** `web.eval("AC.push(" + json + ")")` after bells, ticks and notices while the window is open (throttled to ≤ 4 per second).

Command	Args	Effect
`state.get`	—	Full snapshot
`wizard.queue` / `wizard.pause` / `wizard.resume` / `wizard.cancel` / `wizard.reorder`	session data	Training management
`build.validate`	type, x, y, area	Returns rule violations, adjacency preview, Veil delta (no state change)
`build.place` / `build.cancel` / `build.pause` / `build.reorder`	…	Construction
`room.level` / `room.renovate` / `room.demolish` / `room.set_crop` / `room.toggle`	room id …	Room actions
`furniture.buy` / `furniture.place` / `furniture.move` / `furniture.remove` / `furniture.sell` / `furniture.repair`	…	Furniture
`recruit.enroll` / `recruit.reroll`	applicant id / school	Enrollment (re-validates the tracker server-side)
`character.assign_career` / `character.sabbatical` / `character.enroll_second` / `character.swap_card` / `character.move` / `character.rename` / `character.card_preview`	…	Roster actions
`draft.draw` / `draft.keep` / `draft.reroll` / `hand.play` / `hand.discard`	…	Draft
`alchemy.brew` / `potion.use` / `scroll.craft` / `recipe.research`	…	Crafting
`creature.assign_tamer` / `creature.release` / `creature.recapture` / `creature.rename`	…	Menagerie
`rift.plan`	destination, team, companion, supplies	Returns the chance and modifier breakdown (no state change)
`rift.launch`	same	Launches; fixes the hidden roll
`event.choose`	event id, option	Resolves an event
`spell.cast`	spell id, target	Casts
`veil.reinforce` / `veil.pay_now`	—	Veil actions

Command	Args	Effect
`research.start` / `research.cancel`	research id, researcher	Arcane research such as Veil Theory (Section 18.2)
`egg.place` / `golem.awaken`	egg id, habitat room / workshop id	Eggs and golem cores (Section 27.10)
`term.commence` / `term.start`	…	Modes
`settings.update`	key, value	Updates config or save settings

Every mutating command runs inside one SQLite transaction; on any exception the transaction is rolled back and the error is returned to the UI.

38.17 Performance budgets

Operation	Budget
Review hook (main thread)	≤ 1 ms (append to queue)
`process_pending` per review	≤ 5 ms
Bell	≤ 30 ms (deferred, never inside the hook)
Day tick	≤ 300 ms total, with collection reads in `QueryOp`
Window open to first paint	≤ 500 ms
Save size after a year	≤ 20 MB (ledger rows are compact; prune `notices` older than 60 days; archive ledger rows older than 400 days into `day_log` aggregates)

38.18 Errors, logging, backups and migrations

- Every hook handler and bridge command is wrapped: exceptions are logged to `user_files/<profile>/logs/ankicaster.log` (rotating, 1 MB × 3) and surfaced only inside the game window as “Something went wrong in Ankicaster — your reviews are safe.” Never show a modal during review.
- **Backups:** on the first profile open each day, copy the save to `backups/ankicaster-YYYYMMDD.sqlite`; keep the latest 7. Settings → Data offers export (JSON) and restore.
- **Migrations:** `meta.schema_version`; on open, apply `storage/migrations/NNN_*.sql` in order inside a transaction, after making a backup.
- **Content versioning:** saves store content ids, not content values, so retuning numbers takes effect immediately; removing an id requires a migration that maps it.

38.19 Profiles and sync

- One save per Anki profile, keyed by `mw.pm.name`. Renaming a profile: the add-on looks for the previous folder via a `profile_id` stored in the profile’s own meta (fallback: ask the user to pick the folder).
- **The save is not synced by AnkiWeb.** Mana from mobile reviews still arrives on desktop via the revlog catch-up scan after sync (if enabled). Document: “Your Academy lives on this computer. Export your save to move it.”
- Two desktop computers syncing the same collection would each run their own Academy; the catch-up scan would count each review on both. This is acceptable (each Academy is honest about the shared revlog) and documented.

38.20 Packaging

```
// manifest.json
{
  "package": "ankicaster",
  "name": "Ankicaster",
  "mod": 0,
  "min_point_version": 66,
  "human_version": "1.0.0",
  "homepage": ""
}
```

Build a `.ankiaddon` by zipping the add-on folder contents (not the folder itself), excluding `user_files/`, `tests/` and `__pycache__/`.

39. Build Milestones and Acceptance Criteria

Each milestone must be playable, must keep all guardrail tests green (Section 40.1), and must meet its acceptance criteria before the next begins.

#	Milestone	Scope	Acceptance criteria
M0	Skeleton	Add-on loads; Tools menu opens an empty AnkiWebView window; save DB created per profile; logging; content loader with validation	Installs on Anki 2.1.66 and the latest release without errors; opening/closing the window 20× leaks nothing; `test_readonly_guard` passes
M1	Honest mana	Ledger with H1–H7; live hook + deferred revlog check; undo reconciliation; catch-up scan; toast; top-bar mana link	(a) Pre-install revlog grants 0. (b) Again/Hard/Good/Easy each grant exactly 10. (c) Undo removes the mana. (d) A 300 ms answer grants 0. (e) The 7th answer to one card in a day grants 0. (f) Reviews synced from another device are counted once. (g) No collection writes: a scripted 200-review session produces identical dumps of the `cards`, `notes`, `revlog`, `decks` and `config` tables with the add-on enabled and disabled
M2	Wizard	Creation; Study; orb sessions with queue/pause; 25×n; buffs 6–10; level unlock table	Level n completes after exactly 25×n honest reviews; total to L10 = 1,375; buffs read correctly in the modifier breakdown
M3	Building	Grid, placement rules P1–P7, reachability BFS, construction slots, standard plans for MVP rooms, required furniture, efficiency formula, demolish	All placement rules have unit tests; an Unreachable room shows the badge and runs at 0%; construction completes after its review count
M4	Students	Linking algorithm; enrollment tracker; applicants; beds/seats; constellation UI; skill; mastery-ready; teacher delay (wizard); master title; link maintenance (missing/suspended/new)	Linking picks the youngest free young cards and never shares a card (DB constraint + test); tracker blocks enrollment below 50; skill equals the formula from `cards.ivl`; a forced lapse lowers skill; mastery requires all >30 and the delay

#	Milestone	Scope	Acceptance criteria
M5	Days, Veil, breaks	Day tick; Veil upkeep, strength, bands; away floor; arrival summary; Welcome Back	Simulated 40-day break matches the Section 18.5 example exactly; nothing in the save other than bank, Veil ledger and day log changes during a break
MVP	First playable	M0–M5 + Dormitory, Refectory, classrooms, Faculty Quarters, Storage, Woodlot, Quarry, Vegetable Patch; basic needs and meals	A tester can play a week; the “workable and fun” definition in Section 1.5 is met for days 1–7
M6	Living Academy	Full needs, morale, food chain, greenhouse and crops, cleanliness, adjacency rules, Common Room, Kitchen, Bathhouse	Food balance (Sections 21.4–21.6 and 33.4) holds in the simulator: a 10-student Academy with two Vegetable Patches and one Kitchen never enters shortage at 150 reviews/day; morale responds to services as specified
M7	Careers	Careers, proficiency, faculty limit, teacher delay from teaching cards, career switching rules, sabbatical	Faculty limit equals min(A,B,C,D) in all tests; switching blocked for single-school masters; old career cards kept
M8	Furniture	Full furniture catalog, quality score and tiers, slots, damage/repair, patterns	Q and tier match hand calculations for 10 sample rooms
M9	Draft	Decks, prices, rarity weights, pity, guaranteed offers, hand, Known Plans	Price follows ×1.15^k with 7-day window; pity triggers on the 11th draw
M10	Creatures	Habitats, visitors, lures, taming, food, bond, escapes, recapture	Escaped creatures always return; hidden resources are returned
M11	Trouble	Hazards, breaches, wards, misbehavior, detention, injuries, Infirmary, healers, warders	No hazard or incident deletes anything (property test over 10,000 random bells)
M12	Alchemy	Lab, recipes, discovery, research, potions, scrolls	All 18 recipes brew; no potion grants mana

#	Milestone	Scope	Acceptance criteria
M13	The Rift	Opening at 3 masters, Gate, Yard, adventurers, party skill from adventurer career cards, planner with chance meter, resolution, “when it goes wrong” table, loot pools, eggs and golem cores, Fortune, companions	For every destination, chance equals Section 27.5 (safe odds, 5-day steps, team, elixir, Thin Veil, best-odds cap, 5% floor) in table tests; outcome fixed at launch; 100,000 simulated resolutions match the Draft 8 percentages within ±0.5%; Fortune never changes the chance; no character is ever removed
M14	Life and meta	Events, spells, achievements, Renown, alumni, spellbooks, Timed Term and Endless, term reset	Reset carries wizard levels, alumni bonuses and spellbooks and resets characters and creatures (and releases all card links)
M15	Polish	Art pass, animations, accessibility, Codex, tutorial, settings, performance budgets	Meets Section 38.17 budgets; tutorial completable; Reduce motion works

40. Testing Plan

40.1 Guardrail tests (must always pass; run in CI on every change)

Test	What it proves
`test_readonly_source_scan`	Greps the whole add-on source (excluding `tests/`) for every forbidden API in Section 38.4 (`col.db.execute`, `update_card`, `update_note`, `sched.answer`, `set_due_date`, `suspend`, `bury`, `forget`, `reschedule`, `add_note`, `remove_notes`, `col.set_config`, `mw.col.decks.save`, …) and fails on any hit outside `adapters/readonly.py`
`test_readonly_guard`	`ReadOnlyCollection` rejects any SQL that is not a single `SELECT` (including `WITH … DELETE`, `PRAGMA`, `;`-chained statements and `ATTACH`)
`test_collection_unchanged`	Scripted review session in a test profile; table dumps identical with the add-on on and off
`test_no_backfill`	A collection with 10,000 pre-install revlog rows grants 0 mana on install and after a catch-up scan
`test_equal_buttons`	Ease 1, 2, 3 and 4 each grant exactly `mana_per_review`
`test_no_count_rewards`	Static check: no content file has an effect keyed on review counts per day, streaks, speed or cards added; no achievement condition references `honest_review_count`
`test_breaks_delete_nothing`	Snapshot the save; simulate breaks of 1, 7, 40, 365 and 2,000 days; assert every row in `characters`, `card_links`, `rooms`, `furniture`, `creatures`, `wizard_levels`, `spellbooks`, `alumni_bonuses` still exists and only `wallet`, Veil, `day_log`, `notices` and applicants changed
`test_cards_never_shared`	Property test: 1,000 random enroll / career / retire / release sequences never produce a duplicate `anki_card_id` (also enforced by the primary key)
`test_no_deaths`	Property test over 10,000 random bells and 1,000 Rift resolutions: no character or creature row is deleted or set to a terminal state
`test_hook_budget`	The review hook returns in ≤ 1 ms (median) and process_pending ≤ 5 ms on the reference save

40.2 Unit tests (core, no Anki needed)

Core modules take plain data, so they run under `pytest` without Anki.

- **Mana:** each honesty check H1–H7 accept/reject; multiplier cap; debt repayment; undo reversal with and without spending.
- **Day mapping:** revlog id → Anki day around the cutoff (one millisecond before/after), across DST changes.
- **Wizard:** 25×n per level; cumulative 1,375; queue behaviour; pause keeps progress; cancel refund rules; each school's buff at levels 6–10.

- **Linking:** youngest-first order; honest interval rule; sibling avoidance and relaxation; not enough cards → None; deck scope with filtered decks (`odid`).
- **Skill & mastery:** skill formula; lapse lowers skill; mastery-ready only when every card > 30; teacher delay for 20 teaching cards at intervals 0, 15, 30 and mixed; ready_days pauses not resets; title permanent after a later lapse.
- **Careers:** faculty limit min(); switching blocked for 1-school masters; old cards kept; sabbatical.
- **Retirement:** triggers only when every card of the character (all sets) exceeds `retirement_age_days`.
- **Veil:** cost formula; reductions capped at 50%; band thresholds; the 40-day example of Section 18.5 reproduced exactly; floor at 10% of `bank_at_last_review`.
- **Hazards/misbehavior:** probability formulas and caps; every outcome reversible.
- **Construction:** placement rules P1–P7; BFS reachability; build slots; level-ups; demolition refunds.
- **Quality:** Q, tiers and efficiency for sample rooms.
- **Draft:** price ×1.15^k over a rolling 7 days, cap 2.5×; rarity weights; pity.
- **Rift:** party skill = mean honest interval of all members' adventurer career cards (uncapped); safe odds per destination; −10 per full 5 days below and +5 per full 5 days above; team −10/−25; Rift Elixir +10; Thin Veil −5; best-odds cap; 5% floor; Draft 8 "when it goes wrong" percentages; half rewards rounded up; roll fixed at launch; Fortune affects rewards only.
- **RNG:** named streams reproduce the same sequence after save/reload.

40.3 Integration tests (inside Anki)

Use a throwaway profile and a generated collection (500 notes, varied intervals, some suspended cards, a filtered deck).

1. Install, open the window, create the wizard, review 30 cards, check 300 mana and the toast.
2. Undo the last review: mana −10, session progress −1.
3. Sync a review from a second test client; check catch-up counts it once.
4. Enroll a student; delete one linked card in the browser; next day tick replaces it.
5. Suspend a linked card; after 31 simulated days it is released and replaced.
6. Close Anki mid-bell (kill the process); reopen; the save is intact (transactions) and no review is counted twice.
7. Switch profiles; each has its own Academy.
8. Upgrade from the previous schema version with a real save; migrations run and a backup is created.

40.4 Simulation harness (`tests/sim/simulate.py`)

A headless simulator that drives the core with synthetic review streams. It is the main balancing tool.

- **Inputs:** reviews per day (constant, or a distribution such as 60–250), new cards per day, a card-maturation model (interval growth ×2.5 on pass, reset on lapse at a configurable lapse rate), break schedules, and a **policy** (a simple greedy bot that trains the cheapest useful level, builds rooms from a priority list, enrolls when possible, assigns careers, launches the safest Rift tier).

- **Outputs (CSV + summary):** day of each milestone in Section 34; mana income vs spending by category per week; idle mana (unspent bank) over time; Veil strength; food balance; number of hazards and incidents per week; Rift outcomes.
- **Balance checks** (fail the run if violated):
  - First master between day 40 and day 75 at 150 reviews/day (Section 34.2).
  - Rift open between day 48 and day 90 at 150 reviews/day.
  - After the tutorial, the bank never stays above 5,000 mana for more than 7 days with nothing to buy (the user should always have a goal).
  - Veil upkeep stays between 10% and 35% of daily income for a policy that builds steadily.
  - A 30-day break followed by return never leaves the Veil below Thinning on the day of return for a user who had at least 10 days of upkeep banked.
  - 100 vs 200 reviews/day: the 200 user reaches milestones sooner, but **no reward grows faster than linearly** with reviews (no escalation).

40.5 Manual QA checklist

- Toast does not steal focus or block the answer buttons; it is click-through.
- Every number with a modifier has a working source tooltip.
- Reduce motion stops all looping animation; text scale 1.5 keeps every screen usable at 1280×720.
- Keyboard navigation reaches every button; focus is visible.
- Light and dark Anki themes both render the window legibly.
- No string implies the user studied “too little”, “too slowly” or should add cards.

41. Appendix A — Formula Reference

One place for every formula, so the builder can implement `core/formulas.py` from this page. All numbers are starting values to tune (Section 33) unless the formula is binding (marked **BR**).

#	Quantity	Formula	Section
F1	Mana per honest review	`mana_per_review × (1 + min(0.25, arcane_buff + other_mana_mults))`; Arcane buff = 0.02 × max(0, L_arcane − 5) (**BR-09**)	5, 7
F2	Revlog id → Anki day	`today − floor((day_cutoff_ms − 1 − revlog_id) / 86_400_000)`	38.10
F3	Reviews for wizard level n	`25 × n` (**BR-07**); cumulative to L10 = 1,375	7
F4	Orb cost for level n	`80 × n` mana	7
F5	Buff steps in a school	`max(0, L − 5)` (0–5); per-step values per **BR-09**	7
F6	Honest interval	`max(0, min(cards.ivl, ivl of last type 0–3 revlog))`	9.2
F7	Skill	`100 × mean over linked cards of min(honest_ivl, 30) / 30`	10
F8	Mastery-ready	every linked study card has honest_ivl > 30 (**BR-15**)	10
F9	Teacher delay (teacher)	`max(0, 30 − mean(honest_ivl of the teacher's 20 teaching cards))`; 0 once all 20 are over 30 days, × (1 − 0.03 × Light steps), rounded up (**BR-16**)	10
F10	Teacher delay (wizard stand-in)	`20 − 2 × (L_school − 5)`, × Light factor, rounded up; unstaffed classroom 30	7, 10
F11	Career proficiency	`0.75 + 0.5 × career_skill / 100`	11.3
F12	Faculty limit	`min(1 + floor(total_levels/4), faculty_beds, 1 + 2 × classrooms, 1 + floor(students/3))` (**BR-22** inputs)	11.6
F13	Retirement	every card of the character in every set has honest_ivl > `retirement_age_days` (default 90) (**BR-21**)	13

#	Quantity	Formula	Section
F14	Alumni bonus scaling	`bonus × clamp(retirement_age_days / 90, 0.67, 2.0)`	13
F15	Need update per bell	`need += (target − need) × 0.25 − decay × multipliers`, clamp 0–100; decay Hunger 4, Rest 3, Leisure 3, Composure 1	14.4
F16	Morale	`0.25 Hu + 0.25 Re + 0.20 Le + 0.20 Co + 0.10 He + bonuses`, clamp 0–100	14
F17	Morale factor	`0.6 + 0.6 × morale / 100` (0.6–1.2)	8, 14
F18	Essence per bell (classroom)	`0.20 × efficiency × morale_factor × (1 + teaching_buff)` per student	8
F19	Room quality	`Q = Σ primary-stat points + 0.5 × Σ secondary-stat points`; tiers Bare <10, Plain 10+, Cozy 25+, Fine 45+, Splendid 70+	15
F20	Quality factor	`1 + min(Q, 80) / 160` (1.0–1.5)	15
F21	Room efficiency	`level × quality × (0.7 + 0.3 × cleanliness/100) × staff (1.0 or 0.5) × adjacency (clamp 0.8–1.3)`; 0 if Unreachable	15
F22	Build time	`build_reviews × (1 − 0.03 × Alchemy steps)`; materials × the same factor (**BR-09**)	15
F23	Veil daily cost	`(40 + 3 × size_points) × (1 − min(0.50, reductions))`; reductions from Shadow levels (−3%/step), warders (−3% each, max 3), creatures, Veil Theory research (−3%/rank) and draft cards (**BR-24**, **BR-09**)	18
F24	Veil strength	paid day +5; unpaid day `−20 × unpaid_fraction`; Reinforce +15; clamp 0–100; bands Strong 80–100, Thinning 50–79, Thin 20–49, Torn 0–19; warders +5 and Standing Stones +2 per active day	18

#	Quantity	Formula	Section
F25	Away payment	each away day pay `clamp(bank − 0.10 × bank_at_last_review, 0, cost)`; stop at the floor; strength never changes on an away day (**BR-26**)	18.5
F26	Hazard chance per bell	`min(0.75, 0.02 × band_mult × (1 + size_points/200) × (1 − hazard_reduction))`; band_mult 1 / 1.5 / 2.5 / 4	19
F27	Misbehavior chance per student per bell	`0.004 × personality × traits × morale_mult × (1 − reduction) × room_order_factor`	20
F28	Taming progress per review	`1.0 × tamer_proficiency × (1 + nature_buff) × modifiers`	22
F29	Healing per review	`1.0 × place_factor × healer_factor × (1 + light_buff) × modifiers`	23
F30	Potion yield	`2 × (1 + 0.03 × Alchemy steps + creature and item bonuses)`; fractions accumulate per recipe; an alchemist adds +1 potion every 4 brews	24
F31	Draft price	`min(2.5, 1.15^k) × base`, k = draws from the deck in the last 7 days; Known Plan 1.5 × plan price	26
F32	Rift party skill	mean honest_ivl (uncapped, days) over all adventurer career cards of the team members (**BR-31**)	27.4
F33	Rift safe-return chance	`max(5, min(best, safe + skill_mod + team_mod + elixir + veil_mod))`; skill_mod = −10 per full 5 days below needed, +5 per full 5 days above; team −10 (two) / −25 (alone); Rift Elixir +10; Thin or Torn Veil −5 (**BR-32**)	27.5
F34	Rift outcome	roll 1–100 at launch: ≤ chance → safe return (full rewards); else second roll on the Easy–Medium (60/25/10/5) or Hard–Deadly (40/30/20/10) table: minor injury (full) / serious injury (half) / curse (half) / retreat (none) (**BR-33**)	27.6

#	Quantity	Formula	Section
F35	Rift Fortune	each point +10% to reward quantities, max 5 points; never changes the chance	27.9
F36	Timed-term reset	bank → 0; buildings and materials → starting layout and stock; all characters and creatures removed and their card links released; wizard levels, alumni bonuses and spellbooks kept (**BR-30**, **BR-34**)	30.4
F37	Egg hatching	20 / 40 / 60 / 80 / 100 honest reviews (Common → Legendary) in a matching habitat	27.10

42. Appendix B — Design Decisions Made in This GDD

The approved concept (Draft 8) fixed the rules listed in Section 2 (BR-01 to BR-35, checked against the Draft 8 page). Everything below was decided while writing this document, to fill gaps. Each item can be changed without breaking a binding rule. The owner should review this list.

42.1 Time, mana and honesty

1. **Bells.** One bell = 10 honest reviews. Needs, production, meals (every 5 bells), dirt, hazards, misbehavior and escapes all run on bells, so the Academy "lives" only while the user studies and sleeps while they are away.
2. **10 mana per review**, with every mana multiplier combined capped at +25%.
3. **Honesty checks H1–H7** beyond the binding ones: answers faster than 800 ms earn nothing; at most 6 counted answers per card per Anki day; reviews from other devices are counted through a catch-up revlog scan after sync (on by default, can be turned off).
4. **Undo** reverses a review's mana and progress; if the mana was already spent it becomes a debt paid from future reviews. Things a review completed stay completed.
5. **Honest interval** = the smaller of the card's current interval and the interval recorded at its last real answer, so "set due date" or manual rescheduling cannot fake maturity.
6. **Five school essences** (Insight, Herbs, Radiance, Umbra, Reagents) are produced by students' lessons and spent on school-flavoured items, so mana stays the one neutral currency.
7. **Renown** is a separate prestige stat from rooms, masters, achievements and alumni. It never affects mana.

42.2 Wizard

8. **Orb vs grimoire:** pondering an orb raises a school level; studying a grimoire teaches one of 25 wizard spells. One active session, plus a queue of 1, 2 or 4 by Study level; pausing keeps progress; cancelling refunds only if no progress was made.
9. **Orb cost** 80 × n mana for level n.
10. **The wizard can teach one classroom** per teachable school; teacher delay for the wizard = 20 − 2 × (level − 5) days; an unstaffed classroom's delay is 30 days.

42.3 Students, mastery and careers

11. **New (never-studied) cards are not linked** by default; "young" means an honest interval of 30 days or less; siblings are avoided when possible.
12. **Link maintenance:** deleted cards are replaced at the next day tick; cards suspended, or reset to new, for more than 30 days are released and replaced; if more than 25% of links vanish at once, the game stops and asks before changing anything.
13. **Career card sets** take the 20 youngest free *young* cards first (BR-18), and fall back to the youngest free older cards only when there are not 20 free young cards. The career screen shows the split.
14. **Skill** = average of min(interval, 30)/30 over linked cards, shown as 0–100.

15. **Mastery countdown pauses** (does not reset) when a lapse makes a student not ready.
16. **Faculty limit** = the smallest of: 1 + total wizard levels ÷ 4; faculty beds; 1 + 2 × classrooms; 1 + students ÷ 3.
17. **Career proficiency** 0.75–1.25 from career-card skill.
18. **Sabbatical** lets a faculty member pause a career and resume the same one later (keeping its cards).
19. **Enrollment fee** of 60 mana, plus a bed and a classroom seat; the applicant board shows 3 applicants per school, rerolled for 100 mana.
20. **Titles** go up to Paragon (masters of all five schools, +1 Rift Fortune), and a resident master can tutor (a small bonus to classmates).
21. **Alumni bonus size** scales with the retirement age setting (×0.67–×2.0) so a shorter retirement age is not a shortcut; alumni visit occasionally (3% per day).
22. **Eight personalities, about 34 traits, five needs** (Hunger, Rest, Leisure, Composure, Health). Needs never tick while away; there is no starvation, and shortages only lower morale.
23. **Changing cards per student** resizes every existing student's set at the next day tick (it applies to all students, BR-12). Growing sets take the youngest free young cards; shrinking sets release their most recently linked cards. Titles already earned are kept.

42.4 School, rooms and furniture

24. **Grid model:** castle starts 8×5 cells and expands to 14×11; grounds start at 6 plots and expand to 18; floors connect only by stairs; rooms must be reachable from the Entrance Hall.
25. **Room levels 1–3**, by a duplicate blueprint or renovation.
26. **Five room stats** (Comfort, Focus, Order, Aura, Charm), quality tiers Bare → Splendid, efficiency from level, quality, cleanliness, staff and adjacency.
27. **56 room types**, **114 furniture entries (123 distinct pieces)**, 6 legendary Rift furniture pieces, outdoor decor.

42.5 Veil, trouble and creatures

28. **Veil formula:** daily cost (40 + 3 × size points) mana, reductions capped at 50%; strength 0–100 with four bands that multiply hazards; paid day +5, unpaid −20 × unpaid share. Auto-pay by default; "ask" is an option. Strength never changes on an away day, paid or not (BR-26).
29. **Veil reductions for Draft 8's five kinds:**
    - Warders: −3% each, up to 3.
    - Creatures: Umbral Wyrmling −5%, Velvet Bat Swarm −2%.
    - Arcane research: *Veil Theory* I–III at the Library, −3% per rank.
    - Draft cards: *Veil Thread* (−10% for 7 days) and *Warding Primer* (permanent −3%).
    - Shadow wizard levels follow BR-09.
30. **Hazards and breaches never delete anything:** they lock a room for some reviews, hide resources that are later found, damage furniture (repairable), or cause injuries. Consumables can lose at most 5%, and only during active play.
31. **Misbehavior** comes in 16 incidents, each with a silver lining, plus detention.
32. **Creatures:** 40 species in five rarities. Three were added to match Draft 8's Rift rewards: the Fen Sprite and Mistfeather Owl ("owls and sprites") and the Geode Golem (awakened from "golem cores"). taming takes 40/90/160/260/400 reviews by rarity; three Bond levels that never decrease; escaped creatures always come back.

33. **Injuries:** five types healed by reviews, faster with a healer; no disease and no permanent harm.

42.6 Crafting, draft and Rift

34. **18 fixed potion recipes** of three ingredients each, discovered by brewing or research, optionally shuffled per Timed Term; no potion grants mana.
35. **22 artifacts** that work only while displayed in the Trophy Hall; 4 scrolls. Artifact, rare-material, seed and legendary-furniture sources were re-mapped to the five Draft 8 destinations: minor artifacts come from the Ashen Wastes or the Shattered Spire, and major ones only from the Spire.
36. **Card draft:** 6 decks; 120 mana for the Foundation deck and 150 for school decks; price × 1.15 per draw in the last 7 days (cap × 2.5); a pity rule; draft cards are consumed when played; Known Plans can be rebought at 1.5×.
37. **Rift details inside the Draft 8 rules** (the destinations, odds, modifiers and outcome table themselves are binding, BR-31 to BR-33):
    - **Mana costs:** launch costs of 150 / 300 / 500 / 800 / 1,200 mana.
    - **Logistics:** supplies of 5–30 Provisions per member, loot rolls of 2–6 and supply slots of 1–3.
    - **Party skill:** the mean honest interval of all members' adventurer career cards, not capped at 30 days. Empty slots count as 0.
    - **Skill steps:** the "each 5 days" modifiers use whole 5-day steps.
    - **Thin Veil:** means the Thin or Torn band (strength below 50).
    - **Odds floor:** the chance never drops below 5%.
    - **Curse rewards:** a curse brings half rewards. Draft 8 does not say.
    - **Injury mapping:**
        - Minor injury: Bruised (Easy–Medium) or Sprained (Hard–Deadly).
        - Serious injury: Sprained at Mossy Hollow, Broken Bone elsewhere.
        - Curse: Hexed (Easy–Medium) or Rift-Touched (Hard–Deadly).
        - Retreat: no injury, a short morale dip.
    - **Half rewards:** half the loot rolls, rounded up.
    - **Access:** all five destinations are open as soon as the Gate is built. The old "unlock the next tier" rule was dropped.
    - **Destination flavour and loot pools:** "Rare herbs" are Herbs plus rare herb seeds. "Rare cards" are Rare, Epic or Legendary draft cards.
38. **Rift Fortune.**
    - **Purpose:** every pre-existing expedition bonus (companions, personalities, traits, Training Yard, Gate levels, Rift Map, artifacts, spellbooks, alumni, spells, events) was converted into Fortune. Fortune is +10% reward quantity per point, up to 5 points.
    - **Odds untouched:** only Draft 8's six modifiers change the odds. The Rift Elixir replaces the old Clarity Philter as the one odds item.
    - **Recovery effects:** severity-shifting effects (Courage Draught, Brave, Steadfast, Daredevil, Ember Phoenix) now change recovery time, because injury types come from the binding table.
39. **Eggs and golem cores:**
    - Eggs hatch after 20–100 reviews in a matching habitat and never spoil.
    - Three Golem Cores awaken a Geode Golem. Two are used in the Rift Gate level-3 renovation.

40. **The Rift roll is made at launch** and stored hidden, with the second “when it goes wrong” roll, so reopening the game cannot reroll either.

42.7 Modes, meta and interface

41. **Timed Term:** default length 120 days, exam timer off by default, charter goals for stars. Within the binding reset (BR-34: buildings, materials, unspent mana, students, faculty and creatures), Renown, Known Plans, patterns, the draft history and research ranks also reset. Achievements and the Alumni Gallery are kept as meta records. Masters at term end become alumni. The new term starts with **0 mana**. The earlier “Term Purse” was removed because Draft 8 resets unspent mana, and because a free purse would conflict with BR-01.
42. **Endless School** earns spellbooks at Renown milestones.
43. **21 spellbooks**, about 40 achievements (none tied to review counts), 30 random events, 25 spells and 7 passives.
44. **Interface:** a non-modal window from Tools ▸ Ankicaster, a mana link in the top toolbar, Ctrl+Shift+M, an optional dock panel, a click-through toast after each bell by default; sound off by default.
45. **Art:** a flat, hand-drawn storybook look built from SVG and CSS so no large art pipeline is needed.
46. **Technical:** SQLite save per profile in `user_files/`, seeded named random streams, 7 daily backups, and all content in JSON files.